The Osborne 1 Compendium

A complete reference, assembled from the project's own research

o1nextgen

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DRAFT. This book is assembled from the project's working notes, which are explicitly partial: chapters carry a status and some are not written yet. Where a chapter is incomplete, the source document says so and says why. The repository is the authority.

The machine

Overview and history

(not yet written)

Mainboard and the schematic set

`research/osborne1/OCC1_1A2011-00_Schem_RevE.pdf` is the primary source for the O1′s logic board. Until now it was read by eye, page by page, in a PDF viewer: the scan is dense, the PDF's own text layer is unusable, and the 12 pages are actually **two different drawing sets**, which is easy to get wrong. This document is the map of that archive item: what is in it, how to read it, and what has been extracted from it so far.

1. What the file actually contains

12 pages. Each page is a single 1-bit JBIG2 scan at roughly 600 dpi (6582 × 5137 px on a 785 × 610 pt sheet) — genuinely high quality; every pin number is legible at native resolution.

Each page *also* carries a text layer, but it is an early-1990s OCR pass and is garbage: the disc-controller sheet's title reads `J'U~ I 'IIS IO/OO-tOOnl`. Ignore it. (It is left in the PDF; the fresh pass in §3 is the one to search.)

The 12 pages are **two drawing sets**:

Pages	Drawing	Rev	Sheets
1–3	**DISK ELECTRONICS**, DWG 1A3004-00/01	C	1–3 of 3
4–12	**MAIN PC BOARD**, DWG 1A2011-00	E	1–9 of 9

This distinction matters. The disk-electronics set (1A3004) is the *drive's own* board — read amplifier, data separator, write driver, head connectors. The mainboard set (1A2011) is the machine this project extends. Notes in this repo have cited "DWG 1A3004" for mainboard facts, including the floppy connector pinout; that citation is wrong (§5).

`research/README.md` describes the split as "sheets 1–3 are the disk analog board (DWG 1A3004), sheet 4+ the main logic board" — correct in outline; the exact drawing numbers and sheet counts are as tabulated above.

2. Sheet index

Titles are the drawing's own title-block text (sheet 1 of the mainboard set has no block title; its function is read off the circuit).

PDF page	Drawing / sheet	Function	Devices read off the sheet
1	1A3004 1/3	Disk read electronics: head pre-amp and peak detection, ALC, head connectors P1/P2	U9, U10, U15, U16, U18, U19, U20
2	1A3004 2/3	Disk data separator: index/sector, read data, drive connector P2	U8, U12–U22
3	1A3004 3/3	Disk write driver, erase, motor control, ±12 V/+5 V rails	U1, U2, U3, U6, U11, U12, U13
4	1A2011 1/9	CPU clock (4 MHz), reset, data buffers, power entry	UD11, UD13, UE3, UE20, UC7, **UC11 = 74LS161** clock counter, UA6, UA8, J5, P6 (battery)
5	1A2011 2/9	Address decode, ROM/RAM/I-O select, WAIT generation	UC1, UC2, UD1, UD2, UD3, UD9, UD12, UE3, Q1/Q2 (reset timing)
6	1A2011 3/9	**CPU**: Z80-A, monitor EPROM, address latches	**UC11 = Z80-A**, **UD11 = 2732**, UD4, UD6–UD10, UE6, UE10, UE11, UA4
7	1A2011 4/9	**RAM** array, address multiplex/buffer, **character generator ROM and its serialiser**	**UA15 = 2716** chargen ROM and **UA14 = 74166** pixel serialiser (§4a), **UA18 = 74LS273** character latch (drives UA15 `A0–A6`), UA17/UA18 and the UA23–UA48 array (2114/2112 static RAMs per sheet 1′s parts list), UE13–UE15 (74LS244), UE16–UE18 (74LS158),
10	1A2011 5/9	[illegible]	UE25 (74LS[illegible]), UD[illegible] (74LS374), [illegible]

Sheets 4–12 are numbered 1–9 of 9 in their title blocks; the mainboard set is complete in this scan.

> **Read the device lists at native resolution before citing them.** The *functions* in the last column are reliable; the part numbers came from a whole-sheet pass and at least two are known wrong (§3). `UC11` in particular appears twice — as the Z80 (sheet 3/9, verified at 600 dpi) and as a 74LS161 (sheet 1/9, overview pass); only one can be right.

3. How to read a sheet

Whole-sheet views are useless — scale a 6582 px sheet onto a screen and the pin numbers vanish. Read it one region at a time at native resolution. Two tools:

```
# browse: an overview plus a grid of native-resolution tiles for every sheet
tools/schematic_atlas.sh research/osborne1/OCC1_1A2011-00_Schem_RevE.pdf
#   -> build/atlas/index.md          tile map: page -> tile -> pixel box
#   -> build/atlas/sheet-05/r1c0.png native-res tiles you can just open

# then re-render any region exactly, once the index says where it is
tools/schematic_render.sh research/osborne1/OCC1_1A2011-00_Schem_RevE.pdf /tmp/s
\
    --sheet 5 --dpi 600 --crop 2600 1250 1450 1050
```

`tools/schematic_atlas.sh` writes into `build/atlas/` — gitignored, because the atlas is regenerable and the *coordinates* are the durable part. Its index maps every tile to the pixel box it covers, so a region found by eye once can be re-rendered exactly later. `--info` on the render tool prints the sheet size in pixels at a given dpi. Both need poppler (`brew install poppler`), which `tools/ocr_pdf.sh` already depends on.

Read part numbers at native resolution, never from an overview. At 300 dpi `LS00` and `LS08` are indistinguishable, and so are `LS138` and `LS139`. The first pass over this scan was done from whole-sheet views and got both of those wrong on sheet 2: at 600 dpi it plainly reads **`UC1 LS00`** (quad NAND) and **`UB1 LS139`** (dual 2-to-4 decoder — its pin numbers 15/14/13 for enable/A/B and 12/11/10/9 for Y0–Y3 are exactly the second half of a ‘139). The §2 device lists came from that first pass: treat the *functions* as reliable and the part numbers as needing a native-resolution check before anything is built on them. One conflict is already visible in them — `UC11` is listed as both the Z80 (sheet 3/9, read at 600 dpi, and it matches a standard DIP-40) and a 74LS161 (sheet 1/9, overview pass). The Z80 is the verified one.

`research/osborne1/OCC1_1A2011-00_Schem_RevE.ocr.txt` is a fresh OCR pass (per the `research/README.md` OCR convention) so the scan is greppable for component references. It is useful for *finding* things — “which sheet has the MB8877?” — and not trustworthy for pin-level detail. Everything in §4 and §5 was read by eye at 600–1200 dpi instead.

`research/osborne1/OCC1_1A2011-00_Schem_RevE.ocr.txt` is a fresh OCR pass (per the `research/README.md` OCR convention) so the scan is greppable for component references. It is useful for *finding* things — “which sheet has the MB8877?” — and not trustworthy for pin-level detail. Everything in §4 and §5 was read by eye at 600–1200 dpi instead.

4. Extracted: the Z80 socket (UC11) is a standard Z80 DIP-40
Read from sheet 3 of 9 (PDF page 6) at 600 dpi. The symbol is drawn as **UC11, Z80-A**.

Pin	Signal	Pin	Signal	Pin	Signal	Pin	Signal
1	A11	11	+5 V	21	RD	31	A1
2	A12	12	D2	22	WR	32	A2
3	A13	13	D7	23	BUSAK	33	A3
4	A14	14	D0	24	WAIT	34	A4
5	A15	15	D1	25	BUSRQ	35	A5
6	CLK	16	INT	26	RESET	36	A6
7	D4	17	NMI	27	M1	37	A7
8	D3	18	HALT	28	RFSH	38	A8
9	D5	19	MREQ	29	GND	39	A9
10	D6	20	IORQ	30	A0	40	A10

Read directly off the sheet: A1 = 31 … A10 = 40, A11 = 1 … A15 = 5, A0 = 30, D0 = 14, D1 = 15, D2 = 12, D3 = 8, MREQ = 19, RFSH = 28, IORQ = 20, WR = 22, M1 = 27, RD = 21, +5 V = 11, GND = 29, BUSRQ = 25, NMI = 17. The unread rows are the standard Zilog DIP-40 pinout, consistent with every pin that was read.

This closes the Z80 half of `docs/research-gaps.md` gap #3 from the primary source. The O1's socket is wired as a plain Z80 DIP-40, so any board that mates with it can use the datasheet pinout — which is what `tools/check_schematics.py` asserts and what the interposer generators were corrected to use (#62). It also corroborates that fix: the generators had the pin table off by one across all 40 pins and were corrected to +5 V = 11 / GND = 29 / A0 = 30 from the databook — the sheet says exactly that.

Correction — both halves of that paragraph were wrong. It read *"the char-gen socket (UA15) half of gap #3 is still open. UA15 is a 74S244 on sheet 5 of 9."* UA15 is neither a 74S244 nor on sheet 5, and that mis-direction is why #24's second tap point sat unresolved. UA15 is the **2716 character-generator ROM on sheet 4 of 9** (PDF page 7, title block `RAM`). Read it at 1200 dpi — §4a.

4a. Extracted: the character-generator ROM socket (UA15) — gap #3 closed
Found at 600 dpi on sheet 4 of 9 (PDF page 7), read at 1200 dpi. The socket is `UA15`, silkscreened **2716**, drawn with the address pins on the left and the data pins on the right. The pinout is the **standard Intel 2716**, pin for pin:

Pin	Signal	Where it comes from / goes
1–8	A7–A0	A0–A6 = character code (7 bits, 128 chars) from **UA18 (74LS273)**; A7 = SCAN0
9, 10, 11, 13–17	O0–O7	the pixel row, to the parallel inputs of **UA14 (74166)**
12	GND	**bussed with pins 18 and 20, then to ground**
18	/CE	**grounded** — the ROM is permanently selected
19	A10	SCAN3
20	/OE	**grounded** — the ROM always drives its outputs
21	(A11 on a 2732)	**strapped to +5 V**
22	A9	SCAN2
23	A8	SCAN1
24	Vcc	+5 V

Address map

SCAN0–SCAN3 arrive as labelled nets (5C1 – SCANn): they are the outputs of the scan-line counters **UD17/UD18 (74LS161) on sheet 5 of 9**, which the video PIA UC15 presets from PB0–PB3. With A7 = SCAN0 and the character code on A0–A6, the address is

```
address = (scan << 7) | char       scan = {SCAN3,SCAN2,SCAN1,SCAN0}
        = 128 characters x 16 scan rows = 2048 bytes = exactly one 2716
```

The displayed cell is 8 wide x 10 high (24 rows x 10 = the 240-line video all three MAME drivers produce), so scan rows 10–15 of every character are stored but never shown.

Output path — the assumed one was wrong too

O0–O7 do **not** leave the sheet as eight parallel CHAR0–CHAR7 lines. They drive the parallel inputs of **UA14 (74166)**, an 8-bit shift register on the same sheet, which serialises the pixel row and exports it as the single net VIDEO → 5D4 to the video sheet. UA14 is clocked by DOT CLK (from sheet 1/9) and its SI and /CE are wired together — the usual trick that shifts in 1s as the inter-character blank. The video sheet's own 4D1 → VIDEO reference (sheet 4, zone D1 — where UA14 sits) corroborates the direction.

This matters to #24: the char-gen socket is a **font source, not a pixel bus**. Reading the font through the tap is enough — the pixels are re-rendered from shadow VRAM.

What the strapping means for a tap

/CE (18) and /OE (20) are **hard-grounded on the mainboard**, so the ROM is always selected and always driving O0–O7. A socket tap may buffer and *read* those pins freely, but it cannot tristate the ROM to inject its own font without intercepting pins 18 and 20 between socket and ROM. Any future "supply your own font" mode needs that interception, and the pin-21 strap held at +5 V (the socket is wired 2732-style, so a 4 KiB part would fit physically).

4b. Extracted: the ACIA's clock and the baud divider (sheet 7 of 9)

Read from sheet 7 of 9 (PDF page 10) at 600–1200 dpi, prompted by two sources that describe the same modification from opposite ends — *The Portable Companion* #13 (Hogan, from inside OCC) and the Technical Manual's own Figure 4.4.

The 6850 (UC4) pin numbers as drawn. The symbol omits Vcc/Vss (pins 12 and 1), which is this schematic's house style — every symbol drops its power pins:

Pin	Signal	Pin	Signal	Pin	Signal	Pin	Signal
2	RXDATA	5	RTS	8	CS2	11	RS
3	RX (clock)	6	TXDATA	9	CS1	13	R/W
4	TX (clock)	7	IRQ	10	CS0	14	E
23	DCD	24	CTS	15–22	D7…D0	12 / 1	Vcc / Vss

The clock is strapped. Pins **3 and 4 are wired together** under a bracket labelled CLK — the ACIA runs both halves from one clock — and that net is driven by **UC3′s QD (pin 11)**.

UC3 is a preset-load counter, not a plain divider. It is a 74LS161 at board position **C3**, clocked (CK, pin 2) from a 4USEC/2USEC tap off the video counter through jumper **J1**, with **RCO (pin 15) inverted by UB3 (74LS04) into /LD (pin 9)** — the classic preset-load ÷N. /CLR (pin 1) is tied to +5 V and the preset inputs carry the divisor as solder-select pads. So the stock 19.2 kHz ACIA clock is (250 or 500 kHz) ÷ 13, and the 6850′s own ÷16/÷64 gives 300/1200 — or 600/2400 with J1 out.

It agrees with the Technical Manual pin for pin. Figure 4.4, "Baud Rate Hardware Modification" (manual p. 22 = **PDF page 36**; the manual's +14 offset holds, docs/o1-chargen-graphics.md), draws the same LS161/LS163 at C3 and documents the factory's faster-rate modification: **pin 2 moved** between a *"FOR 9600/2400 BAUD"* and a *"FOR 4800/1200 BAUD"* pad, jumpers **J2** and **J3**, **J1** marked **"DON'T CARE"**, and the caption box

```
300/1200 NO MOD - J1 IN
600/2400 NO MOD - J1 OUT
1200/4800 MOD W/O PIN 2 (J1 = X)
2400/9600 MOD W/PIN 2 (J1 = X)
```

bracketed *"SPEED SELECTED W/IN 6850"*. The consequence for the serial work is in docs/o1-modem-serial.md; the register-level view is in docs/o1-memory-io-map.md §"6850 ACIA detail".

The same corner of the board is also where a third-party product attaches: Advent Products' **Osbaud** solders to the ACIA's clock and supply pins (4 and 12) and two ground pins, and substitutes the clock outright — `docs/o1-modem-serial.md`.

5. Extracted: the P8 34-pin floppy connector — and a correction
Read from sheet 9 of 9 (PDF page 12, DISC CONTROLLER) at 600–1200 dpi. Every pin label on the P8 symbol was read individually:

Pin	Signal	Pin	Signal	Pin	Signal	Pin	Signal
1	GND	10	DRV SEL 1	19	GND	28	WRT PROT
2	GND	11	+12 V	20	STEP	29	GND
3	GND	12	DRV SEL 2	21	+5 V	30	READ
4	**TG43**	13	+12 V	22	WRT DATA	31	GND
5	GND	14	*(not drawn)*	23	+5 V	32	SIDE SEL
6	**EARLY**	15	+12 V	24	WRT GATE	33	GND
7	*(not drawn)*	16	CLOCK (4 MHz)	25	+5 V	34	LATE
8	INDEX	17	+12 V	26	TRACK 00		
9	GND	18	DIR	27	GND		

Pins 4 and 6 are not ground. Pin 4 is **TG43** and pin 6 is **EARLY**; both are driven by the FDC (UB7) through 7406 open-collector buffers (U9/U10) and routed to the connector. Pins 7 and 14 are not drawn on P8 at all.

This contradicted `docs/floppy-adapter-design.md` §2, which listed `1,2,3,4,5,6,7,9,19,27,29,31,33` as GND "per DWG 1A3004". Two things were wrong with that:

- **The citation.** DWG 1A3004 is the *disk electronics* drawing (§1). P8 is on the mainboard drawing, 1A2011-00 sheet 9. The right source was being overlooked and the wrong one cited.
- **The content.** The transcription that went into the generator and the doc came from the archived Wayne Visser adapter (`research/osborne1/floppy-adapter/FloppyAdapter.sch`), which ties pins 4, 6 and 7 to GND. That is safe for a passive Gotek-only adapter — a 7406 open-collector output pulled low is not damaged and a Gotek ignores TG43 and EARLY — but it is not what the machine does, and this adapter switches the OEM cable through to the physical drives, so it must not clamp two live signals.

Fixed in `hardware/floppy-adapter/gen_floppy_adapter_sch.py`: pins 4 and 6 are now `FLP_TG43` / `FLP_EARLY` and pass straight through J1 ⟶ J2 like every other signal. Grounding TG43 is not merely untidy: it is the FDC's track > 43 output, which is the drive's *write-current-reduce* input on 96 tpi drives, so clamping it low would assert reduced write current permanently.

The 2-versus-7 discrepancy (harmless, recorded for completeness)
DWG 1A3004 sheet 1 shows connector **P2** — the drive end of the same ribbon cable — with `GND = 1,3,5,7,9,19,27,29,31,33` and `CASE = 32`, i.e. **pin 7** grounded and pin 2 not listed. Mainboard P8 shows **pin 2** grounded and pin 7 not drawn. The digits were checked at 1200 dpi against the 27/29/31/33 glyphs in the same font to rule out a misread, so the two drawings really do disagree.

It does not matter electrically: neither pin carries a driven signal on the O1 (every signal the interface has is accounted for on other pins), so both are safe to tie to ground. The adapter ties both, which is correct whichever drawing is right.

6. Running it as a machine
A scan cannot be "run": there is no vector netlist in it, only pixels, so there is nothing to hand to a simulator. Reading a schematic and *executing* the machine it documents are two different jobs. The machine itself, however, runs very well — and it is the better oracle anyway, because it exercises the design rather than our reading of it.

MAME has an `osborne1` driver, and this machine already has everything needed:

- MAME (`brew install mame`)
- the O1 ROMs in `~/Documents/Osborne1/roms/`, including `3a10082-00rev-e.ud11` — the **Rev E** monitor ROM, i.e. the same board revision as this schematic (`ROM_DEFAULT_BIOS("ver144")` in MAME's driver)
- a headless harness in the sibling project: https://github.com/Myrcurial/o1prsnt/tree/main/utilities/headless-testing-harness

That harness is a small Lua library MAME runs via `-autoboot_script`. It injects keystrokes through MAME's natural-keyboard device and reads the **screen straight out of video RAM at `0xF000`** (128 bytes/row × 32 rows), so a test asserts on what the machine is *displaying*. Run it with:

```
cd .../headless-testing-harness
O1_ROMPATH=/tmp/o1roms O1_RUNDIR=/tmp \
  ./run-test.sh mytest.lua ~/Documents/Osborne1/floppies/<image>.imd 90
```

`O1_RUNDIR` keeps MAME's `cfg//snap/` litter out of the repo. `O1_ROMPATH` must point at a directory laid out the way MAME expects — `rompath/osborne1/` or `rompath/osborne1.zip` — not a directory of loose ROM files:

```
mkdir -p /tmp/o1roms/osborne1 && cp ~/Documents/Osborne1/roms/* /tmp/o1roms/
osborne1/
```

Verified working, 2026-10-10. A three-line script (`H.boot_cpm()`, `H.dump()`, `H.finish()`) booted CP/M headless and reached `A>` in 9 emulated seconds with this on screen, read out of VRAM:

```
PIP     .COM   8k        Disk A:  1K blocks
SETUP   .COM   6k        Size=  185K, 13 Files, Used=  140K, Space=   45K
SYSGEN  .COM   2k        A>
XDIR    .COM   3k        Loading CP/M
XREF    .COM   7k        Extended Directory version 3.5
```

With `-nothrottle` the 4 MHz Z80 runs at ~20–25× real time, and because MAME counts *emulated* seconds the timings a test measures are valid 4 MHz Z80 timings. That is what makes this useful for #29: it is a machine-level oracle for the memory map, the I/O decode and bus timing, available before any hardware exists.

7. Cross-checks against the running machine

The emulator is not just a way to run the O1; it independently confirms parts of the extraction. MAME's driver (`src/mame/osborne/osborne1.cpp`) encodes the hardware, and the running machine agrees with `docs/o1-memory-io-map.md`:

- **Banks.** Driver memory map: `0x0000–0x3fff` a ROM view whose low half is the 4 KB monitor ROM `.mirror(0x1000)` and whose `0x2000–0x2fff` window is the I/O area `.mirror(0x1000)`; `0xf000–0xffff` an attribute-RAM view. That is exactly §2 of the map doc (ROM 0000–0FFF mirrored into 1000–1FFF, I/O 2000–2FFF mirrored into 3000–3FFF, video at F000).
- **Bank switching.** Driver I/O map is `map.global_mask(0x03)` with a single `0x00–0x03` write handler, dispatching exactly as §4 of the map doc says: 0 ⟶ select ROM+I/O bank, 1 ⟶ back to main RAM, 2 ⟶ map attribute RAM at F000, 3 ⟶ unmap it. The doc's four I/O writes are the four driver cases.
- **I/O decode.** The driver's window tests — floppy `(off & 0x900) == 0x100`, IEEE-488 `(off & 0x900) == 0x900`, keyboard `(off & 0xa00) == 0x200`, serial `(off & 0xa00) == 0xa00`, video PIA `(off & 0xc00) == 0xc00`, ScreenPac `(off & 0xc00) == 0x400` — match the doc's `2100/2900/2200/2A00/2C00` bases and show the decode is *partial*: peripherals alias widely and their outputs are ANDed together when more than one responds, which is why the doc calls it a "memory-mapped I/O window" rather than a decoded bus.
- **Floppy controller variant (gap #11).** The doc says "Fujitsu MB8877 ≡ WD1793 (UB7)". MAME's driver carries the same uncertainty as a source comment: *"Schematics specify a WD1793 floppy controller, but we're using the Fujitsu equivalent MB8877 here. Is it known that the original machines used one or [the other]?"* Two independent readings of the same drawing, same conclusion — the schematic's symbol is a WD1793-compatible part, the fitted chip may be either.
- **Rev E alignment.** The BIOS MAME loads by default for this machine is `3a10082-00rev-e.ud11` (BIOS 1.44), the Rev E monitor ROM — so the emulator, the archived ROM and this schematic are the same board revision.

7a. Asking the machine directly — `tools/check_o1_map.sh`

Reading the driver is still reading someone else's model of the hardware. `tools/check_o1_map.sh` boots the machine headlessly and interrogates it over its own bus, then asserts what `docs/o1-memory-io-map.md` claims. RAM is told from ROM by writing back the complement of the byte that was there and restoring it — a RAM cell follows, a ROM cell does not. Eleven checks, all passing:

```
$ tools/check_o1_map.sh
program space: mask=0xFFFF width=8   io space: mask=0x0003 width=8
booted to A> after 9.0 emulated seconds
PASS  B1    CP/M reaches the A> prompt in the emulated machine
```

```
PASS  C1    0xF000 behaves as RAM (video RAM is in the RAM bank)
PASS  C2    0x4000-0xEFFF is RAM (176/176 pages)
PASS  C3    I/O write to port 0x00 selects the ROM bank
PASS  C4    I/O write to port 0x01 selects the RAM bank
PASS  C5    the data byte on a bank-switch write is ignored
PASS  C6    I/O ports 0x04/0x05 alias 0x00/0x01 (only A0/A1 decoded)
PASS  C10   the selected bank is re-derived on instruction fetch, not latched
PASS  C7    I/O port 0x02 maps the 1-bit bank-3 plane over 0xF000
PASS  C8    ROM at 0x0000-0x0FFF mirrors 0x1000-0x1FFF
PASS  C9    nominally-dead bank-2 addresses alias real devices (sloppy decode)
== 11 passed, 0 failed ==
```

The page maps it prints are §2 of the map doc in machine form: in bank 2, ROM across `0000-1FFF`, devices across `2000-3FFF`, RAM from `4000` up; in bank 1, RAM across all 64K. Four things it established that the documents do not currently say:

- **The bank latch is real but does not persist.** `OUT (1),A` puts the machine in the RAM bank immediately, and 10 ms of emulated time later it is back in the ROM bank — the bank is re-derived on every instruction fetch from the M1/IRQACK flip-flops, and the video PIA interrupts at 60 Hz. The map doc §4 says the selection is "qualified by M1 and IRQ-acknowledge conditions"; this is what that means in practice, and it is why a naive peek at `0x0000` while CP/M runs returns ROM bytes. Interposer logic that assumes a static bank state is wrong.
- **The I/O space decodes two address bits** (`io space: mask=0x0003`), so all 256 Z80 I/O ports are `0x00-0x03` repeated 64 times — the map doc's "only the two low address bits are decoded", measured.
- **The ROM is live while CP/M runs.** Sampling the Z80's PC during a working `DIR` finds it inside the ROM (`0x0377-0x0387`, `0x0EF7-0x0EFC`) as well as in RAM, so the ROM is paged in and out during normal operation; "the ROM is only used at boot" is not true.
- **Bank-2 aliasing is measurable, not theoretical.** `0x2500` reads the floppy register that `0x2100` reads, `0x2601` the keyboard row that `0x2201` reads, `0x2c04` the video PIA port that `0x2c00` reads. Addresses that decode to two devices at once return the AND of both — `0x2301` is keyboard *and* floppy, `0x2f00` is video PIA *and* serial ACIA.

8. Can the schematic be turned into something runnable?

Automatically, no — and that is a property of the file, not of effort. Every page is a single 1-bit JBIG2 raster at 600 dpi:

```
$ pdfimages -list research/osborne1/OCC1_1A2011-00_Schem_RevE.pdf
page   num  type   width height color comp bpc  enc  x-ppi y-ppi size
   5     0 image    6582  5137  gray    1   1  jbig2  600   600 69.1K
   6     0 image    6592  5150  gray    1   1  jbig2  600   600 86.2K
   ...
```

No vector data at all — no lines, no symbols, no netlist, nothing to walk. The only text object is a `HiddenHorzOCR` layer from an early-1990s pass that renders a sheet title as `J'U~ I 'IIS IO/OO-tOOnl`. Whatever "extraction" means here, it starts from pixels.

Re-OCRing the labels does not work either. Tesseract on native-resolution tiles mangles exactly the tokens that matter — `UC1 LS139` came back as `uct LSiss`, `LS32` as `L$3e2`, `LS00` as `tS0e` — and a 1200 dpi pass returned nothing usable at all. A harvested label index would mislead more than it helped, so the reading stays manual and §3 is about making that fast and accurate.

The honest route to a netlist is manual capture, and the only part worth capturing is the discrete glue. Most of these sheets are Z80, DRAM, FDCs and PIAs whose behaviour is already known; the unknowns are the small-scale logic — above all the **wait-state generator on sheet 2/9** (`UC1`, `UC2`, `UD1–UD3`, `UD9`, `UD12`, `UB1`, `UB7`, `UB12`, `UE1–UE3`, `UA4`, `UA5`, Q1/Q2), which is what MAME does *not* model and what issue #29 needs. Capturing that one sheet into KiCad gets ERC and netlist checking from the tooling `hardware/` already has; capturing it into Verilog additionally lets it be *run*.

What "run it as a machine" can mean, in order of fidelity:

Approach	Status	What it validates
MAME, booting CP/M headless	works (§6)	the machine, behaviourally
`tools/check_o1_map.sh` asserting the documented map	works (§7a)	our *reading* of the map, against the machine
Hand-captured glue logic as Verilog, simulated	not started	the schematic itself — the wait states MAME omits
The above, driven by bus traces logged from MAME	not started	the same, against real Z80 traffic
Whole-board gate-level simulation	not sensible	MAME already is the machine

The last two rows are a design, not a wish: MAME's Lua can log every read and write through an address space, so a captured decode/wait model can be driven with real traffic and its chip-select and `WAIT` outputs compared against what the machine actually did. A disagreement is then either a misreading of the schematic or a MAME shortcut, and both are worth knowing. It is a multi-hour careful job, and doing it badly would produce a wrong document — the failure mode this whole exercise exists to avoid.

9. What is not extracted yet

- **What the `CHAR0–CHAR7` nets carry.** They cross to the video sheet carrying a `4C1` reference (sheet 4, zone C1 — the RAM array), so they are *not* the char-gen ROM's outputs (§4a). Presumably the character-code bus out of video RAM; not yet read.
- **What UC16–UC21 do on the video sheet.** The pixel row is serialised by UA14 on sheet 4 (§4a), so the six shift registers §2 lists on sheet 5 need a second look before #24 relies on the phrase "character shifters".
- **Sheet 4 (1A2011 1/9) parts list.** That sheet carries the board's REF/DESCRIPTION table. It is partly legible in the scan and is the fastest route to a full IC inventory.

- **Per-sheet component inventories** for sheets 4, 5 and 8, which §2 lists only from a first pass — see the part-number caution in §3. (Sheet 7′s serial area is now read at pin level — §4b.)
- **The wait-state generator (sheet 2/9)** as a captured, simulatable model (§8).
- **A net-level extraction** of the whole board — not achievable from this scan without manual capture.

Memory and I/O map

Authoritative memory map, I/O decode, and bank-switch behavior extracted from the Osborne 1 mainboard schematic (OCC `1A2011-00` Rev E) and cross-checked against the Osborne 1 Technical Manual (`2F00153-01`, 1982) and the MAME `osborne1.cpp` driver.

Closes issue #40. Unblocks: #11 (hardware), #16–#24 (firmware), the memory-resident vs pure-I/O decision (gap #15), 6850 ACIA modem (#21, gap #10), WD179x floppy interposer (#20, gap #11), and ScreenPac personality (#24).

---

1. CPU and clocking

Item	Value
CPU	NEC μPD780C (≡ Zilog Z80A), DIP-40, location **UC11**
Crystal	15.9744 MHz (X1)
CPU clock	4 MHz (÷4 via UD3 LS161)
Dot clock	8 MHz
Memory / char clock	2 MHz / 1 MHz
Bus width	8-bit data (DATA0–7), 16-bit address (ADR0–15)

Wait-state generation: RAM accesses insert wait states (≈375 ns for non-M1 Wait-state generation: RAM accesses insert wait states (≈375 ns for non-M1 cycles, 188 ns for the first M1); ROM executes without added delay.

1a. One crystal, no free lunch — and the 2× "business turbo"

Per Lee Felsenstein's rule of thumb — *"the number of crystals in a machine design is equivalent to the number of designers"* — the O1 has **one** oscillator, and the whole timing economy (video, ACIA baud, floppy data separator, DRAM refresh, software delay loops) is divided down from it. Consequences:

- **Serial quirk already visible:** the J1 600/2400 jumper is ~8.3% fast (650/2600 actual) because the video-tap ÷13 ÷(16|64) chain can't land exactly on 600/2400 from 15.9744 MHz. The factory's own alternate baud rate shows the seams.
- **No clean general turbo:** 3× has no integer relationship to the video chain (breaks video), and any CPU speedup drags the ACIA baud rate, the FDC data separator, and software timing loops along with it.

- **The clean-ish 2×:** doubling the CPU clock to 8 MHz keeps the ÷2 dot-clock tap coherent (15.9744 ÷ 2 = 8 MHz), so **video timing survives**. It is viable *for business apps* if the three side effects are engineered around:
  1. **Floppy** — the WD1793 data separator is timed for 4 MHz; in turbo mode physical floppies are out. Use the interposer's storage instead (#20/#22: image-library / IEEE-488 storage answers on the bus at whatever speed the host runs), or gate turbo off when the FDC is selected.
  2. **Timing loops** — accept that games/delays run double-speed; scope the feature to calculation-bound work (SuperCalc recalc, dBASE II, compilers, assemblers). A classic PC-era **"TURBO" switch** selects 4 MHz ↔ 8 MHz.
  3. **Serial** — no clean fix at the ACIA. Mitigation: build a **baud-rate buffering corrector** into the WiFi modem (#21). The modem's RP2350 talks to the far end at a rock-solid standard rate (2400/9600) and tolerates whatever the O1 ACIA is actually doing: on **send** it buffers until a batch accumulates before transmitting; on **receive** it buffers a frame, then re-clocks it out to the O1 at the rate the O1 expects. The O1-side rate drift becomes the modem's problem, not the link's.
- **Net:** a 2× business turbo is a *documented, buildable option* (mostly on the replica #49, where SRAM removes the DRAM-refresh constraint); it is not a general speedup and doesn't pretend to be.

cycles, 188 ns for the first M1); ROM executes without added delay.

2. Memory banks

The 64 KB address space is overlaid by **three banks**, selected by the `ROM MODE*` signal and the `BIT 9` latch (see §4).

Bank 1 — main RAM ("normal mode")

64 KB of 4116 dynamic RAM. The top 4 KB is dual-purpose as video RAM.

Range	Contents
`0000–3FFF`	RAM (boot ROM overlays the low 16 KB when bank 2 is active)
`4000–EFFF`	RAM — CP/M TPA, BDOS/CCP, BIOS live here
`F000–FFFF`	Video RAM low 8 bits (128 cols × 32 rows char data)

Page zero. CP/M owns `0000–00FF` while a program runs, but the *Osborne 1 User's Guide* "Memory Allocation" chart (p. R-243) marks **`0050H–005BH` — 12 bytes — as "not used"**, and at least one published program relied on that: the date facility in *The Portable Companion* #7 keeps the system date there in both binary and ASCII form (`docs/rt60a-analysis.md`). Treat it as documented free scratch space — worth remembering for the interposer (#12).

Bank 2 — BIOS ROM + memory-mapped I/O ("shadow mode")
Selected when `ROM MODE*` is active (low). Only `0000–3FFF` is affected; `4000–FFFF` still reads bank 1 RAM.

Range	Contents
`0000–0FFF`	Monitor ROM (4 KB, 2732 at **UD11**; earlier 2× 2716)
`1000–1FFF`	Unused — reads as mirror of the ROM
`2000–2FFF`	Memory-mapped I/O window (see §3)
`3000–3FFF`	Unused — reads as mirror of `2000–2FFF`
`4000–FFFF`	Mirrors bank 1 RAM

Bank 3 — video attribute bit ("dim bit")
Selected when `BIT 9` latch is set. A 9th bit per video cell, stored in a dedicated 4116. Only `F000–FFFF` is affected.

Range	Contents
`F000–FFFF`	Video attribute RAM, 4K×1 (dim attribute = bit 8 of each cell)

2a. Video memory geometry — 128 columns wide, 52 displayed (#97)
The 4 KB at `F000–FFFF` is **32 rows of 128 characters**, of which **24 rows of 52 characters are displayed at a time**:

> Memory from 0F000 hex to 0FFFF hex is used for the video memory map. That memory is considered to be 32 rows of 128 characters, even though only 24 rows of 52 characters are actually displayed on the monitor screen at a time. The character matrix is 7 × 9 in an 8 × 10 box. — *Osborne 1 Technical Manual*, §5.2 "Video Memory", p. 23

This is the answer to the "64 vs 52" question in #97. The screen is a *window* onto a buffer 2.5× wider and 1.3× taller than what is shown, and the displayed width is a property of the window, not of the memory. That is why the same 4 KB shows 52 columns on stock hardware, 80 on Nuevo and 104 with the ScreenPac (`mame-emulation.md` §2) — the text layout is the CBIOS's, the window is the machine's.

It is also why the scrolling feels different from a terminal's. Holding Ctrl and using the arrow keys slides the window over the memory rather than moving the memory, which is a register change and not a rewrite of the buffer — the same mechanism `mame-emulation.md` §6 calls a ring buffer. A terminal scroll *does* move the lines, which is why that stays chunky while the viewport slide is smooth. (The Ctrl+arrow slide is a hardware observation, reported on the real machine; the geometry above is the manual's.)

Bit fields (§5.2, figure): bits 0–6 are the **ASCII character**, **bit 7 is the underline attribute**, **bit 8 is dim**. Bits 0–7 are the byte in the video RAM proper; bit 8 is the separate 4116 in bank 3. This is why the MAME probe masks with `% 128` to read characters (`tools/o1_mame_probe.lua`).

The window width is software-selectable. The SCREEN-PAC user guide (p. 13) gives the mechanism, and it is in the stock BIOS rather than on the board:

> At BIOS location **E1A5H**, POKE or store **0 for 52 columns / 1 for 104 columns / 3 for 80 columns**. CALL BIOS location **E4E5H**. … In these same formats, SCREEN-PAC provides **horizontal scrolling by 32 column blocks**.

So 52, 80 and 104 are three widths of one window onto the same 128×32 buffer, and the 32-column blocks are why the slide lands on round numbers. Without the SCREEN-PAC only 52 is available, because the wider windows need the board's dual-port video RAM — the ROM contains the routine either way.

Logical width is not physical width — this is what "64 vs 52" turns on. The SCREEN-PAC guide (p. 13) separates the two, and it is the piece the technical manual's geometry leaves implicit:

> **Logical screen size** refers to the number of characters you can enter on a line without automatically inserting a carriage return. Logical screen size can range from 1 up to a **maximum of 128 characters**.
>
> **Physical screen size** refers to the number of columns actually displayed on your screen. With SCREEN-PAC, this will be 52, 80, or 104. — *Osborne 1 SCREEN-PAC User Guide*, p. 13

So a program's line width is a **software** choice up to 128 — the width of the video memory row — while the window is 52, 80 or 104. A drawing 64 columns wide is a *logical* line of 64 characters: it fits the row, and the window slides over it. The guide also warns against the obvious mistake, which is worth knowing before writing anything that sets a width: "logical screen size should NOT be less than physical screen size", or the display scrambles.

The width is set from the configuration screen (Item C, "Screen Size", in the Osborne 1 Configuration Program) or from software by poking `E1A5H` and calling `E4E5H` as above. Both do the same thing.

The cursor is hardware, and it can be reversed (#24). The stock cursor is an **underline**, and OCC knew it was poor — Thom Hogan argued against it while at the company and was refused twice (*The Portable Companion* #13, June/July 1984, "Reverse Video and a Faster Serial Port", p. 24): he could not get the change made permanent, and could not get permission to publish it, *"as the company was worried about the legal implications of publishing modifications"*. The modification makes the cursor a **solid white block**, and the article locates the circuit precisely: a **74LS86 piggybacked on the 74LS00 at E22**, with **pin 1 of the 74LS74 at UE21 lifted**, plus changes around the VIDEO and DIM signals.

So **the cursor is generated in hardware** — not by the character generator, and not by the attribute bit — which matters to a video personality: it has to reproduce a cursor the machine makes outside video RAM, and the block cursor is the variant OCC's own engineer preferred. Extract: `research/technical-articles/TPC-Issue13-1984-pp26-31-reverse-video-and-9600-baud.pdf`.

And it works with the SCREEN-PAC fitted, leaving the board in place. Hogan's version is for a stock machine. A reader's fix, printed in *The Portable Companion* #15 (Jan/Feb 1985, letters, PDF p. 32), makes the block cursor work on a machine with the **SCREEN-PAC 80** installed — *"and it can be carried out without removing the piggy-back board containing the Screen-Pac 80 electronics"*. What changes is **where** the logic is: the modification is done on the *upgrade board's* own chips — **U2 (74HC00)** and **U23 (74HC74)** — not the main board's, and the letter warns in capitals that *"THESE PINS ARE NOT THE SAME ONES LISTED IN THOM HOGAN'S ARTICLE!"*.

That the SCREEN-PAC carries a 74HC00 and a 74HC74 of its own is useful in itself: the cursor logic is **duplicated on the piggy-back board**, which fits the service manual's description of the upgrade as an IC-harness board soldered over the main logic (`2F00040` §6.12, the source used by `firmware/screenpac-video/`). Extract: `research/technical-articles/TPC-Issue15-1985-pp32-block-cursor-with-screenpac.pdf`.

2b. The monitor RAM and the system clock (#7, #81)

The top of bank 1 RAM (`EF00`–`EFFF`, 256 bytes) is the **monitor RAM**: the ROM's own storage, above the CP/M TPA. The CBIOS RAM map (`occram25.asm`) and the ROM's absolute equates (`rom144.asm`) agree on its layout, and it contains the machine's **system clock**:

Address	Cell	Meaning
EF00	TEM	boot-routine scratch
EF08	ROMRAM	ROM/RAM flag
EF60	ESCH	ESC-holding flag
EF61 / EF62	PIAAD / PIABD	last PIA-A / PIA-B data
EF63 / EF64 / EF65	IDAY / IMONTH / IYR	calendar date, binary
EF66	**HOURS**	hours, 2-digit BCD
EF67	**MINS**	minutes, 2-digit BCD
EF68	**SECS**	seconds, 2-digit BCD
EF69	**SEC6**	sixtieths, 2-digit BCD
EF6A	DACTVE	disk-active countdown
EF6B	BELCNT	bell countdown
EF6C	LLIMIT	max columns in a logical line
EF6E	LDTRK	last track used on the non-selected drive
EF6F–EF70	IESTK	saved stack pointer
EFD0	SAVTYP	disk type (density + sector size)
EFD1	RWCMD	sectors to read/write
EFEF	VRTOFF	last vertical offset
EFF0–EFFF	**INTBL**	8-word IM 2 interrupt vector table

HOURS–SEC6 is a real clock. The stock ROM's 60 Hz interrupt handler (GKEY) calls UPTIM, which increments SEC6, SECS, MINS and HOURS in BCD with a 24-hour wrap. It starts at midnight on every boot, nothing sets it and nothing displays it in a stock machine — but OZROM 1E documents the same four cells and adds a setter, a display and an alarm. The EFF0 vector table is where the two clock products hook: INTBL+(4*2) = EFF8 is the keyboard-interrupt vector the RT-60A's CLOCK.COM replaces, while OZROM calls its own hook at EFF2. See docs/rt60a-analysis.md ("The third shape") and docs/ozrom-1e.md.

3. I/O decode (bank 2, "memory-mapped I/O")

All onboard peripherals are **memory-mapped**, reached through the bank-2 window. Chip-selects are generated from address lines within 2000–2FFF.

Address	Device	Chip	Registers
`2100–2103`	Floppy disk controller	Fujitsu MB8877 ≡ WD1793 (**UB7**)	2100 stat/cmd, 2101 track, 2102 sector, 2103 data
`2201–2280`	Keyboard (read)	81LS95 + 74LS05 (**UE12/13/14**)	One address bit per row: 2201=row0 … 2280=row7; columns on DATA0–7
`2900–2903`	IEEE-488 PIA	MC6821 (**UC7**)	2900 PA data/dir, 2901 PA ctrl, 2902 PB data/dir, 2903 PB ctrl
`2A00–2A01`	Serial / modem ACIA	MC6850 (**UC4**)	2A00 status(R)/control(W), 2A01 RX(R)/TX(W)
`2C00–2C03`	Video PIA	MC6821 (**UC15**)	scroll/offset latches, bell, brightness, dim control
`2400` (read)	SCREEN-PAC presence	—	**bit 2 LOW = SCREEN-PAC fitted**, HIGH = stock machine (user guide p. 13). Read-only status bit, not a register

6850 ACIA detail (#21 modem, gap #10)

- Base `2A00`: read = status, write = control. `2A01`: read = RX, write = TX.
- Baud clock from divider **UC3** (÷13) fed by a 2/4 µs tap off the video counter (UA13). ACIA (UC4) divides by 16 or 64 ⟶ **1200 or 300 baud**.
- Jumper **J1** selects 300/1200; Rev E+ can be re-jumpered to 600/2400.
- **Faster rates, from inside OCC** (#21): *The Portable Companion* #13 (June/July 1984), "Reverse Video and a Faster Serial Port" by **Thom Hogan** — the same Hogan whose `AUTOCPM.ASM` is the source of the site wordmark (#97) — written as former OCC manager of software and publications: *"developed internally by the company, [but] not promoted … the primary … was the problem of warranty repair on modified machines."*
  - Removing **J1** doubles 300/1200 to **600/2400** — exactly as above — and he adds a detail the schematic cannot show: **the SETUP program cannot sense J1**, so it still displays 300/1200 after the modification.
  - For **2400/9600** the mod is to the baud divider: the **LS161/LS163 at C3**, with **pin 2 moved to the middle pin of J2** ("for 2400/9600 baud rates rather than 1200/4800") and pins 3, 7 and

10 lifted. Figure 3 in the article has the detail. Extract: `research/technical-articles/TPC-Issue13-1984-pp26-31-reverse-video-and-9600-baud.pdf`.

- **Pin level, from the schematic** (sheet 7/9 of 9, read at 600–1200 dpi): the ACIA's **RxCLK (pin 3) and TxCLK (pin 4) are strapped together** on the board and driven by **UC3′s `QD` (pin 11)**. UC3 is a **74LS161 preset-load counter** — `RCO` (pin 15) is inverted by **UB3 (74LS04)** into `/LD` (pin 9) — so the count length comes from the preset pads rather than being a power of two; that is where the ÷13 of the stock 19.2 kHz clock lives. Pin numbers as drawn: `RXDATA` 2, `RxCLK` 3, `TxCLK` 4, `/RTS` 5, `TXDATA` 6, `/IRQ` 7, `CS2` 8, `/CS1` 9, `CS0` 10, `RS` 11, `R/W` 13, `E` 14, `D7…D0` 15–22, `/DCD` 23, `/CTS` 24 (Vcc/Vss, pins 12/1, are not drawn on the symbol).
- **The factory's own 9600/2400 modification is documented**, which the archive had only from inside OCC: *Osborne 1 Technical Manual* **Figure 4.4 "Baud Rate Hardware Modification", p. 22** (PDF page 36 — the manual's printed-page offset of **+14** holds here too, see `docs/o1-chargen-graphics.md`). It draws the LS161/LS163 at **C3** with **pin 2** moved between a *"FOR 9600/2400 BAUD"* pad and a *"FOR 4800/1200 BAUD"* pad, jumpers **J2** and **J3**, **J1** marked **"DON'T CARE"**, and the caption box `300/1200 NO MOD - J1 IN` / `600/2400 NO MOD - J1 OUT` / `1200/4800 MOD W/O PIN 2 (J1 = X)` / `2400/9600 MOD W/PIN 2 (J1 = X)`, bracketed ***"SPEED SELECTED W/IN 6850"***. That is the same modification Hogan describes from inside OCC in #13 — the 6850 choosing within each pair (÷64/÷16). It is also the "method shown on page 22 in the Technical Manual" that *The Portable Companion* #8 points a would-be 80-column terminal user at (see `docs/o1-modem-serial.md`).
- **A fifth route, and a third-party one:** Advent Products' **Osbaud** (1983, $59.95, 16 DIP-switch rates) is soldered to the ACIA's own clock and supply pins and replaces the clock net outright — see `docs/o1-modem-serial.md`.
- **The ACIA's control-register init byte is in the BIOS at `1FE9H`.** The vendor's own "Programming the serial port" patch (*The Portable Companion* #7, June 1983, PDF pp. 56–57) rewrites `1FE9H`/`1FEAH` to move off the 8-N-1 default, with a table of 6850 control codes (`55H` = 8-N-1 @1200). A second handle on the ACIA besides the port pair — see `docs/o1-modem-serial.md` §"Framing".
- Modem status (DTE): MSB (pin 4), MCB (pin 8), RI (pin 9); BIOS `LISTST` at `0E12D` returns `0FFH`/`00H`.
- **IRQ0** = ACIA interrupt (one of three IRQ sources; see §5).

WD179x floppy detail (#20 floppy interposer, gap #11)

- Controller **UB7**, registers `2100–2103` as above.
- Drive-select / density lines are driven from the **video PIA (UC15)** outputs (DRIVE1/DRIVE2/DDEN), **not** a dedicated latch — see schematic sheet 5. `DDEN` selects double density when the add-on DD board is present.
- Stock format: single-sided single-density, 40 tracks × 10×256-byte sectors (102 KB/drive), FM. DD option board adds 5×1024 Osborne-DD, 8×512 IBM, 18×128 Xerox-820, 9×512 DEC-182 formats.

Keyboard detail

- 8×8 keyswitch matrix; rows driven on address lines A0–A7 (one-hot), columns returned on DATA0–7.
- Read strobes at `2201, 2202, 2204, 2208, 2210, 2220, 2240, 2280` (i.e. `2200 + (1<<row)`).
- 14 programmable keys (Ctrl-0…9, four arrows) via BIOS + SETUP.

`F000–FFFF` | Video attribute RAM, 4K×1 (dim attribute = bit 8 of each cell) |

4. Bank switching (the mechanism)

Banking is controlled by **writes to I/O space**, decoded by **UD10** (LS138) on the two low address bits; the data bus value is **ignored**.

I/O write	Effect
`0x00`	Clear UB4 ⟶ `ROM MODE*` low ⟶ **select bank 2** (ROM+I/O). Also forced by CPU reset.
`0x01`	Preset UB4 ⟶ `ROM MODE*` high ⟶ **select bank 1** (main RAM), if `MREQ` low.
`0x02`	Set `BIT 9` latch (UE10A) high ⟶ map **bank 3** (dim bit) into `F000–FFFF`.
`0x03`	Clear `BIT 9` latch ⟶ map bank 1/2 back into `F000–FFFF`.

`ROM MODE*` is *also* set by hardware: power-on reset, and the NMI flip-flop chain (UB6/UB4). Bank selection between 1 and 2 is additionally qualified by M1 and IRQ-acknowledge conditions via three flip-flops — so interrupt acknowledge and the front-panel RESET reselect the ROM bank automatically.

> **Interposer note:** because only the two low address bits are decoded and data is ignored, *any* `OUT (0),A` / `OUT (1),A` toggles banks. The interposer must **never** drive I/O addresses `0x00–0x03`.

Measured on the running machine (`tools/check_o1_map.sh`, #64). All four writes behave exactly as tabulated — the *port number* selects, the data byte is ignored, and ports `0x04–0x07` alias `0x00–0x03` (MAME's I/O space reports `mask=0x0003`, i.e. only A0/A1 reach the decode). The M1/IRQACK qualification above is not a footnote: write `0x01` and the RAM bank is selected *immediately*, but 10 ms of emulated time later the machine is back in the ROM bank, because the bank is re-derived on every instruction fetch and the video PIA interrupts at 60 Hz.

Consequences for an interposer:

- The bank state **cannot be latched and assumed** — it has to be sampled per bus cycle, and a bank-2 I/O access is only meaningful while the CPU is actually running with bank 2 selected.

- At the idle `A>` prompt the machine is parked in a ROM polling loop with **bank 2 live**, which is why a peek at `0x0000` during CP/M returns ROM bytes, not CP/M's page zero.

5. Interrupts

Three IRQ sources into the Z80: - **IRQ0** — serial ACIA (UC4) - **IRQ1** — video PIA IRQA (UC15) - **IRQ2** — IEEE-488 PIA IRQA (UC7)

NMI is generated by the front-panel **RESET switch S1** (debounced by UE20, sequenced through UB6/UB4 on `M1`): NMI restarts the ROM program at `0000` and forces bank 2. There is **no periodic/60 Hz NMI** — the "60 SEC" signal is a video-tick line into the video PIA CA1 (used by the console routine's real-time clock), not an NMI.

6. Connector / socket pinouts (mechanical tap points)

Z80 CPU — DIP-40 (UC11)

Standard Z80A pinout (Zilog Z80 Microprocessor Family Databook; the O1 fits an NEC μPD780C, pin-compatible). **Corrected 2026-10-10** — the previous table here was wrong (it claimed pin 11 = GND, which is +5 V). Enforced from now on by `tools/check_schematics.py`, which asserts this table against every connector that mates with the O1′s Z80 socket.

Pin(s)	Signal
1–5	A11, A12, A13, A14, A15
6	CLK (4 MHz)
7–10	D4, D3, D5, D6
11	+5 V
12–15	D2, D7, D0, D1
16	INT
17	NMI
18	HALT
19	MREQ
20	IORQ
21	RD
22	WR
23	BUSAK
24	WAIT
25	BUSRQ
26	RESET
27	M1
28	RFSH
29	GND
30–34	A0, A1, A2, A3, A4
35–40	A5, A6, A7, A8, A9, A10

Character generator ROM — 24-pin (UA15)
2 KB (2716) character ROM at **UA15** — on **sheet 4 of 9** (PDF page 7, the RAM sheet), *not* the video sheet; see `o1-mainboard-schematic.md` §4a for the extraction. 128 chars × 8×10 in an 8×10 box (7×9 visible), addressed by (char code + scan line), output to the video shift register (UA14, 74166). This is the second tap point the ScreenPac personality (#24) uses.

Pin	Signal
1–8	A7–A0 (`A0`–`A6` = char code from UA18; `A7` = `SCAN0`)
9, 10, 11, 13–17	O0–O7
12	GND — **bussed with 18 and 20 to ground**
18	`/CE` — **grounded**
19	A10 (`SCAN3`)
20	`/OE` — **grounded**
21	strapped **+5 V** (A11 on a 2732)
22	A9 (`SCAN2`)
23	A8 (`SCAN1`)
24	VCC (+5)

> The earlier version of this table grouped pins `13`–`21` as "O3–O7, A8–A10" and put `OE`/`CE` on 22/23. That was wrong — exactly the "never cross-checked" error gap #3 flagged. The socket is the plain Intel 2716 pinout, and because `/CE` and `/OE` are grounded the ROM **always drives** `O0`–`O7`; a tap cannot tristate it without intercepting those two pins.

Other onboard connectors (for reference)

Every connector on an Osborne 1 is on the FRONT of the machine — there is nothing to plug into at the back. That is not a detail: it is why fitting a period peripheral meant unclipping the keyboard, which docks to seal that same face (the O1A design note in `hardware/o1ng/`), and it is why every standalone device in this project attaches at the front (#47, #56, #84).

- **P3** — IEEE-488, 26-pin edge (pins 25/26 unused)
- **P2** — RS-232, DB-25 (DTE)
- **P1** — Modem
- **P4** — Keyboard, 20-pin IDC (1=GND, 2–9 rows 4,0,3,6,2,5,1,7, 10–17 cols 0–7, 18 NC, 19=+12V, 20 GND). On the later **O1A ("blue/grey")** machines the coiled-cable plug has **24 holes but only the inner 20 mate with pins**. Pin 19 carries **+12V only if jumper J6 is bridged** (a 2-pin link, open from the factory) through **R21 (22Ω)** — the intended way to power an accessory off the keyboard cable. Rows are driven low (open-collector inverters, one-hot); columns are pulled up to +5V via a 3.3K resistor pack and buffered onto the data bus. Reference: O1 Technical Manual; pinout per pinouts.ru (verified against a PCB), archived in-repo at `research/osborne1/keyboard/osborne_keyboard_pinout.html` (provenance alongside) — see #47 for the full matrix map.
- **P8** — Floppy drive, 34-pin
- **P5** — Video, 10-pin inline

7. Free I/O windows & recommended virtual-peripheral base

The bank-2 I/O window is sparsely decoded; many sub-ranges alias the real devices. Conservatively **free** (no onboard device) within `2000–2FFF`:

Range	Status
`2000–20FF`	Free
`2104–21FF`	Free (above floppy)
`2300–28FF`	Free (above keyboard, below IEEE) — **largest clean block**
`2904–29FF`	Free (above IEEE PIA)
`2A02–2BFF`	Free (above serial, below video PIA)

> **Caution:** the O1 decode is "sloppy" (MAME) — chip selects do not check all address bits, so large regions alias real devices. Treat only page-aligned blocks *between* the listed devices as safe, and validate on hardware before committing.

Recommendation

- **Primary virtual-peripheral block: `2300–23FF`** (256 bytes). Clean gap between keyboard (`2201–2280`) and IEEE-488 (`2900–2903`). Suggested sub-allocation: CoPower-88 emulation, network/modem FIFO, RTC, status regs.
- **Secondary: `2B00–2BFF`** (between serial `2A01` and video PIA `2C00`) if a second disjoint window is needed.
- Avoid `0x00–0x03` (bank switch) entirely.
- The interposer should decode **bank-2 I/O space** (memory-mapped) rather than Z80 `IN/OUT` port space, since onboard devices are memory-mapped and the plain I/O port space `0x00–0xFF` is almost entirely consumed by the bank-switch decoder aliases (only low 2 bits decoded → `0x00–0x03` repeats every 4 bytes across the whole low port page).

8. Decisions this unblocks

- **Gap #15 (memory-resident vs pure I/O): → pure I/O**, memory-mapped in bank 2 at `2300+`. No need to steal RAM or fight the bank flip-flops.
- **Gap #10 (modem ACIA ports):** → `2A00/2A01` confirmed; the WiFi modem can either emulate the 6850 there or live at a `2300+` FIFO with a small BIOS hook.
- **Gap #11 (floppy variant):** → MB8877 ≡ WD1793 confirmed; registers at `2100–2103`; drive/density selects are PIA-driven (emulate via PIA shadow).
- **#24 (ScreenPac):** → two tap points confirmed: Z80 DIP-40 (UC11) and char-gen 24-pin (UA15).

Sources

- OCC schematic `1A2011-00` Rev E (`research/osborne1/OCC1_1A2011-00_Schem_RevE.pdf`), sheets 1–5 (CPU, RAM, video, IEEE-488, serial/disk).

- Osborne 1 Technical Manual `2F00153-01` (1982), ch. 3 (memory), ch. 4 (interfaces), ch. 5 (video), ch. 7 (disk), ch. 8 (keyboard), ch. 11 (theory of operations: ROM MODE/bank select, NMI).
- MAME `src/mame/osborne/osborne1.cpp` (bank/I/O map cross-check).

ROM and firmware

Issue: #72. **Status:** first pass. The source has been indexed and its port/vector tables extracted; every one of them **agrees with** `docs/01-memory-io-map.md` §3, which had been derived from the schematic and from the running machine. That is the headline result: the map survives contact with the primary source.

This is Osborne's own source for the machine's firmware — the thing the rest of the project had been inferring. It is a safety copy of material already public in several places; see "Provenance" at the end.

The two components

The machine's firmware is **two separate programs**, and the source tree has one of each:

Component	Lives in	Source here	Rev
Monitor ROM — boot, video, keyboard, disk, IEEE-488, interrupt handlers	the 2732 at UD11	`rom144.asm`	1.44
CBIOS — CP/M 2.2 character and disk BIOS	loaded into RAM from the system disk	`occbio*.asm`	1.41

They version independently, so **do not assume the 1.41 CBIOS source is what ships inside the 1.44 ROM.** The monitor ROM is the 4 KB in the socket; the CBIOS is a RAM-resident overlay assembled to a specific system size (below).

File index

`bios141.gen` is the build recipe, and it lists exactly the files present here, so the set is complete:

File	Lines	What it is
`rom144.asm`	4655	The monitor ROM, Rev 1.44, `ORG 0`. Z80. Roger W. Chapman, 2/4/1983. Title: "DOUBLE DENSITY ROM". Contains the memory-mapped I/O equates, RAM storage locations, IEEE/disk equates, monitor main loop, video, keyboard, disk drivers, the IEEE-488 interface and drivers, the parallel (Centronics) printer path, and the interrupt handlers.
`occtxt6.ast`	374	The shared equates file (`S=OCCTXT6.AST` in the build): port bases, ROM vectors, monitor RAM bases, character codes, disk-type constants. `occbio05.asm` says "*FOR USE WITH OCCTXT6.AST ONLY*".
`occbio05.asm` … `occbio95.asm`	47, 145, 80, 70, 341, 141, 246, 60, 505, 66	The CBIOS listing, split into ten sections. `05` is the title block and layout; `45` holds the disk control blocks (`DPBASE`, `DPHGEN`); `85` the I/O-byte dispatch and character devices.
`occram15.asm`, `occram25.asm`	31, 175	"Monitor RAM Storage" — the RAM-resident data area, `ORG MRAM`, assembled into both the ROM and the CBIOS.
`bios141.gen`	56	The generation procedure: needs ACT 3.5e, DDT 2.2, SYSGEN 2.1 and MOVCPM, and the twelve files above.
`tran.txt`	61	The 1.44 ROM release memo (Sai Kit, 2/4/1983): how the HEX and print files were generated, on a DD machine with a Corvus hard disk.
`release.txt`	53	The 1985 Advance letter granting Lee Osborne permission to reproduce the code.
`bios144.asm`	5123	Enhanced CBIOS 1.44 (Bartholomew, 1985). Retained verbatim — see `research/README.md`.
`bios144.mac`	27	Macro file for the BIOS listing.

Revision lineage

Monitor ROM 1.44 — the header carries the delta from 1.43, which is the useful part (it is a change log for the DD machine):

```
CBOOT  - added seek 10 tracks and home drive : tsk
GKEY   - fixed bell bug                      : tsk
SENDEN - changed the number of retrys to NRETRY
PSEKC  - added disk head settle delay
SELDRV - changed start up delay to 500 ms    : tsk
```

`NRETRY = 10` (disk retry count), `SLD_RCT = 3` (slide-key repeat), and `BRTBIT`/`DIMBIT` (`80h`/`00h`) are defined at the top of the file. The 1.44 image is `research/roms/os1-144.rom` (CRC32 `c0596b14`) — see `docs/o1-rom-variants.md`.

CBIOS 1.41 — from `occbio05.asm`/`occbio15.asm`:

```
MSIZE = 59              ; 59K system
CCP   = 0CB00H          ; location of the CCP
BIOS  = ccp+1600h
BDOS  = ccp+806h
DDT R (relocation value) = 1F80H-BIOS
```

So the CBIOS is relocated for a 59K system, and `occbio15.asm` carries the CP/M disk-type tables — `HSTSIZ = 1024` (blocking/deblocking buffer), `FPYSIB = 2048/128` (SD block = 2K), `FPYDIB = 1024/128` (DD block = 1K), `DSKS1 = 5` (single density, 256-byte sectors, single sided).

The memory-mapped I/O map, as the source states it

Straight out of `rom144.asm` (`;MEMORY MAPPED I/O`, around line 83) and `occtxt6.ast`:

Address	Name in source	What it is
2100H	D.CMDR / D.STSR	Floppy: command (write) / status (read)
2101H	D.TRKR	Floppy: track
2102H	D.SECR	Floppy: sector
2103H	D.DATR	Floppy: data (read/write)
2200H	H.KEY	Keyboard
2900H	CPDRA / PA.DTA / PA.DIR	IEEE-488 PIA: port A data / direction
2901H	CCRA / PA.CTL	IEEE-488 PIA: control A
2902H	CPDRB / PB.DTA / PB.DIR	IEEE-488 PIA: port B data / direction
2903H	CCRB / PB.CTL	IEEE-488 PIA: control B
2A00H	H.SCTRL (W) / H.SSTS (R)	6850 ACIA: control / status
2A01H	H.SXMT (W) / H.SREC (R)	6850 ACIA: transmit / receive
2C00H	H.VIO	Video memory controls (the ROM also uses +1, +2, +3)

occtxt6.ast names the same bases — H.FDC = 2100h, H.KEY = 2200h, H.IEEE = 2900h, H.SIO = 2A00h, H.VIO = 2C00h — which is worth noting because the CBIOS refers to the IEEE port as H.IEEE, not as a PIA.

Cross-check against docs/o1-memory-io-map.md §3

Map doc §3	Source	Verdict
2100–2103 floppy, 2100 stat/cmd, 2101 track, 2102 sector, 2103 data	D.CMDR/STSR/TRKR/SECR/DATR	**agrees**
2201–2280 keyboard, one address bit per row	H.KEY = 2200H, read with A0–A7 one-hot	**agrees**
2900–2903 IEEE-488 PIA, MC6821 (UC7)	CPDRA/CCRA/CPDRB/CCRB — a PIA with A/B data+direction+control, i.e. 6821-style	**agrees**
2A00–2A01 serial ACIA, MC6850 (UC4)	H.SCTRL/H.SSTS at 2A00, H.SXMT/H.SREC at 2A01	**agrees**
2C00–2C03 video PIA, MC6821 (UC15)	H.VIO = 2C00H, ROM touches +0…+3	**agrees**

No contradictions. The I/O map in `docs/o1-memory-io-map.md` §3 is now sourced from the firmware as well as the schematic and the running machine.

Free space within the window. The firmware touches exactly five bases (`2100`, `2200`, `2900`, `2A00`, `2C00`), so the clear gaps *between* the device blocks are `2300–28FF`, `2B00–2BFF` and `2D00–2FFF`. That is where a virtual peripheral belongs — but confirm the *decode granularity* against the schematic before relying on an address *inside* one of the used blocks, because several decode more than the registers the firmware names (the keyboard decodes A0–A7 one-hot, for instance). This narrows the recommendation in the map doc §7 and closes most of gap 1 in `docs/research-gaps.md`.

Monitor RAM and vectors

The ROM's scratch area sits immediately below video RAM, and the emulator, the video personality and any bus-level assertion need to know about it:

Address	Name	Note
0EA80H	MRAM	Base of the monitor RAM area (`ORG MRAM` in `occram*.asm`)
0EF00H	TEM	Used in the boot routines
0EF05H	RTRY	Retry counter
0EF08H	ROMRAM	ROM/RAM flag
0EF09H	DSTSB	Six bytes of disk info
0EF0FH	DMADR	Disk DMA address
0EF13H	SEKDEL	Disk step delay
0EF14H	SAVSEC / SAVTRK / SDISK	Saved sector / track / disk
0EF50H	HSTACT, UNASEC, LOGSEC	Host active, unallocated/logical sector
0EF59H	KEYLCK, CURS, LKEY, ESCH	Keyboard lock, cursor, last key, escape
0EF61H	PIAAD, PIABD	Saved PIA A/B data — the IEEE port's shadow
0EF6AH	DACTVE, BELCNT, LLIMIT	Disk active, bell counter, line limit
0EFF0H	INTBL	Interrupt vector table
0F000H	FWAVM	First address of video memory

Also from `occtxt6.ast`: `ROMVEC = 100H` (start of the ROM vector table), `NMIA = 66H` (NMI vector), `BKPI = 0CFH` (breakpoint interrupt = 1), `SEKTMO = 0FEH` (seek time-out status).

Bank switching: what the ROM actually does

The ROM defines exactly two bank-related macros:

```
ENADIM: MACRO          DISDIM: MACRO
        OUT 2                  OUT 3
        ENDM                   ENDM
```

`OUT 2` / `OUT 3` set and clear the `BIT 9` latch (the "dim bit", bank 3) — matching `docs/o1-memory-io-map.md` §4. There is **no `OUT 0` or `OUT 1` anywhere in the ROM**: the firmware never selects the RAM bank itself, so bank 1 ↔ bank 2 must be driven by hardware (reset and the NMI flip-flop chain), exactly as §4 describes. This is a useful negative result for the interposer: the firmware gives us no example of a software bank switch to model, and the "never drive `0x00`–`0x03`" rule in §4 stands.

Floppy command set

The WD179x commands the BIOS issues (`rom144.asm`, `;DISK EQUATES`):

Constant	Value	Command
`D.SEK`	`010H`	Seek
`D.STP`	`020H`	Step
`D.STPI`	`040H`	Step in
`D.STPO`	`060H`	Step out
`D.RDS`	`080H`	Read sector
`D.WRTS`	`0A0H`	Write sector
`D.RDA`	`0C0H`	Read address
`D.FINT`	`0D0H`	Force interrupt
`D.RDT`	`0E0H`	Read track
`D.WRTT`	`0F0H`	Write track

These are the type I/II/III/IV encodings of a WD179x-class part, which is what the floppy interposer (#20) has to speak. The MB8877-vs-WD1793 part question in `docs/o1-memory-io-map.md` §3 remains schematic-level and is unaffected.

IEEE-488 — and the parallel printer is the same port

The ROM carries a complete IEEE-488 implementation (`;IEEE-488 INTERFACE.`, line 2606, drivers from 3113), and it is the register-level contract for Drive C (#6/#22) and for the virtual printer (#19).

Three things worth extracting immediately:

The bus is inverted. Line 2668: *"ALL SIGNALS ARE LOW ON THE IEEE BUS WHEN PIA REGISTER CONTAINS '1'."* An emulated device that assumes positive logic will be wrong.

GPIB command bytes (`;IEEE control codes`):

Constant	Value	Meaning
IE_TALK	40H	make talker
IE_UTLK	5FH	make untalk
IE_LSTN	20H	make listener
IE_ULST	3FH	make unlisten

The PIA handshake programming (;IEEE EQUATES), including PA.CDT = 00101110b — *"to address port a data and set port a in input program handshake mode"* — plus the direction values (PA.DRO = 0FFH, PA.DRI = 00H, PB.DR = 0BFH, PB.DTO = 00000010b, PB.DTI = 00001011b) and the ready/strobe bits (PP.ORDY = 01000000b in port B, PP.IRDY = 10000000b in the PIA control register, STRB = 00100000b in port B). The current PIA state is shadowed in RAM at PIAAD/PIABD (0EF61H/0EF62H).

And one behaviour that matters more than the register values: line 3133 says *"IEEE always appears to be ready"* — the BIOS's device-status routine is effectively a stub. **The O1 will not poll a GPIB device for readiness**, so an emulated Drive C must be ready when addressed.

Finally, line 3249: *"The Parallel port is actually the IEEE port driven with the centronix [protocol]"*. The MX-80 personality (#19) is therefore an IEEE-488 device speaking Centronics on the same PIA, not a separate port.

Video, as the ROM drives it

- H.VIO is written as **pairs** — STO A,H.VIO+1 then STO A,H.VIO ("send data"), and STO A,H.VIO+3 then STO A,H.VIO+2 (lines 382–410).
- **Reading H.VIO clears the video interrupt** (line 2025, LD A,H.VIO ;clear interrupt) — the video PIA is one of the three IRQ sources at 60 Hz (docs/o1-memory-io-map.md §5).
- ENADIM / DISDIM toggle the dim bit through OUT 2 / OUT 3 (above).

Individual register semantics stay in docs/o1-memory-io-map.md §3; this pass records what the ROM does with them, not a re-derivation.

What this closes, and what it does not

Gap (`docs/research-gaps.md`)	Status after this pass
1 — exact memory & I/O map, free decode windows	**Substantially closed.** Every address the firmware touches is now sourced, and the gaps between the five device blocks (`2300–28FF`, `2B00–2BFF`, `2D00–2FFF`) are clear. Decode *granularity* is still schematic-level.
3 — char-gen socket pinout	**Not addressed** — that is a hardware pinout, not firmware. Still open, still needed by #24.
10 — 6850 ACIA location for the WiFi modem	**Closed** — `2A00`/`2A01` confirmed by an independent source.
11 — floppy controller variant, port map, DD changes	**Port map and command set closed**; the DD changes are now documented (1.44 *is* the DD ROM, DD block = 1K, disk-type tables in the CBIOS). MB8877-vs-WD1793 remains open.

Two things this pass adds that no existing doc had:

- the **monitor RAM map** (`0EA80–0EFFF`) and the interrupt vector table at `0EFF0H`;
- the **GPIB control codes and PIA handshake constants** for the IEEE port.

Open questions / next steps

1. **Reassemble `rom144.asm` and compare against `research/roms/os1-144.rom` (CRC32 `c0596b14`).** A byte-for-byte match would prove the source *is* the shipped 1.44 ROM — and would validate the toolchain well enough to attempt **ROM 1.3** from `OSROM13.IMD`, which MAME carries as `NO_DUMP` (see `docs/o1-rom-variants.md`). Blocker: the ACT 3.5e assembler named in `bios141.gen` is not in the repo. It may be runnable **inside MAME** on a CP/M system disk (`research/shipped-software/`), which is the cheapest route.
2. **Does the 1.44 ROM's CBIOS match the 1.41 source?** The CBIOS is disk-loaded, so extract it from a shipped system disk (`OS1SYSD.IMD` / `OS1SYSS.IMD`) and diff it against the 1.41 build.
3. **Read the IEEE driver section (lines 3113–3400) against `docs/drive-c-protocol.md`** to confirm the exact handshake the Drive C firmware expects (#6/#22).
4. **Pin down the video register semantics** from the ROM's write pairs (`+1`⟶`+0`, `+3`⟶`+2`) against the schematic (#24).

Provenance

- The source tree: `github.com/BrettHallen/Osborne_1` `ROM/Source_Code_v144/` (all 18 file sizes match). See `research/README.md`.
- The 1.44 binary used for the CRC comparison: `research/roms/os1-144.rom` (Don Maslin's archive) — identical to MAME's `3a10082-00rev-e.ud11`.
- The OCC engineering source disks (`research/source-code/*.IMD`, bitsavers) are the *disk* copies of the same material and have not yet been extracted — that is #2's pipeline work.

- Licensing: `release.txt` retains Osborne's 1985 FOG letter verbatim; the terms are recorded in `research/README.md`.

ROM variants and third-party ROMs

Issue: #70. **Status:** first pass — the stock and third-party images we hold are identified by CRC32 and cross-referenced against MAME; the open questions at the end are the work that remains.

The machine has two ROM sockets in play:

Socket	Package	Size	Role
UD11	2732 (24-pin)	4 KiB	Monitor / BIOS — the Z80 code that boots the machine
UA15	2716 (24-pin)	2 KiB	Character generator (the 6×9 font the video logic reads)

Both were replaced or re-programmed by the third-party upgrade vendors, so "the Osborne 1 ROM" is really a family. This document is the map.

Method

Every image is identified by **CRC32 over the raw bytes** plus size. CRC32 is what MAME's ROM definitions and the archive indices both publish, so it is the only key that cross-references cleanly. Two independent sources agreeing on a CRC is treated as positive identification; a CRC agreeing under two different *names* is treated as a finding, not an error.

Stock monitor ROM (UD11, 4 KiB)

Rev	Our copy	CRC32	MAME name	MAME BIOS slot
A (1981)	—	—	`osba.bin`	`vera` — **NO_DUMP**
1.2	—	—	`osb12.bin`	`ver12` — **NO_DUMP**
1.2.1	—	—	`osb121.bin`	`ver121` — **NO_DUMP**
1.3	—	—	`osb13.bin`	`ver13` — **NO_DUMP** (source disk held: `research/source-code/OSROM13.IMD`)
1.4	`tools/emulator-setup/roms/rev1.40.ud11`	`3d966335`	`rev1.40.ud11`	`ver14`
1.43	`research/roms/os1-143.rom`	`91a48e3c`	`rev1.43.ud11`	`ver143`
1.44	`research/roms/os1-144.rom`	`c0596b14`	`3a10082-00rev-e.ud11`	`ver144`

Corroboration: BrettHallen's `ROM/OCC1_v140.BIN`, `OCC1_v143.BIN` and `OCC1_v144.BIN` are byte-identical to the three we hold (same CRCs), and `research/source-code/BIOS-144-source/rom144.asm` is the original ACT80 source for 1.44 (#72).

Two things worth noting:

- **MAME's "ver144" is the part `3a10082-00` Rev E**, not a file named "1.44". The revision and the part number are two names for one image.
- **BIOS A, 1.2, 1.2.1 and 1.3 have never been dumped** (MAME carries them as `NO_DUMP` placeholders). We hold the **ROM 1.3 source disk**, so 1.3 is potentially reconstructible — see "Open questions".

Character generator (UA15, 2 KiB)

There are **two distinct images** in circulation, and which one a machine has tracks the BIOS revision:

Image	CRC32	Paired with	Our copies
Early	`5297c109`	BIOS A … 1.4	`tools/emulator-setup/roms/char.ua15`
Late	`6c1eab0d`	BIOS 1.43, 1.44, and every ScreenPac / Nuevo machine	`research/roms/os1vid8.rom`, `research/roms/os1vid80.rom`, `research/roms/OCC1_ScreenPac_RevA.BIN`

MAME's driver carries the early image five times, once per BIOS slot 0–4, with the comment *"this is CHRROM from v1.4 BIOS MB"*, and the late image for slots 5 and 6 (1.43 / 1.44). That is the evidence for the pairing, and it is the "early font / late font" distinction that #44 asks about — **both variants are now captured in this repo.**

The socket itself (UA15, on mainboard sheet 4 of 9 — see `o1-mainboard-schematic.md` §4a) is wired **2732-style**: pin 21 is strapped to +5 V, and `/CE` (18) and `/OE` (20) are grounded. Both variants are 2 KiB 2716s, so either drops straight in with no strapping change, and whichever is fitted is what a socket tap reads.

How the font is stored

The layout is **not** glyph-contiguous, and it is not obvious from the dump — it took MAME's driver to settle it (`src/mame/osborne/osborne1.cpp`):

```
u8 const gfx = m_p_chargen[(ra << 7) | (chr & 0x7f)];
```

so the byte for scanline `ra` of character `chr` is at `rom[ra * 128 + chr]`: all 128 characters' row 0 are contiguous, then all of row 1, and so on. Ten rows × 128 characters = **1280 bytes**, and the remaining 768 bytes of the 2 KiB are `0xFF`. Each row is one byte, **bit 7 = leftmost pixel**, and the cell is 8×10.

(Guessing this cost an hour: reading it as 128 glyphs of 16 bytes, or as 8 bytes per glyph, produces plausible-looking noise and no recognisable letters. The driver had it in one line.)

That layout matters to the video personality (#24): a socket tap reads the font as `rom[ra * 128 + chr]`, and the Nuevo ROM adds an underline by forcing row 9 to `0xFF` when bit 7 of the character code is set.

The two fonts really are different — 93 glyphs of 128

Comparing the two images glyph by glyph (`tools/romfont.py` decodes both):

	Differing characters
0x00–0x1F graphics set	**0 of 32** — byte-identical
0x20–0x7E printable ASCII	**90 of 95**
0x7F and the rest	3

Only three printable characters are shared: space, the apostrophe and the backtick. So this is not a tweak — the late font is a different typeface, and the graphics set was carried over unchanged. That settles the open question below, and it means "which font is fitted" is a visible difference, not a subtle one.

Finding: one late image, five names

The bytes `6c1eab0d` appear under five different names across three sources:

Source	Name
Maslin's archive	`os1-vid8.rom`
Maslin's archive	`os1vid80.rom`
BrettHallen	`OCC1_char.BIN`
BrettHallen	`OCC1_ScreenPac_RevA.BIN` ("Screen Pac BIOS Rev. A")
MAME	`7a3007-00.ud15` and `character_generator_6-29-84.14`

So the stock late character generator, the ScreenPac's 2 KiB ROM and the Nuevo-era date-stamped character generator are **the same 2 KiB image**. The ScreenPac's 80/104-column capability therefore does not live in this ROM, and MAME models the ScreenPac as a *clone of `osborne1` with no ROM differences at all* (`osborne1sp` is a `COMP` of `osborne1`, while `osborne1nv` is a separate machine with its own ROMs). That matches the picture that the Nuevo 80-column implementation is a different design from the OCC one.

Third-party ROMs

ROM	Size	CRC32	Our copy	What it is
Micro Management **OZROM 1E**	4 KiB	`ba1f6131`	`research/roms/OZROM-1E/OZROM_1E.BIN`	Third-party replacement monitor ROM, © 1984 Micro Management. Strings: `OZROM 1E #00187`, `(c) 1984 Micro Management`, `Slip disk in drive and press RETURN.` Manual in the same folder.
Nuevo Electronics **Rev 1.51**	4 KiB	`298da402`	`tools/emulator-setup/roms/monrom-rev1.51-12.ud11`	Nuevo's replacement monitor ROM. Strings: `OSBORNE 1 COMPUTER, ROM Rev 1.51, COPYRIGHT 1983, OSBORNE COMPUTER CORP.`, `COPYRIGHT 1984, NUEVO ELECTRONICS CORP.` — i.e. an OCC ROM with Nuevo's DD/video patches. MAME name: `monrom-rev1.51-12.ud11`.
SWP co-processor	4 KiB	`948556db`	`research/roms/swp-p88.rom`	The SWP co-processor board's own ROM. **This is 8088 code, not Z80** (`8C C8 8E D8` = `MOV AX,CS` / `MOV DS,AX`), and it carries the `WD1793` disk-BIOS error table (`Clock Error`, `Data CRC Error`, `Missing DAM`, `Lost Data`, `Seek Error`, `Write Protected`, `Drive Not Ready`).
OCC 1.44 + OZROM combined	8 KiB	`edc31846`	*not copied*	Bret Hallen's 2764 image: the 2732 OZROM 1E + the OCC 1.44 daughterboard ROM, loaded, MAME-ready.

Personality-relevant properties

Some interposer personalities are only compatible with some of these ROMs, so the firmware needs to know more than the CRC. The facts below are what the personality gate is built on (#86). Each is recorded with its evidence, because two of them were inference until the bytes were read.

ROM	IEEE-488 driver	Video window	Evidence
OCC 1.4, 1.43	**present**	4 KiB (`F000-FFFF`)	`013F-0156` holds eight `JP` vectors (`c3 36 09 …`), and the printer entry points `0187-0190` are `JP`s. Byte-identical between 1.4 and 1.43
OCC 1.44	**present**	4 KiB	Same structure with shifted targets (`c3 49 09 …`), and the implementation is in source we hold: `rom144.asm` `;IEEE-488 INTERFACE.` line 2606 (`o1-rom-source.md`)
OZROM 1E	**removed**	**3 KiB** (`F000-FBFF`)	Manual (`OZROM_1E_Manual.ocr.txt`), verbatim: *"The OZROM 1E does not support the IEEE-488 functions"*; *"013F-0156 WERE IEEE-488 ROUTINES / CALLING THESE FORMER JUMPS RESULTS IN A CONTROLLED SYSTEM CRASH (NO DAMAGE TO DISK DATA)"*; *"IEEE-488 AS IOBYTE DEVICE 3 NOW NULL DEVICE"*. Confirmed in the bytes: `013F-0156` is `ef f3 00 f7 …` — not jumps — and `0187-0190` is `f6 ff c9 f6 ff c9 af c9` `c9` = `OR 0FFH; RET` ×2 then `XOR A; RET; RET`, exactly the documented printer stubs. Video: *"only 3K of RAM is used for the video (F000-FBFF hex) since the last 1K is used for the redefinable keyboard and extra function keys*
OCC Nuevo + OZROM (8 KiB)	**present, unverified**	unknown	The sheet blocking is interwoven with the live routines, so I cannot tell from the CRC alone — but nothing shows

Three consequences, all of which the gate encodes:

1. **Drive C (#6, #22, #84) and the virtual printer (#19) both need the IEEE-488 driver**, so both are refused under OZROM 1E. The printer is the less obvious of the two: the ROM's printer entry points `0187-0190` are precisely what OZROM stubs out.
2. **The video personality (#24) must cope with a 3 KiB window** under OZROM 1E, where `FC00-FFFF` belongs to OZROM's keyboard features.
3. **An unknown ROM must not be assumed good.** BIOS A/1.2/1.2.1/1.3 have never been dumped, so a machine fitted with one is unidentifiable — the gate refuses rather than guesses.

MAME cross-reference

MAME carries three Osborne 1 machines; the relationship between them is itself evidence about the upgrade boards:

MAME machine	Year	Parent	ROM set
`osborne1`	1981	—	BIOS A/1.2/1.2.1/1.3 (NO_DUMP) + 1.4 + 1.43 + 1.44, chargen early + late
`osborne1sp` "Osborne-1 with SCREEN-PAC"	1983	`osborne1`	**no ROM differences** — a clone, so the ScreenPac is modelled as hardware only
`osborne1nv` "Osborne-1 (Nuevo Video)"	1984	`osborne1`	`monrom-rev1.51-12.ud11` + `7a3007-00.ud15` + `character_generator_6-29-84.14` — its own BIOS and chargen

`tools/check_o1_map.sh` already selects a driver from the boot image's `52-`/`80-`/`104-` prefix (`osborne1` / `osborne1nv` / `osborne1sp`), so this table is also the map from video mode to ROM set.

Open questions

1. ~~**Do the two character generators actually differ in the font?**~~ **Answered — yes, substantially.** 93 of 128 glyphs differ, and the 32 graphics characters are byte-identical (see "The two fonts really are different" above). Only space, apostrophe and backtick are shared. `tools/romfont.py` decodes either image to a font, so the comparison can be repeated.
2. **What is BrettHallen's "Screen Pac BIOS Rev. A"?** It is byte-identical to the stock late character generator, so the label may be wrong — or the ScreenPac genuinely ships an unmodified char-gen and does its 80-column work entirely in hardware. Settle against the ScreenPac install and service pages (`research/screenpac/`, `research/osborne1/`). Affects #24.

3. **ROM 1.3 can probably be reconstructed.** We hold the 1.3 source disk (`research/source-code/OSROM13.IMD`, comment `rom 1.3, cbios .3 osborne source 12/16/81`) while MAME carries 1.3 as `NO_DUMP`. Reassembling it from the ACT80 sources would fill a real gap upstream, and would also give us BIOS A / 1.2 / 1.2.1 if those sources exist on the other source disks. Part of #72.
4. **Confirm `swp-p88.rom` is host-independent.** The working assumption is that the co-processor board and interposer are common and only the host driver differs, which makes this the CoPower-88 ROM rather than a Kaypro-only artefact. A disassembly would confirm it and simultaneously yield the host mailbox protocol — #3/#4/#5.

Provenance

- Stock monitor ROMs `os1-143.rom`, `os1-144.rom`, and the character generators `os1-vid8.rom` / `os1vid80.rom` — Don Maslin's ROM archive on retroarchive.org.
- `OZROM-1E/*`, `OCC1_ScreenPac_RevA.BIN` — github.com/BrettHallen/Osborne_1 `ROM/` (the OZROM manual is his cleaned-up copy).
- `tools/emulator-setup/roms/*` — the MAME `osborne1` / `osborne1nv` ROM sets, matching the CRCs MAME's driver publishes.
- ROM 1.44 source — `github.com/BrettHallen/Osborne_1` `ROM/Source_Code_v144/`; the OCC engineering source disks are from bitsavers.
- Cross-reference data — MAME `src/mame/osborne/osborne1.cpp` (`ROM_START( osborne1 )`, `ROM_START( osborne1nv )`, and the `COMP()` lines).

See `research/README.md` for the per-file archive provenance, and #74 for the two images whose source is still unrecorded.

OZROM 1E: a third-party monitor ROM

Status: Solved, as tooling. It boots under MAME on all three Osborne 1 machines (measured, §3), its 58-page manual is OCR'd and indexed here (§4–§8), and its three engineering features — a software 52/80/104-column switch, a chip-level memory tester and a *documented* ROM interface — are available to us without hardware. Its companion **Utilities disk** surfaced in 2026-10 and is catalogued in §10 — the directory is complete, the file bodies are partly erased. One test is still not run: driving the ctrl-shift column keys from the harness (§9).

Source: `research/roms/OZROM-1E/OZROM_1E.BIN` (4096 bytes) and `OZROM_1E_Manual.pdf`, both from BrettHallen's `Osborne_1` repository; `OZROM_1E_Manual.ocr.txt` is the 300 dpi `tesseract` text layer (grep it for the appendices — but not for exact code, it is OCR). Micro Management, Inc., Manual Rev. A 84/09/01. Serves #81, and touches #24 (video), #29 (validation), ##45/#47 (keyboard), #20 (floppy) and #7/#18 (clock).

1. What it is

A replacement for the 2732 monitor ROM at UD11, sold in 1984 by Micro Management (Irving, TX). It is a **ROM swap with capability**, not a font change: the manual's feature list is mostly things the stock ROM lacks. The two engineering features that matter to this project are the *documented memory map and jump table* (§5–§6) — a second, independent description of the

machine's interface to hold against `rom144.asm` (#72) — and the *software column switch* (§7), which is on the design path for the video personality (#24).

2. How to run it under MAME

MAME cannot be told about a BIOS it does not define, but it **will** load a file whose bytes do not match the expected hash (it warns and carries on). So present OZROM under a stock BIOS filename.

Build a romset, do not just drop files in a directory. The simple `mkdir .../osborne1 && cp ...` recipe in `mame-emulation.md` §5 works for `osborne1` and `osborne1sp` but **fails for `osborne1nv`**, because the Nuevo machine needs `character_generator_6-29-84.14`, which MAME satisfies by CRC fallback from `7a3007-00.ud15` *only when the files are loose in the rompath root or inside a zip* — not when they sit in a plain `osborne1/` directory. Build the zip instead:

```
mkdir -p /tmp/ozbuild/osborne1
cd /tmp/ozbuild/osborne1
cp <repo>/tools/emulator-setup/roms/*.ud11 \
   <repo>/tools/emulator-setup/roms/*.ud15 \
   <repo>/tools/emulator-setup/roms/*.ua15 .
cp <repo>/research/roms/OZROM-1E/OZROM_1E.BIN 3a10082-00rev-e.ud11
cp 7a3007-00.ud15 character_generator_6-29-84.14   # satisfies the clone's
chargen
mkdir -p /tmp/ozset && zip -q -X /tmp/ozset/osborne1.zip *

mame osborne1nv -verifyroms -rompath /tmp/ozset     # -> romset ... is good
```

The `character_generator_6-29-84.14` copy is not a hack: it is a byte-identical duplicate of `7a3007-00.ud15` (`docs/o1-rom-variants.md`), and it is what makes the hash fallback deterministic across all three machines.

3. Measured: it boots on all three machines

```
O1_ROMPATH=/tmp/ozset tools/research/mame-boot-probe.sh MACHINE \
    tools/emulator-setup/floppies/52-blank.imd 120
```

Machine	Screen	Columns	Banner	Result
`osborne1`	416×240	52	`OZROM 1E #00187 / (c) 1984 Micro Management`	**BOOTED** to `A>`
`osborne1sp`	832×240	104	same	**BOOTED** to `A>`
`osborne1nv`	640×240	80	same	**BOOTED** to `A>`

So OZROM's *window* is the machine's (52/104/80), exactly like every other disk (`mame-emulation.md` §2), and it boots CP/M off the utility disk to `A>` on all three. MAME prints *"the machine might not run correctly"* because the ROM hash is not what it expects — that warning is the point, not a problem.

4. What it adds, and what it gives up

From the manual (page 1 feature list; Appendix C for the losses):

Gained	Why it matters here
Keyboard-selectable **52 / 80 / 104-column** display	A *documented software* video-mode switch — the mechanism the video personality (#24) is being built on. On an un-expanded machine it can also toggle the left/right 52 columns of an 80-column display.
System memory tester that names the failing chip	A ready-made second oracle for #29, alongside `tools/check_o1_map.sh`.
Software-redefinable keyboard, plus eight locally-redefinable function keys (up to 63 chars)	Relevant to #45/#47.
Time-of-day clock (`TIME`, `ALARM`, `TIMER`, `QUIKTIME`)	#7/#18 — see §8 and `docs/rt60a-analysis.md` "The third shape".
Documented interfaces: interrupt hook, ROM jump table, memory map, keytable	An externally documented O1 ROM interface to hold against `rom144.asm` (#72).
1793/1797 FDC notes; truer Televideo 912/920 emulation	#20 and terminal behaviour generally.
Removes the *"Centronics printer not connected"* lockout on DD systems	#19.

Given up	Consequence
IEEE-488 functions (Appendix C)	**OZROM and Drive C cannot coexist out of the box** — Drive C is an IEEE-488 device. The manual says *"extra software would be required"*, and that software is the driver whose source we hold in `rom144.asm` (#6/#22).
The stock 4 KB video window	OZROM uses **3 KB** (`F000–FBFF`) for video and the last 1 KB (`FC00–FFFF`) for the redefinable keyboard and function keys, so it scrolls by moving the 23 lines above rather than the window. This is the 1 KB price of the redefinable keyboard, and it constrains the 2.0 ROM design (#83).

The IEEE-488 loss is not passive: the jump-table appendix says the entries at `013F–0156` *were* the IEEE-488 routines and **calling one now produces a controlled system crash** (no disk damage

— the machine's two top display lines go "gangbusters hog-wild"), and *"IEEE-488 AS IOBYTE DEVICE 3 NOW NULL DEVICE"*. The parallel/Centronics entries at `0193+` survive unchanged.

5. The OZROM memory map (Appendix C-5)
"Some of the memory locations reserved by the OZROM 1E for its own purposes." The OCR splits the address column from the description column, but the pairing is unambiguous; the clock rows are confirmed independently against the OCC source in `docs/rt60a-analysis.md`.

Address	Purpose
EF07	Parallel-printer control — 0 = normal, non-zero = null device (DD only?)
EF57–EF58	Current BIOS starting address — how to find the start of the O1′s special BIOS regardless of MOVCPM/OS patches
EF5C–EF5D	KTBPTR — keyboard pointer. OZROM sets it to FC00; if (KTBPTR)+2 changes from its initial 0, OZROM repoints it at the ROM table
EF5F	Keyboard matrix value of the last key pressed
EF66	**Start of system clock — hours, 2-digit BCD**
EF67	**Minutes, 2-digit BCD**
EF68	**Seconds, 2-digit BCD**
EF69	**Sixtieths of a second, 2-digit BCD**
EFDF	Time-display toggle byte
EFE0	Keyclick toggle byte
EFE1	Keyclick counter
EFE2	Blink toggle byte
EFE3	Blink counter
EFE4	Blink rate value — gives 30/X blinks/sec; default 20 (decimal) = 1.5 blinks/sec
EFE5	KEYTYP — whether the current key is normal / shift / control / control-shift
EFE6–EFE7	Temporary pointer used by the keyboard-redefinition routines
EFE8	Clock position — F000 + ((value + 43) MOD 256), anywhere on the first two display lines
EFE9	Temporary storage of the character at the cursor (used by the blink routine)
EFF2–EFF4	**Interrupt routine hook** — preset with RET (0C9h); see §8
F000	Beginning of video RAM (3 KB, F000–FBFF)
FC00	Start of the RAM keycode table: FC00–FC37 normal, FC38–FC6F shift, FC70–FCA7 control
FCA8–FCBF	Reserved for future use
FC[illegible]–FD[illegible]	**[illegible]** (64a[illegible])

6. The ROM jump table (Appendix C-6)

Provided "for the programmer's convenience … since actual ROM addresses called by this table may vary according to which version of the OZROM 1E is being used". This is the *same interface* the stock ROM exposes at `0100+`, so it is directly comparable to `rom144.asm`'s vector table (`#72`).

Addr	Entry	Note
0100	JMP COLD_BOOT	
0103	JMP WARM_BOOT	
0106	JMP INPUT_STATUS+01	keyboard
0109	JMP CONSOLE_INPUT+01	keyboard
010C	JMP CONSOLE_OUT+01	display
010F	JMP LIST	serial output
0112	JMP PUNCH	serial output
0115	JMP READER	serial input
0118	JMP DRIVE_RESET	
011B	RET ! NOP ! NOP	**no ROM-level select-disk**
011E	JMP READ	
0121	JMP WRITE	
0124	JMP READ_TRACK_INFO	
0127	JMP READ_SECTOR	
012A	JMP WRITE_SECTOR	
012D	JMP LIST_STATUS	serial-port status
0130	JMP SENSE_DISK_DENSITY	same entry as the stock ROM's SENDEN
0133	JMP ROM_JUMP_1	exit from the interrupt routine for bank 1
0136	RET ! NOP ! NOP	no ROM_JUMP_2
0139	JMP FORMAT	
013C	JMP INIT_SERIAL	
013F–0156	*(were IEEE-488 routines)*	**calling these crashes the machine**
0157–0186	same functions as O1 ROM 1.4	
0187	OR 0FFH ! RET	IEEE printer input status → **null device**
018A	OR 0FFH ! RET	IEEE printer output status → null
018D	XOR A ! RET ! RET	IEEE printer input → null
0190	RET ! RET ! RET	IEEE printer output → null
0193	JMP PARALLEL_OUTPUT_STATUS	**Centronics parallel** output; needs RWCMD (EFD1) and SAVTYP set

7. The keytable (Appendix C-7) and the column switch

OZROM's normal, shift and control keytables are moved up into RAM at power-up or RESET (`FC00+` per §5); the control-shift table stays in ROM and cannot be redefined. The codes are the machine's own, not ASCII: `0x80–0x89` are the ten function keys, `0x8A–0x8D` the four arrow keys, and the control-shift table carries the "process immediately" codes `0xF0–0xFC` (screen positioning; the time, keyclick, blink and auto-horizontal-scroll toggles; the **Screen-Pac 52/80/104-column select keys**; and the assign/reassign-parallel-printer keys). Three codes are permanently assigned: `0xDD` = ctrl-shift-Z, `0xDE` = ctrl-shift-X, `0xDF` = ctrl-shift-R.

The column switch is the part on our design path (#24). OZROM's own words (Appendix C): the ctrl-shift keys include "the Screen-Pac 52-, 80-, and 104-column select Keys", and on a machine with no 80-column upgrade the switch "can also toggle between the left and right 52 columns of an 80-column display".

Measured against the three machines (§3)

OZROM boots at the machine's native width — **52 / 104 / 80** columns on `osborne1` / `osborne1sp` / `osborne1nv` — and reaches `A>`. That is the same result every other disk gives (`mame-emulation.md` §2): the *window* is the machine's, the *text layout* is the software's.

What is **not yet measured** is the switch itself in action: pressing ctrl-shift- and watching the width change. Doing that needs the harness to post a *modifier-qualified* keystroke (`tools/research/mame-boot-probe.lua` only posts plain characters), and the column keys' effect under MAME depends on whether the driver models the width as a PIA-driven horizontal offset or as fixed hardware. It is the one open item from #81 that needs code rather than reading — see §9.

8. The clock, the memory tester, and the error codes

Time-of-day clock. OZROM's `TIME HHMMSS` sets it, ctrl-shift-T shows it (top two display lines, position set by `EFE8`), and `ALARM`/`TIMER`/`QUIKTIME` ride the interrupt hook. Its cells are the machine's own `EF66–EF69` (hours, minutes, seconds, sixtieths, BCD) — **not** a new ABI. Accuracy is "better than three minutes a day" with the drives idle; disk access momentarily stops the clock (the manual is explicit that the clock "has no guarantee" under disk load). The full treatment, and why it matters to our own clock contract, is in `docs/rt60a-analysis.md` "The third shape".

The interrupt hook. On the 60 Hz video interrupt, OZROM's `SYSINT` saves the stack pointer, PC and registers and then calls **`0EFF2h`**; a bare `RET` there continues normally, so a programmer puts a jump to their routine at `EFF2` and ends with `RET` — **no register save/restore needed**. 26 bytes of interrupt stack (13 levels) and 192 bytes of routine RAM (`FCC0–FD7F`) are available. Note this is *not* the RT-60A's `0xEFF8` hook: different `INTBL` slot, different calling convention (see `rt60a-analysis.md`).

The memory tester. Press TAB at power-up or after RESET; it cycles `0 → 0EFFFh`, writing and checking `0`, `0FFh`, `0AAh` and `55h` at each location and naming the failing chip. Video RAM is not tested. A ready-made second oracle for #29.

Error codes (Appendix D): `BOOT ERROR` = "you are not using an OZROM 1E"; `NO SYS!` = a formatted diskette with no system on it (SYSGEN it); `OH, NO!` = same as `BOOT ERROR` (or a bad

system track). Also worth knowing: OZROM **enables CRC checking on disk access** where previous ROMs did not, so copy verification now catches real read errors.

9. What is left from #81

- **Drive the column switch from the harness** (§7) — the one measurement still missing. Needs modifier-aware key injection.
- **Diff the jump table against `rom144.asm`** entry by entry (§6 vs `docs/o1-rom-source.md`). The tables are the same shape; a disagreement would be a finding.
- **Reconcile the `EF66–EF69` clock against ROM 1.44**, whose source no longer ticks the cells (`rt60a-analysis.md`). Which ROM revision dropped `UPTIM`, and did any shipped 1.4x keep it?
- **Find a complete copy of the Utilities disk** (§10). Ours has an intact directory but 290 of 400 sectors erased, so most file bodies are holed. The vendor's own `CRCKLIST.CRC` would verify a replacement immediately.

10. The OZROM 1E Utilities disk — the software half (#81)

`research/roms/OZROM-1E/OZROM_1E_Utilities.IMD` is the disk that goes with the ROM: the utilities the manual documents, as a customer received them. We had the ROM and the manual and not the software, so this is the missing third of the package.

It is a **data disk, not a system disk.** It does not boot — the system tracks carry a stub whose message is "There is no CP/M system on this disk." You boot your own CP/M and put this in drive B, which is why the package shipped as a bare set of files.

10.1 The directory — complete

Readable in full: 20 files, all in user 0, in creation order, allocation intact. Standard single-density O1 geometry (40 cyl × 10 × 256 B FM, 2 KiB blocks, 3 boot tracks, 64 directory entries).

File	Bytes	File	Bytes
ALARM.COM	640	OZPATCH.COM	3328
CLKPATCH.COM	640	QUIKTIME.COM	256
CLKPATCH.DOC	1792	QWERTY.COM	1024
CRASH.COM	256	READ.ME	1280
CRCKLIST.CRC	512	SET.COM	2048
DVORAK.COM	1024	SHOW.COM	512
DVORAK.DOC	1024	TD.COM	896
FK.COM	2816	TD.DOC	1024
MEMORIZE.COM	1280	TIME.COM	640
MYKEYII.COM	640	TIMER.COM	768

The set maps onto the manual's feature list one for one: the column switch (`SHOW`), the keyboard remaps (`DVORAK`, `QWERTY`, `MYKEYII`), the clock (`TIME`, `TIMER`, `QUIKTIME`, `CLKPATCH`, `ALARM`), the

function keys (`FK`, `MEMORIZE`), and `OZPATCH` to patch the ROM image itself. Two things it settles that the manual only implies:

- **The utilities are separate `.COM` files, not ROM calls.** Nothing here is resident; the ROM supplies the interface and the disk supplies the programs. A reproduction does not need to ship this disk to be OZROM-compatible.
- **`CRCKLIST.CRC` is a self-check file** — a name/CRC list, so a 1984 owner could verify their copy. It is the vendor's own integrity manifest, and it is the reason to keep hunting for a complete one: the hashes let a good copy be confirmed in seconds.

10.2 Condition — partial, and that matters

This is a **partially erased disk**. Of its 400 sectors, **110 carry data and 290 read back as uniform `0xE5`** (the format fill byte), and the erased sectors fall *inside* file bodies rather than being confined to free space:

```
     sec: 1 2 3 4 5 6 7 8 9 10
  T0   R R . . . . . . . .      boot stub
  T1   . . . . . . . . . .
  T2   . . R . . . . . . .      the "no CP/M system" loader
  T3   R . R . R . . R . R      directory (S1-S8) + data block 1 (S9-S10)
  T4   R . . R . . . . . .
  ...
  T7   R R R R R R R R R R
  ...
  T21  R . . . . . . . . .
  T22  . . . . . . . . . .      free from here to T39
  ...
```

The directory and the file *names, sizes and allocation* are therefore trustworthy; the file *bodies* mostly are not. `SET.COM` is 100 % present, `FK.COM` 93 %, `MEMORIZE.COM` 86 %; `CRASH.COM` is gone entirely, and `ALARM.COM`, `QUIKTIME.COM` and `READ.ME` have each lost their first sector.

What survives is still worth reading — `READ.ME` is a real note about the rewritten `TIMER` syntax (`TIME TIMER 10.6 WS DOCUMENT.DOC`, and the syntax changed at version 2.00), and `CLKPATCH.DOC` explains the blink-rate patch — but **nothing here should be treated as a byte-exact copy of a 1984 master.**

The `0xE5` is not an artefact of our reconstruction. The IMD holds 110 normal sector records and 290 compressed ones, every compressed record filled with `0xE5`, and **no "unreadable sector" records at all** — ImageDisk read those sectors and got `0xE5` back. Nor is it a de-skew problem: the erased fraction is invariant under rotation, and no block alignment from −1 KiB to +2 KiB produces a consistent layout (§10.4).

10.3 The boot sector is not vendor code

Track 0 sector 1 holds a stub, and it does not look like anything Micro Management shipped:

```
C3 5C D2   JP 0D25Ch
...
+20   "PUT THIS ON T0S1"
```

```
...
+80   "THIS IS A TEST OF THE AMERICAN BROADCAST NETWORK"
```

"PUT THIS ON T0S1" is an instruction addressed to whoever was writing the disk, and the Emergency Broadcast System test line is a fill/verification pattern. The pair reads like a **placeholder written by whoever produced this image**, not product. Two consequences: treat the boot sector as an imaging artefact rather than evidence about the product, and expect that this disk cannot be booted to check anything — that has to be done by copying the files onto a system disk.

10.4 A tooling bug this found

`imdinfo.py --raw` appended sectors in the order they appear in the `.imd`, which equals logical order only when the numbering map is sorted. Single-density Osborne disks are not: they number `(2,3,…,10,1)`, so **every track came out rotated by one sector.**

The failure was silent, and that is the part worth remembering. A rotated CP/M disk still parses — the directory lands somewhere plausible and `cpmfs.py` returns a file list — it is just the *wrong* list. Here it returned 12 files instead of 20, with 8 of them carrying another file's blocks. Nothing errors, nothing looks malformed, and the output is plausible enough to archive.

Fixed in `tools/research/imdinfo.py`; the dump is now byte-identical for sorted maps (checked against `DRIVE_C.IMD`) and correct for rotated ones. Two archived images turn out to be rotated and were never convertible before the fix: `research/diagnostics/DIAG_2.04.IMD` now yields the **15-file Osborne diagnostic disk** (`DIAG.COM`, `FILETEST.COM`, `DDBIT.COM`, `SETUP.COM`, `SYSGEN.COM`, …), and `research/source-code/OCCROM.IMD` its 77-track FM/MFM disk.

10.5 What it changes for #81

Less than the file count suggests, and that is the honest answer: the ROM is the deliverable and this is the optional companion disk. What it does give us:

- The **complete and authoritative file list** of the utilities, so a future search knows exactly what to look for — and `CRCKLIST.CRC` supplies the hashes to verify a good copy against.
- Confirmation that the utilities are **external programs, not ROM entry points**, which bounds the reproduction scope: OZROM compatibility means the ROM's jump table and memory map, not this disk.
- A second, independent description of the feature set to hold against the manual (§4).

Still open: a *complete* copy. Anyone with a working OZROM disk can supply one, and `CRCKLIST.CRC` makes verifying it trivial.

Sources

- `research/roms/OZROM-1E/OZROM_1E.BIN`, `OZROM_1E_Manual.pdf`, `OZROM_1E_Manual.ocr.txt` — BrettHallen `Osborne_1`, `ROM/`; OCR per the `research/README.md` convention (300 dpi, `tesseract`).
- `research/roms/OZROM-1E/OZROM_1E_Utilities.IMD` — the companion utilities disk (§10). Owner-supplied local file, 2026-10-11; source URL not yet recorded. ImageDisk 1.19, label `ozrom`, 40 cyl × 10 × 256 B FM.

- `research/source-code/BIOS-144-source/rom144.asm` and the OCC CBIOS modules (`occram25.asm`, `occbio15/45/95.asm`) — the stock interface and RAM map the jump table and clock cells are compared against.
- `docs/mame-emulation.md` §2/§5, `docs/o1-rom-variants.md`, `docs/rt60a-analysis.md` — machine widths, ROM inventory, the clock.

Video and the character generator

The 2716 at **UA15** holds 128 glyphs of 8×10, and the first 32 of them (`0x00–0x1F`) plus `0x7F` are **graphics characters** rather than letters. They are the most stable thing in the ROM: the two char-gen variants differ in 93 of 128 glyphs, and the `0x00–0x1F` set is byte-identical in both (`docs/o1-rom-variants.md`).

They are block graphics, not box-drawing

The natural assumption is the IBM/CP437 line-drawing set (─│┌┐└┘├┤┬┴┼). That assumption is wrong, and the manual says so in its own section title:

> **5.5 BLOCK GRAPHICS** — *Osborne 1 Technical Manual* p. 25

Looking at them confirms it — the sheet is committed beside this file as `o1-chargen-graphics.png`, rendered by `tools/romfont.py --glyph-sheet`:

- **halves and quadrants** (solid blocks filling half or a quarter of the cell)
- **staircases and triangles** (the diagonal edges, drawn in steps because an 8×10 cell has no other way to do a diagonal)
- a **circle** and a **cross**
- `0x7F`, a **checkerboard/hatch** fill — the glyph the site's wordmark is hatched with

So the candidates to compare against are **Block Elements** (`U+2580–U+259F`) and Geometric Shapes, not the box-drawing block. That changes the mapping work from "guess a line junction" to "measure a fraction of the cell", which is a much better position: a block glyph can be *derived* from its coverage.

What the set is for

It is what the machine used for graphics on a character display — the same purpose as CP437's block and box characters, and the reason the O1 can draw frames and charts at all. Two consequences for this project:

- **A video personality (#24) has to render these correctly**, not just the letters. `docs/o1-memory-io-map.md` §2a has the cell and attribute layout.
- **They are page furniture (#97).** A period site's rules, table borders, footer and bullets can be the machine's own characters instead of CSS borders.

The mapping (open)

The mapping table is **not written yet, on purpose**: the issue asks for each glyph to be identified by looking at it beside its candidate, and recorded with its evidence rather than asserted. The sheet is the evidence side of that — it is generated from the ROM, so it cannot drift from it.

Next: put the sheet beside a rendered Unicode Block Elements chart, identify each of the 33, then add the table to `tools/romfont.py` so the codepoints ship in the font.

Why 64 columns (see also §2a)

The recovered AUTOCPM art is 64 columns wide and the O1′s screen shows 52. That is not a contradiction, and the SCREEN-PAC guide supplies the missing piece: the machine separates **logical** width from **physical** width.

> Logical screen size refers to the number of characters you can enter on a line without automatically inserting a carriage return. Logical screen size can range from 1 up to a **maximum of 128 characters**. … Physical screen size refers to the number of columns actually displayed on your screen. With SCREEN-PAC, this will be 52, 80, or 104. — *Osborne 1 SCREEN-PAC User Guide*, p. 13

A 64-column drawing is a **logical line of 64 characters**: it fits the 128-character video memory row, and the physical window slides over it. On stock hardware you see 52 of the 64 columns at once and scroll for the rest; with the SCREEN-PAC at 80 you see all of it.

The full statement, with the technical manual's geometry and the attribute bits, is in `docs/o1-memory-io-map.md` §2a.

Sources

- `research/osborne1/2F00153-01_Osborne1TechnicalManual_1982.pdf` §5.0–5.5, pp. 23–25 (image-only scan; read by rendering the page range — the manual's page 15 is PDF page 29, so the offset is **+14**).
- `tools/romfont.py` — the decoder and the `--glyph-sheet` renderer.
- `docs/o1-rom-variants.md` — the char-gen variants and what differs.

Interfaces

The floppy subsystem and its formats

The DD upgrade is usually described as "twice the capacity". Its own manual describes something else: a controller that recognises **six disk formats**, five of them other manufacturers'.

> Thus, you'll find that you didn't just purchase double density, you purchased a **multi-density disk option**. Under the current release of double density, the following formats are all **automatically recognized by the computer**.

Format	Capacity
Osborne 1 single density	92k per diskette
Osborne 1 double density	182k per diskette
Xerox 820 single density	**80k** per diskette
Cromemco single density	80k per diskette
IBM Personal Computer (CP/M-86 format)	156k per diskette
DEC VT-180	171k per diskette

— *Osborne 1 Double Density User Manual*, p. 4

That is the primary source for #95, and it **confirms the claim**: the DD upgrade is a format capability, not just density. Note the wording is "automatically recognized by the computer" — the recognition is in the machine, which is what makes it testable in MAME without the hardware.

Correction to the quoted list

oldcomputers.net gives the Xerox 820 entry as **82K**. The manual says **80k**, and the manual is the one to follow — the page-4 list is the manufacturer's own statement of what the option does.

The manual also records what the option deliberately does **not** handle, which matters as much as what it does:

- **Hard-sectored diskettes** (North Star, Micropolis, Vector) — the controller cannot see the sector holes.
- **Non-block formats** (Apple, NEC) — no block-of-bytes structure to read.

and names the formats that were *anticipated* rather than shipped: TRS-80 Model I and III CP/M, and Zenith/Heath double density, "most likely in our user magazine, The Portable Companion".

Why the ROM is part of the answer — and the answer

The upgrade is described as coming with the **1.4 ROM**, and the recognition is a property of the computer rather than of the drive electronics. The 1.44 BIOS source and the overlay CBIOS modules are in the repo (`research/source-code/BIOS-144-source/`), and reading them settles #95 without hardware. The mechanism is **two cooperating parts, and neither is the monitor ROM alone**:

1. **The monitor ROM senses the disk.** `SENDEN` (jump-table entry `0130`, `SENSE_DISK_DENSITY`) tries both densities and issues a WD179x **READ ADDRESS** (`RADR`) on track 0. The address mark the controller finds *is* the disk's geometry: it yields the density (FM/MFM), the sector length and the sectors per track. There is no format table in the monitor ROM — the geometry comes off the disk. `SAVTYP` is set to a code of `density (bit 0) | sector length (bits 2–3)`, where the length field is `0`=128, `1`=256, `2`=512, `3`=1024 bytes.
2. **The DD CBIOS carries the DPB table.** `occbio95.asm` defines a table of Disk Parameter Blocks, and `occbio45.asm` (`GDEN`) converts the sensed physical sector count into a count of 128-

byte *logical* sectors (×1/×2/×4/×8 for 128/256/512/1024) and searches the table for the entry whose sectors-per-track **and** type code both match. On a match it installs that DPB and its sector-skew translation table.

The table is the smoking gun (`occbio95.asm` / `occbio15.asm`, the names are the manufacturer's own):

Entry	Type code	Sectors/ track	Block	Dir	boottrk	`DSM` formula	**capacity**	Manual's figure
Osborne DD (`DSKD1`)	`0Ch` = MFM, 1024 B	40 (=8×5)	1 K	64	3	`((40-3)*8*5)/8` = 185	185 K	182 k
Osborne SD (`DSKS1`)	`5` = FM, 256 B	20 (=2×10)	2 K	64	3	`((40-3)*2*10)/4` = 46	46 blk × 2 K = 92 K	92 k
Xerox (`XEROX`)	`1` = FM, 128 B	18 (=1×18)	1 K	32	3	`((40-3)*1*18)/8` = 83	83 K	80 k
IBM (`IBM`)	`8` = MFM, 512 B	32 (=4×8)	1 K	64	1	`((40-1)*4*8)/8` = 156	156 K	156 k
DEC (`DEC`)	`8` = MFM, 512 B	36 (=4×9)	1 K	64	2	`((40-2)*4*9)/8` = 171	171 K	171 k

`NUMDPB = 5`. **The capacities match the manual's list exactly** — IBM 156 k and DEC 171 k to the kilobyte — which is what makes this the answer rather than a guess. The manual lists **six** formats and the table has **five** entries because **Cromemco shares the Xerox geometry** (both 5.25" single-density, 80 k), so one DPB serves both names.

Each entry also carries a **sector-skew translation table** (`XTAB` in `occbio95.asm`): Osborne DD is 1:1, Osborne SD is 2-to-1, Xerox is 5-to-1, IBM is 1:1, DEC is 2-to-1. "Mostly DPB + FDC parameter tables", as #30's text put it — confirmed, and the *skew* is the part that would be missed by a geometry-only description.

So the DD upgrade's multi-format claim is a **CBIOS DPB+skew table selected by a monitor-ROM density/sector probe** — recognition in the computer, exactly as the manual says, with the *filesystem* parameters living on the diskette's own CBIOS rather than in the ROM. That also explains the G2 answer (#30): a "G2 format set" is more entries in the same table.

What this means for the project

- The floppy personalities (#20, #30, #62) cannot be "an Osborne drive": they have to be able to *present* six geometries, which is a larger requirement than the issue assumed.
- The capacity figures are the acceptance target for the MAME test in #95: each format either reads **and** writes byte-comparably, or is recorded as failing, with the failure attributed to the ROM, the controller model or the image tooling.

A second vendor made the same claim, and a bigger one: Nuevo

OCC is not the only multi-format claim of the period. Nuevo's four-way product line, advertised in *The Portable Companion* #15 (Jan/Feb 1985, PDF p. 39), escalates from density to format:

Nuevo product	Price	Claim
Double Density Upgrade	$150	"a full **183K** of data storage per disk" — and *"upgradable to double-sided double density"*
2 x 2 Upgrade (780K)	$595	double-sided double density, "nearly **400K each**", *"Format compatible with the new ACC Vixen"*, **128 directory entries**
4 x 4 Upgrade (1580K)	$745	"both drives are replaced with **double-sided 96 track-per-inch drives**"

The 4 x 4 text is the important one:

> This system will **automatically recognize single-sided disks that are either single or double density and double-sided disks that are either 48 TPI or 96 TPI**. You can **read or write, even boot**, any of these discs automatically. Directory Capacity 256 File Names.

So Nuevo claims four formats beyond the six in OCC's DD table — 96 TPI, both densities, both sidednesses — and claims to **boot** them. Two things make that credible rather than just advertising:

- **The stock O1 interface already carries what it needs.** P8 pin 4 is **TG43**, the drive's write-current-reduce input for 96 tpi media, and pin 32 is **SIDE SELECT** — both documented in `docs/floppy-adapter-design.md` from the schematic and the OEM cable. The controller is not the limit; the drives and the format table are.
- **The format table is the CBIOS's job, and Nuevo replaces it.** The installers on `OS1NUEVO.IMD` are DDT patches inserting `BIOS15.OBJ` at `1F00` and `BIOS15BM.OBJ` at `29E0` (`docs/o1-upgrade-boards.md`), and the advertised directory capacities — 128 and 256 entries — are disk parameter

block values, exactly what such a patch changes. The 2 x 2 and 4 x 4 prices include **new drives**, so the media capability is hardware and the *recognition* is the BIOS's.

Consequence for #95 and #30: the multi-format requirement is not an OCC quirk. Two vendors shipped it, with overlapping but different format lists, and the O1′s own floppy connector carries the signals for both. A floppy personality (#20) has to be able to present more geometries than the six in the OCC table.

Nuevo's own BIOS 1.4 → 1.5 list: the formats are disk parameter blocks

Nuevo's DD manual has a passage that settles the mechanism behind all of this. Its BIOS 1.5 differs from the factory BIOS 1.4 in ways it enumerates itself:

> 2) For BIOS Version 1.5 the **disk parameter blocks for IBM, Xerox and DEC formats have been removed** to provide space for code to allow **auto configuration at boot time via NEWSETUP.COM**. Space remains for two more DPB's to be inserted.
>
> 3) BIOS Version 1.5 **checks for the presence of the Nuevo 80 Column Upgrade**. If not present, the configuration routine for the 80 Column Upgrade will be skipped. Thus, the Version 1.5 BIOS will boot and run correctly on machines that don't have the Nuevo 80 Column Upgrade.
>
> 4) The jump table configuration and the jump table start address (**E100h**) is the same in BIOS Version 1.5 as in BIOS Version 1.4. — *Nuevo Double Density Upgrade Manual*, "differences between the two BIOS versions" (OCR: `research/nuevo/Nuevo_Electronics_DD_Upgrade_Manual.ocr.txt`)

Five things follow, and the first two correct an assumption worth recording:

- **"BIOS 1.4" and "BIOS 1.5" here are the *CBIOS on the system tracks*, not the monitor ROM.** Nuevo's installers load `BIOS15.OBJ` at `1F00` and `BIOS15BM.OBJ` at `29E0` (`docs/o1-upgrade-boards.md`), and the manual compares 1.5 against *"Version 1.4, the most recent **factory** version"* — so the DPB table it describes lives in the bootable CBIOS image, and *that* is the file to read for #95, not the 4 KiB monitor ROM.
- **The monitor ROM contains no literal DPBs, and the factory BIOS does not need them.** A structural scan of all four ROMs the project holds — factory 1.4, 1.43, 1.44 and Nuevo 1.51 — finds **zero** CP/M DPB-shaped structures. The reason is in the 1.44 source: the loader *"will load single or double density and any number of sectors per track or bytes per sector"*, and `SENSE THE DENSITY` **probes** — it reads an address mark at one density, flips the density bit, and retries, with a second pass after a home command (`rom144.asm` lines 3666–3712). Geometry is discovered by trying, not by table lookup.
- **Nuevo is not a superset of the factory BIOS.** Nuevo *removed* IBM, Xerox and DEC to make room for its own code and re-added formats through **auto-configuration at boot via NEWSETUP.COM**. Two vendors, two mechanisms, overlapping lists — the same conclusion as above, now with the reason.

- **"Space remains for two more DPB's to be inserted"** — the table is extensible, which is how the 2 x 2 and 4 x 4 geometries fit into the same scheme.
- **Nuevo's BIOS checks for its own video board's presence** and skips the 80-column configuration if it is absent — the same *pattern* as the SCREEN-PAC's `2400H` flag (`docs/01-upgrade-boards.md`), and the reason its 1.5 ROM still boots on a machine with no Nuevo hardware. The jump table sits at **E100h**, unchanged from 1.4, which is the region the SCREEN-PAC's width variable (`E1A5H`, `docs/01-memory-io-map.md` §2a) lives in.

A third vendor sold the format list itself: Osmosis OSMOS 4

Osmosis advertised the capability as a **standalone product**, in *The Portable Companion* #11 (October 1983), PDF p. 32 = printed p. 30:

> **OSMOS 4** — Disk Formatting & Conversion: Allowing your Osborne to **read, write and format over 20 other manufactures CP/M formats**. $225

That is the strongest form of the claim in the archive: not a capability bundled into a density upgrade (OCC), and not a side effect of a BIOS rewrite (Nuevo), but a **$225 product sold on its own**, priced *above* the $195 double-density board it was usually bought with. The same page bundles them — "**OSMOS 1, 4 AND 5 ALL FOR $385**", $670 separately — which is a fair measure of how much of the value was seen to be in *formats* rather than in *density*.

Its density line, for the record: **OSMOS 1** 184k per drive on existing drives ($195), **OSMOS 1.5** single-sided DD now, 370k later with new drives ($335), **OSMOS 2** double-sided DD, 370k ($895), **OSMOS 3** double-sided DD **double tracking**, 750k per drive, "an incredible 1.5MB" ($1050). The full product table is in `docs/01-upgrade-boards.md`.

The tally is four independent format mechanisms: OCC's DPB table, Nuevo's boot-time auto-configuration, Osmosis's separate conversion program, and the factory BIOS's density/sector probing. A floppy personality (#20) that knows only the stock geometry fails on any of them.

What the DD upgrade costs in memory: the Corvus link program

A user note in *The Portable Companion* #11 (October 1983), PDF p. 54 = printed p. 52 — "**Doubling hard disk memory**" by Bob Gras, reprinted from the OSUG newsletter (Anchorage) — is a patch to Corvus's link program, and it says plainly why it is needed:

> If you had a single-density Osborne computer, but now have a double-density and are using the Corvus hard disk drives, this one is for you. … With the new double density system, **more memory is required for managing the Osborne floppy drives**, so this value has to be reset to accommodate the double density version of the Osborne.

The value is a CP/M offset in `CLINK2K2.ASM` (the source for the `CLINK2` program that links the machine to the Corvus drive):

```
IF NOT (ABOVE)      ;
DELTA  EQU  500H    ; OFFSET FROM STANDARD CP/M *
```

and the fix is to "type a **900H** in place of the **500H**", then assemble and link with `ASM.COM` and `LOAD.COM` (the article points at the Osborne manual's assembly instructions, pp. 253–255 of the "Rev: 2/28/82" edition). So the constant grows by `400H` — 1 KB — between the single- and double-density versions of the same machine, and the author's stated reason is the DD system's larger floppy- management code. **The DD upgrade is not memory-neutral**, which matters for anything that assumes a fixed TPA base, and it is the kind of third-party consequence the DD documentation itself does not mention (#95).

The acceptance test, run in MAME (#95)

The manual's claim is a capability, and the capability is testable without hardware because the recognition is in the computer. The test: **boot ROM 1.4+ with a DD CBIOS, put a disk in drive B, move files on and off it, and read the result back byte-for-byte.**

Tooling

Two small stdlib helpers were added for this (`tools/research/`):

- **`imdmake.py`** — writes a blank Osborne 1 disk in ImageDisk form (the writer half of `imdinfo.py`): `--dd` = 250 kbit MFM, 40×5×1024; `--sd` = 250 kbit FM, 40×10×256, filled with `0xE5`. It emits the `IMD 1.x:` version line first, because `disk-analyse` refuses a file that does not start with one.
- **`o1disk-copy.sh` + `o1disk-copy.lua`** — boot a disk in `-flop1`, put another in `-flop2`, wait for `A>`, type a CP/M command, and report the screen. It is the keyboard-injecting counterpart to `mame-boot-probe.sh`.

Results (measured)

Test	Command	Result
ROM 1.4+ boots DD	`mame osborne1 -bios ver14/ver143/ver144` on `52-blank.imd`	**all three boot to `A>`**
Content onto B:	`PIP B:=A:MBASIC.COM`	B: image gains `MBASIC.COM`, **byte-identical** to A:'s (24320 B)
Content off B:	`PIP A:BACK.COM=B:MBASIC.COM`	A: image gains `BACK.COM`, **byte-identical** — full round trip
Foreign density read	DD system in A:, **Osborne SD** disk in B:, `DIR B:`	**lists B:'s files** — the DD CBIOS auto-recognised the single-density disk

The last row is the important one for this issue: it exercises the real mechanism end to end. The monitor ROM's `SENDEN` read the SD disk's address marks (FM, 256-byte sectors, 20/track), the

DD CBIOS matched `DPBS1`, and `DIR` read a foreign-density directory. **Osborne SD is one of the manual's six formats, confirmed on the emulated machine.**

A MAME gotcha worth writing down

`os.exit(0)` in the Lua script silently discards guest disk writes. MAME flushes a modified image when the device is torn down at a *clean* shutdown; `os.exit()` skips that. A `SAVE 1 HELLO.TXT` under `os.exit()` left the image byte-identical; the same command with the script simply ending (MAME running on to `-seconds_to_run`) persisted. `o1disk-copy.lua` therefore does **not** call `os.exit()` — the opposite of `mame-boot-probe.lua`, which only reads.

What the test does *not* yet cover

- **The other four formats** (Xerox 820, Cromemco, IBM CP/M-86, DEC VT-180) — we hold no images for them, so the DPB-table proof above stands on the source rather than on a run. The way to close it is to synthesise an image per DPB entry with `imdmake.py` (or `disk-analyse`) and `DIR` it; that is the next unit of work, and it is cheap now that the geometries are known.
- **Real hardware** — see below.

IMD → HFE for the real machine

`disk-analyse` converts both ways, and the round trip is lossless on the raw sector data:

```
disk-analyse blank-dd.imd blank-dd.hfe      # write to a Gotek/HxC
disk-analyse blank-dd.hfe blank-dd-back.imd # read it back
tools/research/imdinfo.py --raw a.raw blank-dd.imd
tools/research/imdinfo.py --raw b.raw blank-dd-back.imd
cmp a.raw b.raw                             # identical
```

Measured: a `--dd` blank and the `52-blank.imd` system disk both convert to HFE (2 039 296 bytes each), and IMD→HFE→IMD reproduces the raw sector image exactly. `disk-analyse`'s HFE output is what a hardware floppy emulator wants, so the images built and validated above are directly testable on a real Osborne 1.

Sweep of the archive (measured 2026-10-11). Every `.imd` we hold converts — **22 of 22, no failures** — including the ones that are *not* Osborne-format: the double-density system disks, all fourteen Dunfield shipped-software images (SD and DD), `DRIVE_C.IMD`, `OS1NUEVO.IMD`, `OCC1_HARDDISK.IMD` and the two diagnostic disks. A spot round trip on the single-density 80-column `OS1XUTLS.IMD` reproduces the raw sector image byte-for-byte. So the "foreign" images in the collection are ready to write to a hardware emulator.

```
for f in research/shipped-software/*.IMD research/drive_c/DRIVE_C.IMD; do
    disk-analyse "$f" "$(basename "${f%.*}.hfe")"
done
```

- `research/osborne1/o1doubledensityusersguide.pdf` — *Osborne 1 Double Density User Manual* (15 pages, image-only; OCR'd per the convention in `research/README.md` as `o1doubledensityusersguide.ocr.txt`). https://www.kev.pulo.com.au/osborne1/documents/o1doubledensityusersguide.pdf

- `docs/o1-memory-io-map.md` §3 (WD179x detail) — the formats already recorded there from the schematic side.

The floppy adapter design

Design spec for issue **#62**. Tracks the floppy interposer personality (#20) and G2 multi-format support (#30). Derived from `research/osborne1/floppy-adapter/` (Wayne Visser passive adapter + provenance), Richard Loxley's Osborne Restoration parts 5 and 12–15, and `hardware/floppy-adapter/README.md`.

> Status: **design settled 2026-10-10; Rev A schematic captured.** Open questions are resolved in §8 with the evidence behind each decision. Rev A KiCad capture lives in `hardware/floppy-adapter/` (`floppy-adapter.kicad_sch`, generated by `gen_floppy_adapter_sch.py`).

1. Purpose

Let the user choose, from a front-panel switch, whether the **drive A** position is served by the physical drive or by a **Gotek** (FlashFloppy) emulator, with **drive B always present** in both modes. The card sits between the O1 mainboard/DD-board floppy header and the OEM floppy cable; nothing in the machine or its cable is cut.

Prior art, deliberately **not** copied:

- Wayne Visser's `Osborne1FloppyAdapter` is a **passive** tap — it puts a Gotek on the bus alongside the physical drives and switches nothing. We keep its pin/rail mapping as the O1 interface reference.
- Richard Loxley's hand-wired build (parts 12–15) proved the concept but needed a DPDT switch + a 5 V relay soldered onto the Gotek, plus hand-cut Dupont cable. This design moves all of that onto a PCB at the connector end of the cable.

2. Osborne 1 floppy interface (34-pin, Shugart-like)

The O1 uses a Shugart-like 34-pin interface, but with **+5 V and +12 V routed to the drives over otherwise-unused cable pins** — there are no separate drive power connectors. The pin functions below were first transcribed from the archived Wayne Visser adapter (`research/osborne1/floppy-adapter/FloppyAdapter.sch`); they agree with Loxley's signal audit and the standard Shugart ordering for the shared signals. **Re-read from the O1 mainboard schematic itself, 2026-10-10** — DWG 1A2011-00 Rev E, sheet 9 of 9 (DISC CONTROLLER), the P8 symbol — see §8.2. That reading corrected two pins and the GND row.

Pin(s)	Signal	Notes
1,2,3,5,9,19,27,29,31,33	GND	pin 7 is GND on the drive-end drawing instead; neither 2 nor 7 is a driven signal, so both are tied to GND here
4	**TG43**	**not GND** (corrected 2026-10-10) — FDC-driven; the drive's write-current-reduce input on 96 tpi drives
6	**EARLY**	**not GND** (corrected 2026-10-10) — FDC-driven
7	GND (drive-end drawing)	see note on pin 1 row
8	INDEX	
10	DS-A (drive select A)	**the switched line** — Gotek replaces drive A
11,13,15,17	+12 V	power-over-cable (physical drives)
12	DS-B (drive select B)	not switched — drive B stays present
14	*(not drawn on P8)*	
16	4 MHz	O1-specific clock; not used by Gotek
18	DIRECTION	
20	STEP	
21,23,25	+5 V	power-over-cable (physical drives; tapped for Gotek)
22	WRITE DATA	
24	WRITE GATE	
26	TRACK 0	
28	WRITE PROTECT	
30	READ DATA	
32	SIDE SELECT	
34	LATE	

Pins 4 and 6 carry live FDC outputs, so the card **passes them straight through J1 → J2** rather than grounding them. The earlier transcription (from the Visser adapter) tied 2, 4, 6 and 7 to GND;

that is harmless for a Gotek-only adapter, but this card switches the OEM cable through to the physical drives, and clamping TG43 low would assert reduced write current permanently. | 34 | LATE | O1-specific |

The Gotek needs only the static/step/data lines plus its drive-select (Loxley's 11-line audit). The card routes that subset to the Gotek header and leaves 4 MHz and the spare power pins off it.

3. Topology

```
flowchart LR
  MB["O1 mainboard<br/>34-pin floppy header"]
  subgraph CARD["floppy-adapter daughter card"]
    PLUG["J1  2x17 socket (bottom)"]
    CABLE["J2  2x17 header (top)<br/>to OEM cable"]
    GOT["J3  2x17 header (top)<br/>to Gotek signals"]
    PWR["J4  Gotek power<br/>(switched +5V, GND)"]
    SW["J5  panel-switch lead"]
    K1["K1  4PDT relay"]
  end
  DA["Physical drive A<br/>(150 ohm term pack)"]
  DB["Physical drive B<br/>(always present)"]
  GK["Gotek / FlashFloppy<br/>(as drive A)"]

  MB -- "J1 mates header" --> PLUG
  PLUG -- "pass-through (all pins)" --> CABLE
  CABLE --> DA
  CABLE --> DB
  PLUG -- "signals only" --> GOT
  GOT --> GK
  K1 -- "+5V / GND (when Gotek)" --> PWR
  PWR --> GK
  SW -- "coil current only" --> K1
  PLUG -. "DS-A (pin 10) in" .-> K1
  K1 -. "DS-A to physical A" .-> CABLE
  K1 -. "DS-A to Gotek" .-> GOT
```

Everything between J1 and J2 is a **straight pass-through** except pin 10 (DS-A), which is intercepted. The physical drives therefore see an unmodified cable; the only difference is which device the machine's DS-A reaches.

4. Switching design

4.1 Drive-select routing (pin 10)

DS-A from the mainboard (J1.10) is the relay common. De-energised it goes to J2.10 ⟶ the physical A drive; energised it goes to J3.10 ⟶ the Gotek. Never both, so the two devices cannot both answer as drive A. DS-B (pin 12) is untouched, so drive B stays present and selectable in both modes.

4.2 Gotek power switching

Loxley's field notes (part 15) are decisive and are the reason this card exists:

- **Switching only +5 V does not turn the Gotek off** — the O1′s pulled-up signal lines parasitically power the board through its input protection diodes.
- **Switching only GND makes the *other* drive unreliable** — the Gotek's signal lines are still referenced to the shared 5 V.
- **Switching both +5 V and GND works.** That is what this card does.

Consequence: the Gotek's **GND and +5 V must arrive only through J4**, *not* through the J3 header, otherwise switching GND has no effect. J3 carries signals only.

4.3 Switch element — mechanical relay (recommended)

A single **4PDT relay** (3 poles used, 1 spare) does all three switches at once:

Pole	Contact	Function
1	NO	switched +5 V ⟶ Gotek (J4)
2	NO	switched GND ⟶ Gotek (J4)
3	NC / NO	DS-A ⟶ physical A (J2.10) / Gotek (J3.10)
4	—	spare (e.g. front-panel LED)

- Coil: 5 V, ~72 mA (Omron G6A-434P 5 VDC: 69 Ω, ~360 mW), fed from the cable's +5 V through the panel switch J5.
- De-energised = physical drives (**fail-safe**): a broken/unplugged switch lead parks the machine on the physical drives.
- The relay only ever switches a **static DC select line** and the power rails — no stepped/data signal passes through the contacts, so contact bounce/bandwidth are irrelevant. A mid-switch transient is a single floppy error the machine tolerates (per the author's experience).
- Add a flyback diode across the coil and a 10 kΩ pull-down on the coil+ node so a disconnected switch is deterministic.
- The panel switch carries only coil current — never drive power, as the existing README requires.

4.4 Solid-state alternative (optional, smaller/silent)

If the relay is unwanted, the same three functions can be built from silicon: a P-MOSFET high-side load switch (e.g. AO3401) + N-MOSFET low-side switch (AO3400/2N7002) for the rails, and a 2:1 analog/bus switch (74HC4053 or 74CBT-class, 5 V-capable) for DS-A, all driven from one logic input. More parts and more 5 V-level care, but no coil current, no click, and lower profile. Recommendation: **relay for Rev A**, evaluate solid-state for Rev B.

4.5 Termination and pull-ups (why drive A matters)

The O1 terminates the floppy bus with a **150 Ω resistor pack on every signal line, located in physical drive A** — the drive at the far end of the cable. Drive B has no termination (Loxley parts 5 and 12). Two consequences:

- The shared signal lines stay terminated in both modes, because only DS-A is switched: drive A remains connected to every other line, so its 150 Ω pack stays on the bus even when the Gotek is selected.
- **DS-A loses its pull-up in Gotek mode.** The 150 Ω pull-up lives at drive A; routing DS-A to the Gotek disconnects it. The Gotek has **1 kΩ pull-ups on most lines but none on the drive-select lines** (Loxley part 5), so a floating DS-A is exactly the failure Loxley chased for days.

→ **Add an on-card pull-up on the Gotek-side DS-A net** (relay NO contact → J3.10): **1 kΩ default, with a 150 Ω option** (jumper or second footprint) to match the O1′s own termination. This is the single most important addition to the original concept, and it exists only because the Gotek replaces drive A rather than drive B. Loxley tried both 1 kΩ and 150 Ω on the select lines.

5. Reference schematic (Rev A, netlist level)
Components:

Ref	Part	Notes
J1	2×17 socket, 2.54 mm	bottom, mates O1 mainboard header
J2	2×17 header, 2.54 mm	top, OEM cable → physical drives
J3	2×17 header, 2.54 mm	top, Gotek (signals only)
J4	4-pin header (floppy power)	switched +5 V / GND to Gotek
J5	2-pin header	front-panel switch lead
K1	4PDT relay, 5 V coil (e.g. Omron G6A-434P)	3 poles used
D1	1N4148	coil flyback
R1	10 kΩ	coil-node pull-down
R3	1 kΩ	DS-A pull-up, Gotek side, default (§4.5)
R4	150 Ω (DNP)	DS-A pull-up alternative — fit R3 **or** R4 (§4.5)
R2, D2	1 kΩ + LED (optional)	front-panel “Gotek active” indicator

Connections (pin n of J1 unless stated):

- **Pass-through:** J1.n → J2.n for every pin **except 10**. This carries the signals *and* the +5 V / +12 V / GND rails to the physical drives unchanged.
- **DS-A:** J1.10 → K1 pole-3 common. Pole-3 NC → J2.10 (physical A). Pole-3 NO → J3.10 (Gotek).

- **DS-A pull-up:** R3 (1 kΩ default; 150 Ω option) from the J3.10 net to +5 V, on the Gotek side of the relay (§4.5).
- **Gotek signals:** J1 pins 8, 18, 20, 22, 24, 26, 28, 30 (and optionally 32, 34) ⟶ the matching pins of J3. J3 pins 12, 14, 16 and all GND/power pins are **NC**.
- **Gotek +5 V:** J1 +5 V (21/23/25) ⟶ K1 pole-1 common; pole-1 NO ⟶ J4 (+5 V).
- **Gotek GND:** board GND ⟶ K1 pole-2 common; pole-2 NO ⟶ J4 (GND).
- **Coil:** J1 +5 V ⟶ J5.1 ⟶ (panel switch) ⟶ J5.2 ⟶ K1 coil+ ; K1 coil− ⟶ GND. D1 across the coil (cathode to coil+). R1 from coil+ to GND. Optional R2+D2 from coil+ to GND for the indicator.

6. BOM (Rev A)

Qty	Item	Notes
1	4PDT relay, 5 V coil, DIP-14	Omron G6A-434P-ST-US (or equivalent)
1	2×17 socket, 2.54 mm, vertical	bottom mezzanine
2	2×17 header, 2.54 mm, vertical	top (cable + Gotek)
1	4-pin header (floppy power)	Gotek power, right-angle
1	2-pin header	panel switch
1	1N4148	flyback
1	10 kΩ 0805/TH	coil-node pull-down
1	1 kΩ 0805/TH	DS-A pull-up (R3, default)
1	150 Ω 0805/TH (DNP)	DS-A pull-up alternative (R4)
1	1 kΩ + LED	optional indicator
1	PCB, 2-layer, 1.6 mm	ENIG preferred (relay/contacts, mating)

7. Physical design brief

Board (PCB). - 2-layer, 1.6 mm FR4; ground plane on the bottom layer; +5 V/GND traces ≥ 0.5 mm (Gotek ~0.3–0.5 A). - Envelope to be fixed against the mainboard keep-outs (`research/osborne1/OCC1_1A2011-00_Schem_RevE.pdf`) and host photos (`research/pictures/`); target a compact card that clears the DD board. - J1 (socket) on the bottom face, positioned to engage the mainboard 34-pin header; add 2 mounting holes/standoffs so the mezzanine cannot rock. - J2 and J3 on the top face; orient both so the OEM ribbon and the Gotek ribbon exit without fouling. Silk-label each connector, mark pin 1, and label "O1 IN", "⟶ DRIVES", "GOTEK", "GOTEK 5V", "SWITCH".

Gotek mount (3D-printed part). - The Gotek (top cover removed) fits a floppy storage pocket; with its top headers fitted it is too tall, so connect it by a slim pigtail/direct-wire pads rather than plugging a header on top. - Design a printed bracket/faceplate that holds the Gotek in the pocket at the right depth, exposes the USB port, and carries the panel switch. This is the "Modification to support Gotek drives with 3D-printed floppy-disk faceplate" write-up planned

in `site/README.md`. - Cable egress: a full 34-pin ribbon will not pass the pocket vent slots. Run the Gotek on a reduced signal set (the §5 lines + power) on a narrower cable, or route through the "Ext Video" port as Loxley did.

Panel switch. - Plain panel-mount SPST (or SPDT for an indicator). 2-wire lead to J5; no drive power on the lead. Mount next to the Gotek pocket.

Reversibility. No cable or machine modification: unplug the card and the machine is stock.

8. Open questions / decisions (settled 2026-10-10)

All seven items are now closed. Each carries the evidence it rests on; the two that remain bench-verifiable are flagged as such.

1. **Drive topology — CONFIRMED (user, 2026-10-10).** The switch selects **drive A** (physical A ↔ Gotek); **drive B stays present** in both modes.
2. **Connector genders — CONFIRMED.** The O1 logic board's floppy connector is **P8, a 34-pin MALE header**. The Field Service Manual (2F00040) instructs "Attach the disk harness to the 34-pin connection on the logic board … Be careful not to bend any pins", and the double-density adaptor's "sockets … fit onto the logic board disk harness connector (P8)". The harness is a **female IDC socket** and is **not keyed** — "the RED stripe on the harness must be to the RIGHT" fixes pin 1. Therefore **J1 = female 2x17 socket** (card bottom) and **J2 = male 2x17 header** (OEM cable). Pin order is confirmed in §2.
3. **Switch element — DECIDED: electromechanical relay (Rev A).** A single 4PDT relay does all three switches with no 5 V-level silicon and no signal-path parts (§4.3). The contacts only ever carry a static DC select line and the power rails, so contact bandwidth/bounce are irrelevant. Solid-state (§4.4) stays a **Rev B** option if the click, coil current or height become issues.
4. **Isolation depth — DECIDED: Loxley's proven minimum.** Switch **DS-A + both power rails** and nothing else. The shared signal lines are *not* additionally gated: doing so adds series elements to a 4 MHz-class bus for no proven benefit, and the failure Loxley actually chased (a Gotek that would not power down) is fixed by the dual-rail switch alone. Full signal gating is recorded as a Rev B experiment, not Rev A scope.
5. **DS-A pull-up — CONFIRMED 1 kΩ default, 150 Ω option.** The FSM settles the termination location: "The A drive has an 8 pin 150 OHM Terminator resistor pack. B DRIVE DOES NOT." So when DS-A is routed to the Gotek it loses the O1′s 150 Ω pull-up, and the Gotek has no pull-up on its drive-select lines (Loxley part 5). The card therefore fits **R3 = 1 kΩ** as the default and **R4 = 150 Ω** as an alternative footprint (fit exactly one) on the Gotek-side DS-A net.
6. **Gotek drive-select jumper — SETTLED (bench-verify).** The Gotek must answer the O1′s **drive-A** select, which is cable **pin 10**. The Gotek manual labels the jumpers `S0 = Drive Select 0` / `S1 = Drive Select 1`, and FlashFloppy's generic guidance for a Shugart host is "unit 0 → place the selection jumper at **S0**". The O1, however, straps its drive A as **DS1** (FSM 8.1.4.11: "Drive A has a jumper at positions HM and DS1"), and the proven O1 adapter (`WayneVisser/`

`Osborne1FloppyAdapter`) notes "Jumper only **S1** on the GoTek". **Start with S1**; if the Gotek is not seen, move to S0. This is a 30-second bench check and the card is agnostic to it.

7. **Gotek cable route / bracket — DECIDED.** Gotek in the **right-hand floppy storage pocket** (Loxley part 14), driven from a **reduced ~12-line pigtail** that passes the pocket vent slots (a full 34-pin ribbon does not fit), with a 3D-printed faceplate/bracket carrying the Gotek and the panel switch. This is the "Modification to support Gotek drives with 3D-printed floppy-disk faceplate" write-up planned in `site/README.md` (§7).

Evidence added 2026-10-10

- **O1 Field Service Manual 2F00040** (`research/osborne1/`): 150 Ω terminator is on drive A only; logic-board connector P8 is male; harness not keyed, red stripe = pin 1; drive A strapped DS1.
- **O1 mainboard schematic, DWG 1A2011-00 Rev E sheet 9 of 9 (DISC CONTROLLER)** (`research/osborne1/OCC1_1A2011-00_Schem_RevE.pdf`, PDF page 12): the P8 34-pin header — GND = `1,2,3,5,9,19,27,29,31,33`; +12 V = 11,13,15,17; +5 V = 21,23,25; **pin 4 = TG43 and pin 6 = EARLY, neither is GND**; pins 7 and 14 are not drawn at all; DS-A/DS-B on 10/12. Read at 600–1200 dpi; see `docs/o1-mainboard-schematic.md` §5 (#62).
- **O1 disk-electronics schematic, DWG 1A3004-00/01 Rev C sheets 1–3** (PDF pages 1–3 of the same file — a *different* drawing, the drive's own board): head/drive connectors P1/P2; P2 lists GND = `1,3,5,7,9,19,27,29,31,33` and CASE = 32. It disagrees with P8 on pin 2 versus pin 7, which is harmless — neither is a driven signal on the O1, so the card ties both to GND.
- **Gotek SFR1M44-U100K user manual**: jumper table `S0 = Drive Select 0`, `S1 = Drive Select 1`, `MO = Motor On`, `JA = READY on 34-pin interface`.
- **FlashFloppy wiki — Initial Setup / Gotek Models**: Shugart hosts expect unit 0 ⟶ jumper `S0`.
- **Omron G6A-434P** datasheet: 4PDT DIP-14 through-hole relay; 5 VDC coil = 72.5 mA / 69 Ω / ~360 mW; contacts 1 A 30 VDC (ST-US).
- **Richard Loxley, Osborne Restoration parts 5 and 12–15**: 150 Ω pack at the far drive; Gotek has 1 kΩ on most lines but not on drive select; switching only +5 V leaves the Gotek parasitically powered, only GND upsets the other drive — switch both.

9. Rev A schematic capture

Captured 2026-10-10 as `hardware/floppy-adapter/floppy-adapter.kicad_sch` (KiCad 7 format), generated by `gen_floppy_adapter_sch.py` — the same generate-then-normalise pattern as the interposer boards.

- **J1** female 2x17 socket (O1 logic-board P8), **J2** 2x17 header (OEM cable), **J3** 2x17 header (Gotek signals), **J4** 1x04 header (switched Gotek power), **J5** 1x02 header (panel switch), **K1** 4PDT relay, **D1** 1N4148 flyback, **R1** 10 kΩ coil pull-down, **R3/R4** DS-A pull-up, **R2+D2** optional indicator.
- Every J1 pin except DS-A (10) is labelled straight through to J2. DS-A is the relay common (K1 pole 3); its NC contact returns to J2.10, its NO contact goes to J3.10. `+5V/GND` reach the Gotek only via K1 poles 1/2 ⟶ J4; the J3 header is signals-only, and +12 V, 4 MHz and pin 14 never reach the Gotek.

- Nets are joined by labels, so the pin/relay mapping can be edited without re-routing. Verified with `kicad-cli sch export netlist` (netlist matches the net table above) and `kicad-cli sch erc` (0 errors).

Two known follow-ups at layout time: the **relay pin numbers are DIP-14 logical placeholders** (confirm against the chosen relay's terminal arrangement), and no **footprint** is assigned yet (KiCad has no G6A-434P footprint; create one or use a 4PDT socket).

10. References

- Issue #62 (this design); feeds #20 (floppy interposer), #30 (G2 multi-format).
- `research/osborne1/floppy-adapter/` — Wayne Visser passive adapter + PROVENANCE.
- `research/osborne1/OCC1_1A2011-00_Schem_RevE.pdf` — mainboard schematic.
- `hardware/floppy-adapter/README.md` — concept summary.
- Richard Loxley, Osborne Restoration **part 5** (150 Ω term pack at the far drive; Gotek has 1 kΩ on most lines but *not* drive select) and **parts 12–15** (A drive has 150 Ω, B has none; DPDT + relay switching lesson).

Serial and the modem

Source: Osborne 1 Technical Manual, ch. 4 "Osborne 1 Interface Design" — §4.2 SERIAL RS232 INTERFACE, §4.3 MODEM — archived in-repo at `research/osborne1/modem/o1techman-interface-extract.pdf` (manual pages 17–19).

This is the authoritative connector/pinout reference for the WiFi-modem work (#21, extended by #56) and for the severable modem module (#56 §E / COMM PAC variant). The memory-map and ACIA register detail live in `docs/o1-memory-io-map.md` §"6850 ACIA detail"; this document adds only the **physical connector** information that doc does not carry.

4.2 RS-232 serial interface (P2)

- Configured as **RS-232C-compatible DTE**, but *"certain of the RS-232C signals are held at +5 volts since they are not needed to control the Osborne 1."* A **6850 ACIA** drives the port.
- Port addresses: **status `2A00H`, data `2A01H`** (shadow mode). Status bits (Fig 4.2): `RDRF`, `TDRE`, `DCD`, `CTS`, `FE`, `OVRN`, `PE`, `IRQ` — see the 6850 datasheet for full semantics.

DB-25S pinout (Fig 4.2.2)

DB-25	RS-232	Definition
1	AA	Frame ground (optional)
2	BA	Transmitted data (low = 1)
3	BB	Received data (low = 1)
4	CA	Request to send (high or no connection enables)
5	CB	Clear to send (**always high on OCC 1**)
6	CC	Data set ready (**always high on OCC 1**)
7	AB	Signal ground
8	CF	Received line signal detected (**always high**)
20	CD	Data terminal ready (high or no connection enables)

Pins 9–19, 21–25: no connection. Only the listed pins are wired.

4.3 MODEM interface (P1) — TTL levels, not RS-232

> "The modem and RS-232 interfaces are basically one and the same. In addition to the serial port, **TTL-level signals may be directly input into the 6850 ACIA using the modem port connection.**"

The MODEM jack is the clean way to attach a TTL-serial device (an ESP32 WiFi modem, our module) **straight onto the ACIA** without an RS-232 level shifter. Read/write it via the CP/M `IOBYTE` function, same as the serial port.

4.3.1 Signal direction

Osborne 1	dir	Modem device
GND	—	GND
TXDATA	→	TXDATA
RXDATA	←	RXDATA
MSB	←	MSB
CTS	←	CTS
MCB	→	MCB
+12	→	POWER IN
RI	←	CD

4.3.2 DE-9P pinout ("standard numbering of the DE-9P connector")

Pin	Signal	Definition
1	GND	Signal ground
2	TXD	Transmitted data — **TTL logic, 1 = high**
3	—	Not used
4	MSB	Modem status bit — open collector, 50 µA sink = inactive
5	CTS	Clear to send
6	RXD	Receive data — **bipolar input, –0.5 V…–10 V = 1**
7	+12 V	Connected to power supply through **22 Ω**
8	MCB	Modem control bit — TTL, **low suppresses output**
9	RI	Ring indicator — TTL, high-to-low sets flag

Design notes for #56

- **Levels are asymmetric.** TXD/MCB/RI are plain TTL (drive/read directly from a 3.3 V MCU with the #12 level spec). **RXD is a bipolar input** (–0.5…–10 V = mark): on sheet 7 the RXDATA line enters receiver **UE3 (LM1458 op-amp)** with zener clamp **CR1 (1N5231B, 5.1 V)** — a comparator-style front-end, not a plain TTL gate. So driving RXD with 0–5 V (or 0–3.3 V) leaves the mark (1) level at/below threshold. The module should swing TX-toward-O1 **negative** for a clean mark (a small charge-pump –5 V, or bias the comparator reference), and this must be **validated on real hardware** — see the open item below.
- **+12 V on pin 7 is current-limited by 22 Ω** (schematic sheet 7: R21 — the same R21 that feeds keyboard P4 pin 19 through jumper J6). Fine to sense, and marginal as a *sustained* source; the **interposer-onboard** module is therefore powered from the interposer's dedicated 5 V rail (research-gaps #5) rather than from P1. The **severable external** module is the reverse case — it exists because the port carries power at all. Its current budget is deliberately deferred until a board exists: see `hardware/wifi-modem-module/README.md` §Power.
- **MSB (pin 4) is open-collector** and **MCB (pin 8) low suppresses output** — together these are the modem's "hook"/enable handshake. RI (pin 9) falling edge sets the ring flag; the ESP32 can drive it to signal an incoming "call" (AT RING) to the O1.
- **CTS is the only hardware flow-control line** brought out (pin 5).

On-board P1 header — schematic-confirmed

Schematic sheet 7 (SERIAL I/O, `OCC1_1A2011-00_Schem_RevE.pdf`) draws the P1 (MODEM) connector on the right edge with the **DE-9P pin numbers carried through unchanged** — there is no renumbering between the case DE-9 and the on-board P1 header:

P1 pin	Net	P1 pin	Net
1	GND	2	TXDATA
4	MSB	5	CTS
7	+12V (via **R21 22Ω**)	6	RXDATA
8	MCB	9	RI
3	(NC)		

Physically P1 is a **2×5 IDC male header** on the board, cabled to the case DE-9P with a straight ribbon (1:1, no conductor swap). A straight IDC crimp interleaves the DE-9 rows, so on the 2×5 the signals land as: row A (1-2-3-4-5) = DE-9 **1,3,5,7,9** = GND, NC, CTS, +12V, RI and row B (6-7-8-9-10) = DE-9 **2,4,6,8,—** = TXDATA, MSB, RXDATA, MCB, NC.

> The schematic confirms the **signal→pin** mapping (the table above). The **physical orientation** of the 2×5 header on the board (which end is pin 1) is a layout detail not in the schematic — confirm against the PCB silkscreen / a real board before committing a PCB footprint. Same class of in-hardware validation as #47.

Also on sheet 7 (for the record): the ACIA baud clock comes from **UC3 (LS161)** dividing a `4USEC/2USEC` tap (jumper **J1** selects the rate), and **MODEM RI** routes to the video-PIA **UC15 CA2** input (sheet 5) — consistent with `docs/o1-memory-io-map.md`.

Baud-rate options the archive documents (#21)

The stock machine gives 300/1200 (J1 in) or 600/2400 (J1 out), and the factory documents 2400/9600 as a modification — the jumper and divider detail is in `docs/o1-memory-io-map.md` §"6850 ACIA detail". Two further things come from *The Portable Companion* #8 (July 1983).

The factory modification, as the Companion cites it

The issue's "**The CRT Connection**" box (PDF p. 87, printed 85) tells a user who wants an 80-column terminal to *"set your computer for 9600 baud using Osbaud or the method shown on page 22 in the Technical Manual."* That page is the manual's **Figure 4.4 "Baud Rate Hardware Modification"** — the LS161/LS163 at C3, pin 2 moved, J2 fitted, J1 *"DON'T CARE"* — quoted in `docs/o1-memory-io-map.md` §"6850 ACIA detail". So the Companion's pre-SCREEN-PAC answer to "how do I get 80 columns?" was the **serial port plus the factory's own baud mod**, not a video board.

The CRT recipe itself

1. Set the machine for **9600 baud** (as above).
2. Connect a standard serial cable between the terminal and the computer. The cable is only **pins 1, 2, 3 and 7, straight through** (1–1, 2–2, 3–3, 7–7).
3. Set the terminal's DIP switches to **1 start bit, 1 stop bit, 8 data bits, no parity, full duplex**.
4. `STAT CON: = CRT:` to make the terminal the console.

5. *"The end result is a 80-column, 9600 baud refreshed, block cursor screen."* The procedure is *"untested with terminals other than Televideo"* — the reviewer's was a **Televideo 920C**.

Advent Products "Osbaud" — a clock substitution, not a bus interposer

Reviewed by **Brad Baldwin** in the same issue (PDF p. 87, printed 85; extract at `research/technical-articles/TPC-Issue08-1983-pp87-osbaud-baud-rate-generator.pdf`). $59.95, **16 DIP-switch-selectable baud rates**, used by the reviewer at **110** (an ASR33 Teletype), **9600** (the Televideo 920C) and **19200** (a $10,000 high-speed band printer), with mainframe up/downloads at high speed. It is *"a small auxiliary circuit board … soldered to the main logic board **in the area behind the serial port**"*, with the DIP switches reached through **a small slot cut in the venting grill** and a self-adhesive rate chart stuck to the keyboard. Advent even supplies the .050" Allen wrench for the brightness and control knobs, and the package covers both case colours (blue/gray and black/brown). Installation wants a Phillips screwdriver, an **Xacto knife to cut across a circuit trace**, and a low-wattage soldering iron — and, like any such product, **voids the warranty**.

The four solder points, decoded

*"Osbaud is soldered to **pins 4 and 12 of the 6850 UART**, **pin 7 of the 74LS32**, and **pin 7 of the 74LS04**."* Read against the schematic (`docs/o1-mainboard-schematic.md` sheet 10, and §"6850 ACIA detail" in the memory map):

Solder point	Is	Why
6850 **pin 4**	**TxCLK** — strapped on the board to pin 3 (RxCLK)	the clock *input*: one wire drives both halves of the ACIA
6850 **pin 12**	**Vcc**	+5 V, taken from the ACIA's own supply pin
74LS32 **pin 7**	GND	ground
74LS04 **pin 7**	GND	ground — and this one is **UB3**, the inverter inside the ACIA's own baud divider

So OSBAUD does not tap the bus, needs no address, and does not use the LS161′s socket: it **replaces the ACIA's clock**, powered and grounded from the ACIA's immediate neighbourhood. That makes the Xacto knife unambiguous — the trace to cut is the one carrying the factory clock (UC3′s `QD`) to the ACIA's clock pins, so that OSBAUD's oscillator is the only driver on that net.

What the design has to be

The review gives the connections and the rate count, not a schematic, so the rest is inference — but tightly constrained:

- The ACIA still divides by 16 or 64 internally (the factory's Figure 4.4 is captioned *"SPEED SELECTED W/IN 6850"*), so OSBAUD's clock is **16× the marked rate** on the ÷16 setting — 800 Hz … 307.2 kHz across 50 … 19200 baud.

- Sixteen rates off a DIP code means sixteen selectable taps, but **a single binary chain from one crystal cannot produce the odd rates** of the standard 16-rate list (110, 134.5, 2000), so the divider has to be **preset/programmable** — the same trick the factory divider next door uses, its LS161 being a preset-load counter (`RCO` ⟶ inverter ⟶ `/LD`).
- The three rates the reviewer names (110, 9600, 19200) all sit inside the classic 50/75/110/134.5/150/300/600/1200/1800/2000/2400/3600/4800/7200/9600/ 19200 set — the same list Nuevo's optional generator advertises ("16 baud rates from 50 to 19.2K", `docs/o1-upgrade-boards.md`).

Open: the crystal and divider topology, whether the ACIA is left in ÷16 mode, and where the **74LS32** sits on the board (the LS04 is UB3 on the serial sheet; a 74LS32 near the serial port has not been located on the schematic yet). The OSBAUD instruction sheet — which the review calls thorough — or a photo of a fitted board would settle all three; worth watching for on the second-hand market.

Framing — the 6850 control byte in the BIOS (#21, #56)

Baud rate is only half of what a modem personality has to present. The other half is the **frame** — data bits, stop bits and parity — and OCC documented how to change it. "**Programming the serial port — modifying the serial port for scientific test devices**", signed by the **Osborne Computer Corporation System Technical Support Group**, in *The Portable Companion* #7 (June 1983, PDF pp. 56–57 = printed 54–55; extract `research/technical-articles/TPC-Issue07-1983-pp56-57-programming-the-serial-port.pdf`) is a worked BIOS patch:

- The factory default is **8-bit data, 1 stop bit, no parity**.
- The change is a **single control byte written into the BIOS image**: build a relocatable CP/M (`MOVCPM 60 *` for a **1.3** BIOS, `MOVCPM 59 *` for **1.4**), `SAVE 39 CPM60.COM`, `DDT CPM60.COM`, then **`S1FE9`** and put the code in at **`1FE9H`/`1FEAH`**, and `SYSGEN` it to the target disk. So the ACIA's **control-register init byte lives at BIOS offset `1FE9H`** — a second, independent handle on the ACIA beyond the `2A00H`/`2A01H` port pair, and the address to watch for #21/#56.
- **Figure 1** is a "**Table of 6850 control codes**" covering all 16 combinations at 1200 and 300 baud — e.g. **8-N-1 @1200 = `55H`**, **8-N-1 @300 = `56H`**, 7-E-1 @1200 = `49H`, 8-N-2 @1200 = `51H`; arrows mark the factory default. Read as the 6850's control register (word-select bits 2–4, clock-divide bits 0–1) the values are exactly right, which is a useful cross-check on the `docs/o1-memory-io-map.md` §"6850 ACIA detail" claim that the ACIA divides by 16 or 64 — the `55H`/`56H` default sits in the **÷16** field.
- **Figure 2** is a **which-BIOS-do-I-have** test: `DDT`, `DA900`, and look at the third byte after the `490` — `C3 xx E5` is a **1.3** BIOS, `C3 xx E1` a **1.4**.

The article does not touch the baud divider, so it is complementary to the Hogan/Figure 4.4 rate modifications above: **rate from the divider, frame from the BIOS control byte**. For the module (#56) both matter — a WiFi modem that only ever presents 300/1200 8-N-1 will fail on a machine whose owner has patched the frame for a lab instrument, which is exactly the case the article exists for.

The IEEE-488 port

Source: *The Portable Companion* #8, July 1983 — "**Understanding the IEEE-488 port** — Scientific interfacing with the Osborne 1" by **Wayne Kogachi** and **Shahriar Kianersi**, PDF pp. 38–41 (printed 36–39), and "**Interfacing External Devices** — add a game joystick to the Osborne 1" by **Kelly Smith**, PDF pp. 44–49 (printed 42–47). Archived in-repo at `research/technical-articles/TPC-Issue08-1983-pp38-41-understanding-the-ieee-488-port.pdf` and `research/technical-articles/TPC-Issue08-1983-pp44-49-interfacing-external-devices-joystick.pdf`.

Both authors of the first article are **OCC staff** — Kogachi a software engineer, Kianersi a system support specialist — so the article is effectively the vendor's own account of the port, and it points at the *User's Guide* chapter "IEEE-488 implementation" for the same material.

The port's address map and the ROM's view of it live in `docs/01-memory-io-map.md` §2 (`2900`–`2903`, **MC6821 at UC7**) and `docs/01-rom-source.md`; the Drive C: RAM disk that hangs off it is in `docs/drive-c-protocol.md`. **This doc records what the two articles add**: the port's device-level behaviour, the eight ROM drivers as the articles name them, and the one fully-worked application — a game joystick wired straight onto the data lines.

Bearing issues: #22 (Drive C / IEEE-488 storage), #84 (standalone Drive C on P3), #19 (virtual MX-80 / Centronics capture), #12 (interposer spec).

1. What the port is

- **TTL logic level**, driven by a **Motorola 6821 PIA**; "after configuring the pin connection for the IEEE standard, one can simply connect the Osborne to any IEEE compatible device."
- The machine can act as any of the three IEEE-488 device definitions: **Talker, Listener and Controller**. Bus management is: announce the **Talker first, then the Listener**, after which the Controller releases the bus and lets those two devices talk. The Controller may announce itself as either Talker or Listener, and its periodic takeovers "should be done *synchronously* rather than *asynchronously*."
- **Up to 15 devices** on the bus: *"The IEEE-488 port can call up to 15 different devices at one time on the data bus. (From a software point of view, the figure is 29 different devices; however, due to practical hardware limitations only 15 devices are allowed.)"*
- **The Osborne's own address is zero**, *"so no other device with that address port can sit on the data bus"* — a hard constraint on any bus we build.
- The second article's framing is the useful one for this project: *"The IEEE-488 Bus implementation is really just a **parallel data port controller** that can be used for a variety of interfacing possibilities that require byte-wide (8 bits at a time) data. This controller is known as a 6821 PIA … and may be programmed in a variety of different input/output control modes."*

2. The eight ROM drivers

> "Osborne's IEEE-488 port has **eight low-level 'drivers,' which reside in the ROM** and take care of all the IEEE-488 interfacing requirements. These routines are documented in the *User's Guide* in the IEEE-488 implementation chapter. One may invoke any of the routines

simply by calling them and passing the proper parameters through preassigned registers. One may also receive messages through assigned registers."

#	Driver	Parameter registers	Returns
1	**Control Out**	`C` = code to clear the interface (IFC) and/or set/reset REN	—
2	**Status In**	—	`A` = value acknowledging that a device is requesting service
3	**Go To Stand By**	—	lets other devices on the bus take control
4	**Take Control**	`C` = code; synchronous or asynchronous takeover	`A` = error messages, if any
5	**Output Interface Messages**	`C` = message byte (talker or listener address)	`A` = error messages
6	**Output Device Messages**	`C` = data byte; `B` = end-of-message acknowledgement code	`A` = error messages
7	**Input Device Messages**	—	`A` = data byte; `L` = error message
8	**Input Parallel Poll**	—	`A` = the parallel-poll bit pattern

Driver 8 carries a hardware consequence worth recording: *"The devices that are capable of 'parallel poll' will respond by pulling low a preassigned data line so that some kind of code is returned which will correspond to a particular device.* ***If this routine is used, only eight devices can sit on the data bus*** *(one data bit line for each device)"* — parallel poll trades device count for polling speed, 15 ⟶ 8.

The same eight, as a jump table in the ROM

`docs/o1-rom-variants.md` records that OZROM 1E removed the IEEE-488 support and that the machine-readable evidence is a jump table: *"013F-0156 WERE IEEE-488 ROUTINES".* **`013F–0156` is 24 bytes — eight 3-byte `JP` instructions, one per driver above**, which independently confirms both the count and the order: the *n*-th entry sits at `013F + 3n`, so the **eighth** (Input Parallel Poll) is at **`0154`** — and the joystick article calls exactly that address, as `0E154H`, the same routine seen in the ROM's high mapping (see §6). The articles name the drivers; the ROM fixes their addresses and their order.

Checked in the bytes, in every stock image the project holds — all of them carry eight `C3` (`JP`) triples from `013F`, the eighth at `0154`:

ROM	the first seven entries	8th entry at `0154`
factory 1.40 (`tools/emulator-setup/roms/rev1.40.ud11`)	`C3 36 09`, `C3 79 09`, `C3 8C 09`, `C3 98 09`, `C3 D8 09`, `C3 FB 09`, `C3 3D 0A`	`C3 99 0A`
factory 1.44 (`research/roms/os1-144.rom`)	`C3 49 09`, `C3 8C 09`, `C3 9F 09`, `C3 AB 09`, `C3 EB 09`, `C3 0E 0A`, `C3 50 0A`	`C3 AC 0A`
Nuevo 1.51 (`tools/emulator-setup/roms/monrom-rev1.51-12.ud11`)	`C3 3D 09`, `C3 80 09`, `C3 93 09`, `C3 9F 09`, `C3 DF 09`, `C3 02 0A`, `C3 44 0A`	`C3 A0 0A`

The parallel-poll routine moves between ROM revisions — `0AAC` in 1.44, `0999` in 1.40, `0AA0` in Nuevo's 1.51 — which is exactly why the joystick article's source must be assembled at a different origin per monitor version and then hand-patched. Same table, different targets: a usable ROM-variant signature, and the thing OZROM 1E destroys.

3. Errors the drivers report

- **Time Out** — non-fatal: *"a 100 microsecond grace period of time for the device to respond. Alternately, you can call the device repetitively to allow for slow devices to respond."*
- **No Device Present** — *"either the port is not hooked up to any device, or all the devices sitting on the data bus are turned off."*
- **Not taking the data after agreeing to take the data.**

4. A printer or plotter needs none of this

> "if only a printer/plotter is being used on the port there is no need to go through these routines. Using the SETUP program, make sure that the right protocol is chosen (Printers protocols, IEEE-488 compatible) and that the IEEE address port matches the device number. The latter can be simply accomplished with **BIOS 1.4**. With **BIOS 1.3 and earlier** versions there is a patch that is provided in the Technical Tips section of this issue. Double density computers (1.4 BIOS) have a **menu-driven device address patcher**."

That is the emulation surface for #19 (virtual MX-80): the printer path is a SETUP-selected protocol plus a device address, not a driver call sequence.

Two operational details from the same article, both worth keeping:

- **The control codes must be in capital letters.** *"Some manufacturers enforce that; otherwise an error message will be returned."*
- The MBASIC example works by poking a small assembly-language interface plus a register-save area high in RAM and calling it from BASIC (`CALL IEEE0(…)` / `IEEE1(…)`), passing the

driver number and its parameters through the shared locations. That is why the BIOS-1.3-or-earlier caveat exists — the poked routine and the ROM drivers have to coexist with CP/M in high memory.

5. The worked application: a game joystick on the data lines

The joystick article is the project's only fully-worked external device on this port, and it is small enough to reimplement: **a switch-closure box read as five data bits.**

The cable

- One **DB-9 male** (it mates an Atari joystick), one **26-pin (13-pin dual row) card-edge / flat-ribbon connector** that mates with the IEEE-488 connector on the machine, and about 7 inches of **12-conductor ribbon** (14-conductor is standard — strip two conductors off).
- **Only `D0`–`D4` and GND are used.** The cable should leave the IEEE-488 connector going *up* and then loop down under the case for strain relief.
- The physical fact to keep: the machine's "IEEE-488" port is a **26-pin card edge connector**, and the article's Figure 1 is its pin assignment as seen from the front of the computer.

The bit mapping (Figures 5 and 6)

Data bits `D7…D0`	Decimal	Direction	Screen offset
`00000000`	0	center	0
`00000001`	1	north	−128
`00000010`	2	south	+128
`00000100`	4	east	+1
`00001000`	8	west	−1
`00000101`	5	north-east	−127
`00000110`	6	south-east	+129
`00001001`	9	north-west	−129
`00001010`	10	south-west	+127
`00010000`	16	**fire button**	(the program clears the screen)

The offsets are screen-memory deltas for the O1′s **128-character row** — a line is ±128, a character position ±1, so the diagonals are ±127/±129. Diagonals are *"the sum of the binary values (actually two simultaneous switch closures) — combinations such as 3 and 7 are not mechanically possible within the joystick."* Bits 5–7 are spare: *"If you have a joystick with three more switches, they could just as well be assigned to bit positions 5, 6, and 7 and 'sampled-in-software' accordingly."* The fire button has a quirk — *"As programmed, you MUST return to center position on the joystick to detect the closure of the 'Fire Button' ONLY"* — and the article warns that any switch wired in *"must have full ISOLATION from any source of external electrical power"* (its worked example of what not to do is a doorbell button: *"Zorch-City"*).

Reading it two ways

(a) High level. `CALL 0E154H` — the ROM's **Input Parallel Poll** driver (#8 above) — returns the switch-closure bit pattern in `A`. That is the whole interface for the article's "Etch-a-Sketch" checkout program, which walks a pointer around the 128×32 screen by adding the offsets above.

(b) Low level. `JOYSTIK.ASM` programs the 6821 directly:

```
porta0  equ 2900h  ; 6821 pia port-a data/direction register
porta1  equ 2901h  ; 6821 pia port-a control register
portb2  equ 2902h  ; 6821 pia port-b data/direction register
portb3  equ 2903h  ; 6821 pia port-b control register
flagpos equ 0ef08h ; ram/rom flag position address: 0=rom, 1=ram
```

`istick` floats both ports and then sets port A for input; `rstick` reads the joystick as `di` / `out 0` (ROM on, RAM off) / `xra a` / `sta flagpos` (tell the system the ROM is in) / `lda porta0` / `mov b,a` / `out 1` (RAM on, ROM off) / `mvi a,1` / `sta flagpos` / `ei` / `mov a,b` / `ani 01fh` (mask to `D0–D4`) / `sta sdata` / `ret`.

Two machine-level facts fall out of that listing:

- **`EF08H` is the ROM/RAM bank flag** the system reads: `0` = ROM, `1` = RAM. The routine has to keep it consistent with its own `OUT 0`/`OUT 1` bank switch, because it reads the port *while the ROM is banked in.*
- The mask is **`01FH`** — the port is read as a full byte and only `D0–D4` carry the switches.

Version dependence

The source is assembled at a **different origin per monitor ROM version** — `ver2` ⟶ `0EE80H`, `ver3` ⟶ `0ED80H`, `ver4` ⟶ `0EA80H` — because it calls into ROM addresses that move between revisions. The `.HEX` then has to be patched by hand for the machine's version (*"Change to 61"*, *"Change to CBE6"*, …), and the article says to press RESET and read the sign-on message if in doubt. For BIOS 1.4 the labels are `sdata EA80`, `rstick EA81`, `istick EA8B`, `end EAC8`, and the `.HEX` records load at `EA80`. This is the same version-dependence the first article flags from the other side (move CP/M down 1K on BIOS 1.3 or earlier).

The article is **part one of two**; the SPACEWAR game that uses the interface is promised for the next issue. Kelly Smith is with *Wizard of OsZ* and wrote "Monster" in the Companion's inaugural issue; he offered the SPACEWAR program (200 lines) and joystick hardware for a nominal fee (P.O. Box 964, Chatsworth, CA 91311).

6. The P3 edge connector — the pinout, and what the port actually implements

`docs/drive-c-design.md` §7 listed "the pin-level map of P3" as its first unmeasured item. It was never missing: it is in the Technical Manual extract the project already archives, §4.1.2 "IEEE-488 Pinouts", and the manual explains why the connector is not the standard one:

> Because this port is used for more than just IEEE 488 signals, we've declined to use a standard IEEE connector.

Source: `research/osborne1/modem/o1techman-interface-extract.pdf`, PDF p. 2 = printed p. 14 (Figure 4.1.2), and Figure 4.1.1 on PDF p. 1 = printed p. 13 for the directions. **The scan has no text layer** — it is EPSON Scan output, so `pdftotext` returns nothing and the tables below were read off a 250 dpi render.

The connector is a **26-pin card edge**, read looking at the front of the machine, odd pins on one side and even on the other; **pins 25 and 26 are not used** ("provided for compatability"). The IEEE column is the standard 24-pin IEEE-488 assignment, kept here because it is what a GPIB cable or a GPIB device document will name.

IEEE-488	**P3**	Signal	PIA line (ROM)	O1 direction
1	**1**	DIO1	PA0	bidir
13	**2**	DIO5	PA4	bidir
2	**3**	DIO2	PA1	bidir
14	**4**	DIO6	PA5	bidir
3	**5**	DIO3	PA2	bidir
15	**6**	DIO7	PA6	bidir
4	**7**	DIO4	PA3	bidir
16	**8**	DIO8	PA7	bidir
5	**9**	EOI	PB3	O1 drives
17	**10**	REN	CB2	O1 drives
6	**11**	DAV	PB5	O1 drives
12	**12**	signal ground (DAV)	—	ground
7	**13**	NRFD	PB7	O1 drives
18	**14**	signal ground (NRFD)	—	ground
8	**15**	NDAC	PB6	**box may drive**
19	**16**	signal ground (NDAC)	—	ground
9	**17**	IFC	**CA2**	O1 drives
20	**18**	signal ground (IFC)	—	ground
10	**19**	SRQ	CA1	**input**
21	**20**	signal ground (SRQ)	—	ground
11	**21**	ATN	PB4	O1 drives
22	**22**	signal ground (ATN)	—	ground
12	**23**	cable shield + ground	—	ground
24	**24**	signal ground (logic)	—	ground
—	**25**	*unused*	—	—
—	**26**	*unused*	—	—

The PIA column is the ROM's own comment block (`rom144.asm` 2606–2715, quoted in `docs/drive-c-protocol.md` §2); the two sources agree line for line, which is worth saying because they were derived independently — one is a 1983 manual, the other a 1983 ROM listing.

Note what the PIA column does **not** contain: there is no DAV, NRFD, NDAC, EOI or IFC *input*. **The O1 cannot read any of the five handshake/control lines it drives.** Only SRQ (CA1) and NDAC (PB6, and only after CRB bit 2 is cleared) can be read back at all.

6.1 The port is not GPIB, and Figure 4.1.1 shows it

Figure 4.1.1 draws each line solid or dashed, and the split is the whole story:

Drawn	Lines	Meaning
solid	REN (out), ATN (in), SRQ (in), DATA0–7 (bidir)	implemented by the O1
dashed	IFC, NRFD, DAV, EOI, NDAC	*not implemented* by the ROM's driver set

The manual says as much in words — *"a subset of the complete IEEE specification"*, *"no provision has been made for controlling multiple devices"* — but the figure is the useful version, because it names the subset. **The three IEEE-488 handshake lines (DAV, NRFD, NDAC) are the ones missing**, which is exactly why nothing on this port can do an IEEE-488 three-wire handshake, and why every device on it invents its own clock. Drive C: uses CA2/IFC; the printer uses DAV as a strobe (§6.2).

ATN and SRQ are drawn as *inputs*, which is odd for a controller and correct for this machine: the O1 can only ever be told, not ask.

6.2 The same pins, as a Centronics printer port

Figure 4.1.5 (PDF p. 4 = printed p. 16) re-labels the same edge connector for the printer personality that SETUP selects (IOBYTE = BAT:, UR1:, UP1:, LPT:). It is the same wires, with four of them re-purposed:

P3	O1 name	Centronics pin	Centronics name
1	DIO1	2	data 0
2	DIO5	6	data 4
3	DIO2	3	data 1
4	DIO6	7	data 5
5	DIO3	4	data 2
6	DIO7	8	data 6
7	DIO4	5	data 3
8	DIO8	9	data 7
11	DAV	1	**out strobe**
12	ground	19	ground
15	NDAC	11	**busy**
16	ground	29	ground
19	SRQ	13	**select**

So `DAV` is the printer strobe, `NDAC` is *busy* and `SRQ` is *select* — and `NDAC` is the same line the Drive C transport leaves as an input as its optional "ready" (`docs/drive-c-protocol.md` §2, §7). **One pin, two meanings, chosen by which personality is installed.** Anything we put on P3 has to be harmless in both modes, which is a constraint `docs/drive-c-design.md` §2.3 now carries.

7. What this means for o1nextgen

- The port is a **byte-wide parallel port on a 6821**, so a personality can implement it as such (`docs/virtual-peripherals-block-diagram.md`), and the **eight ROM drivers are the behaviour that has to be right**.
- **Address 0 is the machine's own** — anything else we place on the bus needs a different address.
- The port already has three quite different users in the archive: the ROM's driver set (this doc), a printer selected in SETUP (§4), and a dumb switch-closure box read through driver 8 (§5). Drive C: is a fourth, and the manual is explicit that it does **not** speak real GPIB on this connector (`docs/drive-c-protocol.md`).
- The driver table at **`013F–0156`** (eight `JP`s) is a *checkable* signature: present in stock ROMs, destroyed in OZROM 1E (`docs/o1-rom-variants.md`).
- **The pinout is solved** (§6) — it was in the archive all along, in the Technical Manual extract, and the ROM's comment block corroborates it. So `hardware/drive-c/` can be laid out against a documented connector rather than a measured one, and the one remaining electrical unknown on P3 is loading, not mapping.

Keyboard and printer

(not yet written)

Peripherals and storage

Drive C: the IEEE-488 RAM disk

Status: solved. The host side is read out of `DCL.COM`'s disassembly and **confirmed against a trace of the real Drive C software running on the emulated machine** (§9) — and a model of the box, built from this spec, makes the 1983 loader complete (`QUICKPAC Drive C: loading has been completed`) and CP/M start copying to drive `C:` with PIP. What is *not* known is the box's internals (§10); a replacement does not need them, because the contract below is what the host exercises.

Sources: `research/drive_c/DRIVE_C.IMD` (the software), `research/drive_c/extracted/` (`dcl.com`, `dfd.spr`), `research/source-code/BIOS-144-source/rom144.asm` (the ROM's own account of the port), and `research/drive_c/Drive_C_Users_Manual.ocr.txt`. Serves #6 (the protocol) and #84 (the reproduction); `docs/drive-c-design.md` is the hardware + firmware spec built on it.

1. What the device is

A battery-backed RAM disk that hangs off the Osborne 1's **IEEE-488 edge connector** (P3) and duplicates that connector plus the Ext. Video connector on its own front panel, so the printer and a monitor can be plugged through it. It sold as 192 K and 384 K, in two parts: the unit (the RAM) and a software package (`DCL.COM` + `DFD.SPR`) that installs it as a CP/M drive and adds a print buffer. `DCL.COM` is the installer and carries the transport; `DFD.SPR` is a **substitute console command processor** that CP/M reloads on every warm boot in place of `CCP.COM` (`docs/drive-c-executive.md` §4).

The manual is explicit that this is not GPIB:

> **IEEE-488 COMMUNICATIONS PROTOCOL IS NOT SUPPORTED** ON THE Drive C PRINTER/HARD DISK/8088 EDGE CONNECTOR

The connector is used as a **parallel port** — the same trick the ROM itself uses for Centronics printers (§2). That is the single most important fact about the device: it is not a GPIB device and does not decode GPIB handshake semantics.

2. The electrical layer: the PIA, and what each line is

The host drives the box by bit-banging the motherboard's **MC6821 PIA at `2900–2903`** (bank 2). The ROM's own source documents what each pin is (`rom144.asm` §"IEEE-488 INTERFACE", lines 2606–2715), and it is the authority here — these are the vendor's comments, not an inference:

PIA	Line	GPIB name	Centronics use (`rom144.asm` 3250–3263)
PA0–PA7 (`2900`)	DIO1–DIO8	data	**data bus**
CA1 (`2901` bit 7)	SRQ	service request in	"low→high gates input data to port A"
CA2 (`2901` bits 5:3)	IFC	interface clear	**"going low indicates to device that we are busy"**
CB1	—	not used	—
CB2 (`2903` bits 5:3)	REN	remote enable	—
PB0 (`2902` bit 0)	enable data out		**`0` = data bus is output, `1` = input**
PB1	enable NDAC/NRFD		set to 1
PB2	enable EOI/DAV		set to 0
PB3	EOI		`0` output, `1` input
PB4	ATN		not used
PB5	DAV		**output strobe, active = 1**
PB6	NDAC		**`0` = printer busy, `1` = printer ready**
PB7	NRFD		not used

Two facts fall out, and both are load-bearing:

- **PB0 is the direction control for the data lines.** It is how the host says "I am writing" (`PB0=0`) or "you drive the bus" (`PB0=1`), and it is also the bit that distinguishes the two operations (§4).
- **All signals are active-low** — *"ALL SIGNALS ARE LOW ON THE IEEE BUS WHEN PIA REGISTER CONTAINS 1"* — and the O1 drives them through 7406 open-collector gates. A replacement must **release** lines, never drive them high.

Drive C's own use of the port (measured, §3) is a subset: port A for data, PB0 for direction, PB3/PB7 as framing, and **CA2 as the byte clock**.

3. The host's frame — measured

`DCL.COM` installs a relocatable block at `0x0600–0x07BE` (the file holds it at `0x1600–0x17BE`; the internal `JP 0757h`/`JP 0763h` are its *installed* addresses, which is what gives the relocation away). The transport routine is `0x06D5–0x07B4`; a helper at `0x07B5` writes port A and pulses CA2.

Every operation is one frame, and every frame re-initialises and tears down the PIA. A trace of the real software (10 195 PIA accesses, §9) shows this sequence, byte for byte:

```
DI                                  ; interrupts off for the frame
OUT (0),A                           ; bank 2 (the O1's own I/O bank)
CRA=32h ; PA<-FFh ; CRA=36h         ; port A = output
CRB=32h ; PB<-FFh ; CRB=36h         ; port B = output, CB2 (REN) high

PB<-08h                             ; frame open
PA<-h1   ; CA2 pulse                ; header byte 1
PB<-00h                             ; phase separator
PA<-h2   ; CA2 pulse                ; header byte 2
   [read frames only] CRA=32h ; PA<-00h ; CRA=36h   ; port A -> INPUT
PB<-88h  (write) | 81h (read)       ; the operation
   128 x {  PA<->byte ; CA2 pulse  }  ; the data phase

PB<-09h                             ; frame close
CRA=32h ; PA<-FFh ; CRA=36h         ; port A back to output
CRB=00h ; PB<-BFh ; CRB=04h ; PB<-02h ; port B: PB6 becomes an INPUT
OUT (1),A ; EI                      ; bank 1, interrupts on
```

Two details in the teardown matter: `PB<-BFh` is written with `CRB` bit 2 clear, so it lands in the **direction** register — **PB6 becomes an input** — and the final `PB<-02h` leaves the port in a resting state. So after every frame the box is *able* to drive PB6 (NDAC / "ready"), which is the one line the host could read for flow control.

4. Port B: the four states

Port B is written as a **state word**, not as individual handshake lines. Only four values are ever written during a frame, and each has a job:

Value	Bits set	Meaning
`08h`	PB3	**frame open** — EOI asserted low, data drivers disabled
`00h`	—	**phase separator** — between the two header bytes
`88h`	PB7, PB3	**write** — NRFD asserted, EOI low, and **PB0=0 so the host drives DIO**
`81h`	PB7, PB0	**read** — NRFD asserted, and **PB0=1 so the host releases DIO**
`09h`	PB3, PB0	**frame close** — EOI low, data drivers released

So 88h and 81h are the two operations, and the difference that matters is **PB0**: 88h = write to the box, 81h = read from the box. That is confirmed three ways — the op byte, the DDRA the host sets before the data phase, and the direction of the LDI in the transfer loop (§6).

5. Port A: the two header bytes, and the record address

Before the operation byte the host sends **two bytes on port A**, each clocked by a CA2 pulse. They are not a command and a length; they encode **the record address**, and the code says exactly how. The caller (DCL.COM 0x1122–0x115D, installed 0x0122) is entered with:

- HL = the caller's 128-byte buffer,
- DE = the **16-bit record address**,
- A = 0FFh to read, 0 to write;

and it computes

```
h1   = (DE >> 1) & 0FFh          ; sent first
n    = (DE >> 9) & 0FFh          ; high bits of the record address
h2   = f(n) | ((DE & 1) << 7)    ; sent second
```

where f(n) is the bank/size nibble the box wants. **f(n) is computed, not looked up**, and the code is short enough to quote — the installed routine at 0x0702:

```
    ld a,(051Ah)      ; 051Ah holds (DE&1)+1
    rrca
    and 80h
    ld b,a            ; b = (DE&1) << 7
    ld a,(0519h)      ; 0519h holds n
    cp 3
    jr c,+8
    sub 3
    rlca
    rlca
    or 3
    jr +2
    or 0ch
    or b              ; h2 = f(n) | ((DE&1)<<7)
```

so

n	0	1	2	3	4	5	6
f(n)	0Ch	0Ch	0Ch	03h	07h	0Bh	0Fh

h1 and n are not computed in the frame routine at all. The caller (0x0124–0x015F) shifts DE right by one and stores the **whole 16-bit result at 0x0518** — ld (0518h),hl — so 0x0518 receives (DE>>1)&0FFh = h1 while 0x0519 receives (DE>>1)>>8 = n. That is the cleanest confirmation that h1 and n are the low and high parts of *one* address, not two independent fields.

So (h1, h2) encodes DE>>1: h1 is its low eight bits, and its high bits ride in the low nibble of h2 through f. DE is the CP/M record address in 128-byte units, and the DPB says **SPT = 2** (§8) — so

`DE>>1` is the **track** and `DE&1` is the sector within it. The box's unit is a **256-byte track**, and bit 7 of `h2` is the sector.

> **Correction, 2026-10-11.** An earlier revision of this doc gave `f(n)` as `Ch, Dh, Eh, 3h, 7h, Bh, Fh` and called the encoding *bijective*. That table was inferred, not measured, and it is wrong for `n = 1, 2`: the installed code yields `0Ch` for **every** `n < 3`. The trace agrees — across 163 frames of a boot-then-`PIP` run, `h2` took only `0Ch` and `8Ch`, never `Dh` or `Eh`. The consequence is not cosmetic: with `f` constant over `n = 0…2`, `(h1, h2)` no longer distinguishes those three banks. §10 records that as the open question it is.

Measured examples from the trace: the boot-time format writes `h1 = 00h…0Fh` with `h2 = 8Ch` (⟶ `DE = 1,3,5,…,31`) and `h1 = 00h…0Fh` with `h2 = 0Ch` (⟶ `DE = 0,2,4,…,30`); and the first operation is a read at `h1 = E8h`, `h2 = 83h` ⟶ `DE = 0x7D1`.

6. The byte clock, and the data phase

CA2 is the byte clock. The host writes `CRA=3Eh` then `CRA=36h` — CA2 low then high — once per byte, and the helper at `0x07B5` wraps "write port A, then pulse CA2". Nothing else is handshaken: **the transport never reads PB6 and never polls for ready**, so the box has to keep up. The loop is ~52 T-states per byte (about 13 µs at 4 MHz), and a read must be answered within ~6 T-states of the CA2 edge.

The order differs between the two directions, and that is the box's cue:

- **write:** the byte is placed on port A **first**, then CA2 pulses — the pulse says *latch it*.
- **read:** CA2 pulses **first**, then the host reads port A — the pulse says *present the next byte*. (A box that keeps the next byte on DIO and advances on each pulse satisfies this; one that only drives the bus after seeing the pulse has ~1.5 µs to do it.)

128 bytes are transferred per frame in both directions.

7. The device contract — what a replacement must do

This is the normative summary. A box that satisfies it will run `DCL.COM`, `DFD.SPR` and the rest of the software unchanged.

1. **Observe the PIA lines.** Port A (DIO1–8) is the data byte. Port B bits 0, 3, 5 and 7 carry direction and framing. CA2 is the byte clock. CB2/REN is held high and can be ignored.
2. **Decode the frame** from the port-B state words: `08h` opens, `00h` separates the header, `88h`/`81h` is the operation, `09h` closes.
3. **Capture the two header bytes** on their CA2 pulses (after `08h`, then after `00h`) and decode them to a 16-bit record address (§5).
4. **For `88h`,** latch 128 bytes from port A, one per CA2 pulse, and store them at that record.
5. **For `81h`,** present 128 bytes on port A, one per CA2 pulse.
6. **Release the data lines** whenever `PB0 = 1`; drive them only while `PB0 = 0`. Everything is open-collector: never drive a line high.

7. **Keep up with the clock** — there is no ready handshake in the transport. PB6 may be driven (high = ready) and is left as an input at the end of every frame, but nothing in the transport reads it.
8. **Survive the frame boundary.** Each frame is self-contained: init, header, data, teardown. A replacement may be stateless across frames.

8. Capacity — the DPB, the ceiling, and how `C:` appears

The box itself is a block store; the *filesystem* is CP/M's, and the geometry CP/M uses for `C:` comes from the **DPB that `DCL.COM` installs into the CBIOS**. That DPB is now in hand, and it is not a guess.

8.1 The DPB

`DCL.COM` is a .COM loaded at `0x100`, and its payload is relocated on install by `-0x1000` (the internal `JP 0757h` is the loop's own body at file `0x1757h`). At **file `0x142F` = installed `0x052F`** sits a bare 15-byte CP/M 2.2 DPB, and the payload's own pointer table names it (installed `0x0529` holds `2Fh 05h`):

```
file 0x142F:  02 00 04 0F 01 BF 00 7F 00 C0 00 00 00 00 00
```

Field	Bytes	Value	Meaning
`SPT`	`02 00`	2	128-byte records per track
`BSH`	`04`	4	block shift ⟶ 2 KiB blocks
`BLM`	`0F`	15	block mask (2 KiB − 1)
`EXM`	`01`	1	extent mask for `BLS = 2048`
`DSM`	`BF 00`	191	192 blocks × 2 KiB = **384 KiB**
`DRM`	`7F 00`	127	**128** directory entries
`AL0`	`C0`	`1100 0000b`	2 blocks reserved for the directory
`AL1`	`00`	0	—
`CKS`	`00 00`	0	no directory checking (fixed media)
`OFF`	`00 00`	0	no reserved tracks

Every field matches the vendor's own 384K column in the manual (§8.2), which is what makes this a reading rather than a guess. The `AL0` check is the nice one: `C0h` = the top two bits of the allocation vector = 2 blocks × 2 KiB = **4096 bytes** of directory, exactly the "Directory Space: 4096 bytes" the manual prints for the 384K unit.

8.2 The manual's own table — and the 192K unit

Drive C User Manual, §7-3 "Drive C SOFTWARE SPECIFICATIONS", printed p. 134 = PDF p. 142. The OCR in the archive mangles the column alignment, so this is read off a 300 dpi render. The three rows in the middle are **centred across the two RAM-disk columns**, i.e. shared by the 192K *and* the 384K unit, and the TurboPac column has its own values:

CP/M parameter	192K unit	384K unit	TurboPac
Useful storage capacity	187 KB	376 KB	10, 20 or 33 MB
Block size	1K	2K	8K (can be modified)
Directory entries	64	128	512 per logical drive
Directory space	2048 bytes	4096 bytes	16384 bytes per drive
Logical sectors	1536	3072	varies with block size & capacity
Reserved tracks	**0** (both)		612, landing area
Logical sectors/track	**2** (both)		32
Bytes per sector	**128** (both)		256
TPA RAM used	1152 B	1152 B	3172 B

Two independent confirmations fall out: **`SPT = 2`** matches the DPB exactly, and `1536 × 128 = 192 KiB` / `3072 × 128 = 384 KiB` match the two capacities.

The 192K unit's DPB is derived, not found — the archive holds the 384K unit's disk, and no literal `SPT=2, BSH=3, BLM=7, EXM=0, DSM=191, DRM=63, AL0=C0h` exists in any of the extracted files. It follows from the manual's column and the same arithmetic (192 × 1 KiB = 192 KiB; a 2048-byte directory is 2 × 1 KiB blocks):

Field	192K (derived)	384K (found)
`SPT`	2	2
`BSH/BLM/EXM`	3 / 7 / 0	4 / 15 / 1
`DSM`	**191**	**191**
`DRM`	63	127
`AL0`	`C0h`	`C0h`

Note that **`DSM` is 191 in both** — the two units hold the same *number* of blocks, just bigger ones. That is the design signature of the whole product.

8.3 The ceiling

Three separate limits apply, and the smallest one wins:

Limit	Where it comes from	Value
The protocol	`f(n)` is four bits: `((n-3)<<2)|3` has to stay inside the nibble, so `n ≤ 6`	**1792 tracks × 256 B = 448 KiB**
The DPB	`DSM` is 16 bits, so up to 65535 blocks	65535 × 16 KiB ≈ 1 GiB
CP/M 2.2	drive letters `A:`–`P:`	16 drives

The protocol bound is the interesting one, because it is the *host's own code* that imposes it: at `n = 7` the shift pushes bit 4 into the byte, the box's four-bit latch sees `03h`, and the frame collides with `n = 3`. So the shipped driver can address **at most 448 KiB**, and the largest unit the vendor sold (384 KiB) uses tracks `0…1535`, i.e. `n = 0…5` — the driver's `n = 6` branch is already headroom.

A replacement may therefore present up to 448 KiB and still be drivable by the unmodified 1983 software. Anything larger needs a new driver, which is outside what #84 is for.

And the manuals' own capacity figures reconcile with the DPB. The Osborne 1 book quotes 187 KB / 376 KB "useful"; the Executive book, for the same hardware, quotes 190 KB / 380 KB. Both are right, and the difference on the 384K unit is exactly `DFD.SPR`:

	192K unit	384K unit
Raw capacity	192 KB	384 KB
less the directory	−2 KB	−4 KB
= the Executive's "useful"	**190 KB**	**380 KB**
less `DFD.SPR` (Osborne 1 only)	−3 KB	−4 KB
= the Osborne 1's "useful"	**187 KB**	**376 KB**

380 KB is 190 blocks of 2K, which is `DSM = 191` (192 blocks) **minus the two directory blocks** — so the vendor's own figure *is* our DPB, reached independently. `docs/drive-c-executive.md` §3.

8.4 Drive letters, and what TurboPac actually is

- **`C:` is one drive and its letter is renamable.** `DCN.COM` ("DCN version 2.07", shipped on the disk) backs the `DCN x` command, and the manual warns that `COPY` requires the unit to be named `C:` while it runs. The DCL-era command is `DC1 A` (install and name it `A`); the older one was `DRIVEC A: C:`, which named the unit *and* the left floppy in one go. So the unit can take any of the 16 CP/M drive letters — **the limit is CP/M's, not the device's.**
- **TurboPac is not a bigger RAM disk — it is the *hard-disk controller*.** The manual is explicit, and it corrects a reasonable assumption:

> TurboPac software turns your Drive C into a hi-speed **Cache Buffer** for the Trantor hard disk. […] The 192K TurboPac system uses all of Drive C as a Cache Buffer. The 384K

> TurboPac system partitions Drive C into a 256K Cache Buffer and a 128K Fixed Print Buffer. **The hard disk attaches to the Drive C PRINTER/HARD DISK/8088 port** and no mechanical modifications are necessary.

So the Drive C unit is **also the host adapter for the Trantor Winchester**: the hard disk plugs into the Drive C's own pass-through connector, not into the O1. Two consequences: the pass-through connector in §2 is the *controller path* and not merely a duplicated printer port; and **a device that implements this protocol is not by itself a TurboPac** — that needs the hard-disk side too, which is a separate question (and one that overlaps #8).

8.5 What the trace shows about the format

At boot the software writes `0xE5` (CP/M's blank byte) across two regions — `h1 = 00h…0Fh, h2 = 8Ch` and `h1 = 00h…0Fh, h2 = 0Ch`, i.e. `DE = 0…31` — then **reads each block back** to verify. Two regions, 16 blocks each, 128 bytes per block: 4 KiB, which is the *directory and boot area*, not the whole disk. The format is therefore incremental and continues as the disk is used. A later pass of writes touches `h1 = 18h, 19h, 20h, 28h` — the directory region again, at `DE` around `0x30–0x50` — which is `DCL` installing itself.

9. Validation — the spec, run against the software

Two tools in `tools/research/` were written for this, and both are worth keeping:

- **`drive-c-trace.lua`** — write taps on `2900–2903`, logging every PIA access while the real software runs. Stable and reproducible: **10 195 accesses** on a 30-emulated-second run, which is the trace quoted throughout this doc. It is how the frame in §3 was *measured* rather than only read out of the disassembly — including the two facts the disassembly is easy to misread: that the header bytes go to **port A** (not port B), and that the read frame sets `DDRA=00h` before the operation.
- **`drive-c-box.lua`** — a **model of the box** built from this spec: it decodes the port-B states, reconstructs the record address, keeps 128-byte blocks, and answers reads. With it present, the real 1983 software **completes**:

```
QUICKPAC.  Loading Drive C:....
A>DCL
A>DCQ T + 1
A>PIP C:=A:*.*[WVOR]
COPYING -
AUTOST.COM
DCQ.DAT
```

  i.e. `DCL` installs, QUICKPAC loads the driver, and **PIP starts writing to drive `C:`** — so CP/M sees the drive and the protocol is good enough for the shipped software. (`[WVOR]` includes `V` for verify, which reports a mismatch against this simplified model; the addressing is right, the model's `0xE5`-only initial contents are not.)

A MAME limitation, recorded because it cost time: a read tap that returns a *value* eventually segfaults MAME's Lua layer once the guest's resident driver starts polling (the run dies ~20

emulated seconds in; returning a constant instead keeps it alive). The trace tool is unaffected because it only writes taps. Treat the box model as a validation aid for short runs, not a harness.

10. What is still open

1. **The DPB — closed** (§8.1). It is `DCL.COM` file `0x142F`, installed `0x052F`, and it decodes field for field against the manual. What remains is *mechanical* rather than unknown: the payload's pointer table lists the DPB, the transfer buffer (`0x055A`) and two further addresses (`0x053F`, `0x0558`) with no obvious DPH among them, so **the DPH is presumably built at install time** rather than stored. That only matters if we ever want to re-create `DCL.COM`, which we do not.
2. **How the box resolves banks 0–2.** `f(n)` is `0Ch` for `n = 0, 1, 2` (§5), so `(h1, h2)` is *identical* for tracks 0, 256 and 512. Either the box ignores `n` below 3 — in which case the first 192 KiB aliases three ways and a 384 KiB unit could not work — or the nibble is not a bank select and the real bank comes from something not yet identified. This is now the sharpest question in the document. It is testable (write across the whole 384 KiB, read back) and it cannot be settled from software alone: the trace never reaches `n ≥ 1`, because the boot-time format and `DCL`'s own install only touch tracks 0–40. **It does not block a reproduction** — a box that treats `f` exactly as the driver does behaves identically whatever `f` turns out to mean.
3. **The print-buffer path — mostly closed.** The manuals give the budget (`docs/drive-c-executive.md` §5): a **fixed** buffer is carved out of the unit's own RAM disk, user-selectable from **16K to 128K in 16K steps**, and the 384K TurboPac configuration is stated outright as **256K cache + 128K fixed print buffer**. So the DPB in §8.1 is the *no-buffer* case and a unit with a fixed buffer presents a smaller `DSM` — the hypothesis this item was carrying is confirmed. What is still not established is the spooling *protocol* — how the box interleaves buffer traffic with the pass-through printer connector — which only matters if a reproduction offers a buffer.
4. **`DFD.SPR` — closed, and the item was asking the wrong question.** This document already quoted the vendor calling `DFD.SPR` "a substitute console command processor (CCP)" (appendix table, §Appendix), but §10 framed it as the resident transport layer. It is not: it is a **replacement CCP**, loaded by CP/M on every `^C` and every program exit in place of the Osborne 1's `CCP.COM`, living in **user area 15** on the unit and required on logical drive `A:` during installation. `DCL.COM` is the loader and carries the transport. The code-aware disassembly this item asked for would have characterised a command processor rather than a device — and a replacement **inherits** `DFD.SPR` rather than reimplementing it, so the item is removed from the work list. See `docs/drive-c-executive.md` §4.

Sources

- `research/drive_c/DRIVE_C.IMD` — the Drive C software disk (ImageDisk, 40-track FM 10×256); `research/drive_c/extracted/` holds the files.
- `research/source-code/BIOS-144-source/rom144.asm` — the ROM's IEEE-488 and Centronics line definitions (lines 2606–2715 and 3250–3269). The authority for §2.
- `research/drive_c/disasm/dcl.asm` — `z80dasm` of `dcl.com`; §3 and §5 are read from it (the resident block is file `0x1600–0x17BE`, installed `0x0600–0x07BE`). **The committed listing misaligns across the payload**: the installed code has to be read as bytes at file offset − `0x1000`, not from the disassembly, which is how §5's `f(n)` and §8.1's DPB were found.

- `research/osborne1/modem/o1techman-interface-extract.pdf` — Technical Manual chapter 4, including §4.1.2′s **P3 pinout**; transcribed in `docs/o1-ieee488-port.md` §6. Image-only scan: OCR it, do not `pdftotext` it.
- `research/drive_c/Drive_C_Users_Manual.ocr.txt` — the vendor's manual (OCR). §7-2 and §7-3 are transcribed in `docs/drive-c-design.md` §2.2 and §8.2; the OCR loses the spec table's column alignment, so read those pages from the PDF.
- `docs/o1-ieee488-port.md` — the port's device-level behaviour and the ROM's eight drivers.
- `tools/research/drive-c-trace.lua`, `tools/research/drive-c-box.lua` — the measurement and the model (§9).

Appendix — the files on the disk

File	Size	Role
DCL.COM	6656	“DRIVEC version 2.08” — driver installer/loader. Loads DFD.SPR, DCA/DCU. “Cannot install driver *Driverisalreadyinstalled*”. Per the vendor’s review it **replaces the earlier `DRIVEC.COM`**, and when the print buffer is switched on it copies `DCU.COM` into the unit
DFD.SPR	2560	Resident driver (page-relocatable SPR). Contains CCP-style command table (DIR ERA HELP REN SAVE TYPE USER) + BIOS hook code. The vendor describes it as “**a substitute console command processor (CCP)**” that is “**automatically loaded**”
DCA.COM	9984	“DCA version 1.85” — the **Archive-Retrieve** program per the vendor’s own review (it “replaces both `DCARCH.COM` and `DCRET.COM`”). The earlier “assign/configuration” label came from the disassembly and is questioned below
DCU.COM	6272	“DCU version 1.40” — RAM-disk utility (erase all, directory, files/line config; generates DCUN.COM via SAVE 28)
DCQ.COM + DCQ.DAT	896+128	“DCQ version 1.22” QUICKPAC — fast boot-time loader; DCLOADED.SYS marker
DCQINS.COM	7296	QUICKPAC installer (writes boot track via DCL); text by David E. Price
AUTOST.COM / DCQAUT.COM	768	Auto-starter: submits QUICKPAC install/boot via $$ SUB
PIP.COM, XDIR.COM	7936	Print test file (likely exercises the print-buffer feature)
	98	

Author: Kenneth M. Toy, (c) 1983 Drive C. Also supports a print buffer ("Bad print buffer parameter, not loaded", "Not enough space for fixed print buffer"). Osborne-1 only: "cannot be used with the Executive".

The transport lives in DCL.COM, not DFD.SPR

This doc previously implied DFD.SPR held the protocol. It does not: `grep -E '290[0-3]h' dfd.asm` returns **nothing** — DFD.SPR never touches the PIA. DCL.COM holds all 22 PIA references. DFD.SPR's only port access is a single `OUT (0F1h),A` at `0x0803`, still unexplained (`F1h` is not a PIA address; likely the double-density board or a status port).

So the split is **DCL.COM = transport + installer**, **DFD.SPR = the command layer that drives it** (its own command table is DIR/ERA/HELP/REN/SAVE/TYPE/USER). How DFD.SPR reaches the transport is still the one thing missing from the command layer (see §10), because it makes no direct PIA references — so it must be a patched vector or a relocation DCL performs at install time. #### DFD.SPR's layout, and why the committed listing misleads (#6)

`research/drive_c/disasm/dfd.asm` is a `z80dasm` linear sweep from `0x100` over the whole 2560-byte file. Reading the file's own structure (a raw byte scan — the disassembly alone hides this) shows two large **zero-filled regions**, file offsets `0x003–0x0FF` (253 bytes) and `0x10C–0x17B` (112 bytes), plus a header/jump area and the command table:

File offset (= address − `0x100`)	Contents
`0x000–0x002`	`00 C6 07` — SPR header
`0x003–0x0FF`	**zeros** (253 bytes)
`0x100–0x10B`	`JP 0103h` / `JP 00FFh` + `STAT`
`0x10C–0x17B`	**zeros** (112 bytes)
`0x183` / `0x18B`	`COM` / `$$$ … SUB`
`0x1CC`	**command table**: `DIR ERA HELP REN SAVE TYPE USER`
`0x1F0`	command **vector table** (`0591h 0525h 0647h 0666h`)
`0x1F8`	`=_.:;<>`
`0x2A3`	`NO FILE`

Because the sweep starts at `0x100` and walks straight through those zero blocks, its `call sub_01aah` / `sub_01b1h` / `sub_01f5h` / `sub_020fh` … targets all land **inside the zeros** and are spurious — they are not entry points. So the command layer cannot be read off this listing; it needs a disassembly that starts at the code region and treats the zero blocks as data (§10).

What *is* established from the bytes:

- DFD.SPR contains **no access to the PIA** `2900–2903h` (no `32 xx 29` stores) — confirming the split above.
- Its only I/O opcode is the single `OUT (0F1h),A` at `0803`.

- Its command layer is self-contained in the `0x180–0xB00` region, with a real command table at `0x1CC` and a vector table at `0x1F0` whose entries point *within the file* (`0x591`, `0x525`, `0x647`, `0x666`) — code, not relocation slots.

What the unit was like to use (contemporary reports)
Two letters in *The Portable Companion* #15 (Jan/Feb 1985, PDF p. 52) give the user side of the unit. That matters to the reproduction (#84): it has to be worth using, not merely protocol-correct.

> I've been using my **384K unit** for one month and it's turned my single density Osborne 1 into a totally new machine. … it takes **six seconds** to go from the editing no file menu in WordStar to the dot prompt in dBASE II. **Before I had Drive C: I had to boot up a new disk every time** I wanted to edit a program. That was minutes waiting while the Osborne booted up different files. … I had to shell out **$250 to Xerox for an exchange disk drive** two months ago. Now my **disk drives get 90% less wear** and I do 90% more work. — Martin Kilgore, South Pasadena, CA

And the reviewer, in the same issue, standing by a negative verdict:

> I still stand by what I wrote — **Drive C: is not a good buy for the average user**. … both of the readers who responded are doing precisely what I said the unit would be good at — **disk intensive operations for large parts of the day**. — Donald G. Krantz, Minneapolis, MN

So there was a **384K** configuration, and the disagreement was never about whether it worked — it was about who it was for. Extract: `research/technical-articles/TPC-Issue15-1985-pp52-drive-c-letters.pdf`.

The vendor's software set, from a contemporary review
The Portable Companion #14 (Nov/Dec 1984), printed pp. 41–45 = PDF pp. 43–47, carries **"Review of New Drive C: Loader & Archive Programs" by Larry Walker** — and, usefully for provenance, it is a **reprint from *FOGHORN* (First Osborne Group) 9/84**. It documents the second-generation Drive C: software, which is what this disk holds, and it names each program and its job:

Program	What the review says it is
`DCL.COM`	the loader "that makes the Drive C: unit accessible", replacing `DRIVEC.COM`; also turns the print buffer on, and then copies `DCU.COM` into the unit
`DFD.SPR`	"**A substitute console command processor (CCP)**, DFD.SPR is designed to make Drive C: work better and is **automatically loaded**"
`DCA.COM`	the **Archive-Retrieve** program, replacing both `DCARCH.COM` and `DCRET.COM`
`DCU.COM`	the print-buffer utility, loaded into the unit with the buffer; `DCU` gives a menu, `DCU E` erases every file
QuickPac: `DCQ.COM`, `DCQINS.COM`, `DCQ.DAT`, `DCQAUT.COM`	`DCQ.COM` "allows you to store a series of commands in SUBMIT fashion to load Drive C:, name it, copy the contents of a disk (or disks) into it, select a print buffer option, and run a program at the end of the process"; `DCQINS` is the menu-driven installer, writing the ASCII `DCQ.DAT`; startup becomes automatic by renaming `DCQAUT.COM` to **`AUTOST.COM`**
`DCN.COM`	"**automatically generated and loaded into Drive C: when the loader program is run**" — it backs the `DCN` rename command
`PRN.TST`	the print test file, sent to the printer at the end of installation

The archive lineage is spelled out too: `DCARCH.BAS` (needed MBASIC, "very slow") → `DCARCH.COM` (compiled, "bugs that arose when it was called upon to close files") → `DCA.COM`.

Command syntax of this generation

- `DC1 A` — install the unit and name it logical A. **This replaces `DRIVEC A: C:`**, which named the unit A and the left floppy C in one go.
- `DCN x` — rename the unit to logical name `x` at the CP/M prompt.
- `DC1 A16` … `DC1 A128` — as above plus a **fixed print buffer of 16, 32, 48, 64, 80, 96, 112 or 128K**. Substituting `P` installs the **dynamic** buffer — all otherwise unused memory, released back to programs as the buffer drains. Any other letter means no buffer is installed.
- `DCU` — buffer menu (stop/restart the dump, clear the buffer, report dynamic space); `DCU E` erases the unit.
- `DCA RB` — run Archive-Retrieve in **Retrieve** mode from drive B:.

Behaviour, and one gotcha that matters for emulation

- **A printer that is connected but switched off crashes the machine.** The review is explicit: *"If the printer is connected but OFF, the Loader will not work and the Osborne 1 will crash."* Cables must meet OCC's specification, and some printers (the reviewer's Daisywriter 2000) "look funny" to the unit — the fix being to disconnect the cable or leave the printer on. This is a property of the duplicated printer port, so a personality for #84 has to reproduce the failure and not only the success.
- **The reset button does not lose the unit's contents** — "only turning off the computer does that" — with the obvious advice to save first.
- **Archived files are stored contiguously as a single file** and are only readable through Retrieve; the directory shows one entry. A set spanning two disks names the second `filename.AR2`. The program warns when the target disk is not blank, shows its directory, and asks twice before erasing.
- **Speed**, as measured by the reviewer: `WSMSGS.OVR` (34K) retrieved into the unit in **10 seconds** versus **17** with PIP; a 272K WordStar + Spellguard set in 115 seconds. Single-density owners need two disk changes and take longer.
- **Cabling**: the unit mounts in the pocket below the right drive; the longer cable goes to the **IEEE-488 connector on the left side**, the shorter to the **External Video port** directly below the unit. Both O1 connectors are duplicated on the Drive C: panel (Centronics printer cable and shorting plug), and the unit has its own external video jack, so no adapter is needed.
- **Capacities sold**: 192K and 384K (the reviewer bought 384K).
- **Terms**: the new package was free to customers who bought after **31 December 1983**, $39 to earlier ones; the manual runs to nearly 150 pages.
- **BackPac** — a battery pack inside the keyboard enclosure that holds the unit's memory for **8–10 hours** after power is lost, "deliberately or through power failure". Announced but not shipping at the time of writing, manual still being written, target price **$150**.
- The same vendor resold **Trantor** hard disks at a discount. The Trantor advertisement itself (*The Portable Companion* #5, February/March 1983, PDF p. 28; extract `research/technical-articles/TPC-Issue05-1983-pp28-trantor-tsl-hard-disk-ad.pdf`) shows what the product was: the **TSL-5** and **TSL-10**, **5 MB $2995 / 10 MB $3395**, *"unpack your disk, **plug it into the IEEE port**, run a simple 5 minute installation procedure"*, supplied *"complete with BIOS, back-up, retrieve utilities and cable"*. So Trantor's Winchester is a **third IEEE-488-attached** storage device on this machine alongside Drive C — one more claimant on the port that #84 has to arbitrate.

Extract: `research/technical-articles/TPC-Issue14-1984-pp43-47-drive-c-loader-and-archive-programs.pdf`.

Disassembling

```
z80dasm -a -l -g 0x100 -o research/drive_c/disasm/dcl.asm research/drive_c/
extracted/dcl.com
z80dasm -a -l -g 0x100 -o research/drive_c/disasm/dfd.asm research/drive_c/
extracted/dfd.spr
```

z80dasm 1.2.0 (`brew install z80dasm`) — the same version and flags as the committed CoPower listings, so the two sets are directly comparable. It is a linear sweep, so it misaligns around data tables: DCL.COM has a data and jump-table region from roughly `0x0840`, and absolute `JP`/`CALL` targets that land there are table bytes or loader-patched relocation points, not code. (The `jp pe,00757h` / `jp pe,0763h` inside the data-transfer loop are the clearest example: `0x0757 + 0x1000` is the loop's own body, so the loader is adding a page offset when it installs the resident copy.) Read the listing with that in mind rather than trusting it end to end.

cpmtools `cpmls`/`cpmcp` with the `osb1sssd` diskdef worked perfectly, once the IMD was converted to raw with disk-analyse.

The Drive C reproduction: hw + firmware

Status: design — buildable. The protocol it must speak is solved and validated (`docs/drive-c-protocol.md`), the **DPB** that sets the drive's size is found (`drive-c-protocol.md` §8.1), and the **pin-level map of P3** is transcribed from the Technical Manual (`docs/o1-ieee488-port.md` §6). Nothing in the electrical or logical design is waiting on a measurement any more; what is left is a layout and firmware (§7).

Tracks: #84 (the device), #6 (the protocol). Sibling of `hardware/usb-keyboard-adapter/` and `hardware/wifi-modem-module/` — a period device reproduced on modern parts, hanging off a port, reversible.

1. What we are building

A standalone replacement for the **Drive C** IEEE-488 RAM disk:

Element	Choice	Why
Bus	**P3**, the O1′s 26-pin IEEE-488 card edge	the device's own interface
Power	**P5**, the 10-pin video connector, via a passive shunt	EXMON precedent; keeps the machine unmodified
Compute	**RP2040** (a stock Pico)	bit-banging at the required rate is well inside PIO/GPIO
Storage	**external flash**, 512 KiB+	the largest Drive C was 384 KiB
Bonus	composite video out from the same shunt	the EXMON shunt already does this, passively

It is a **reproduction**, not an adapter: the machine sees a device on the port speaking the same protocol, so period software drives it unchanged. Look closely and you find an RP2040 where the original had a handful of TTL — which is the point.

1.1 The envelope, from the vendor's own spec sheet

§7-2 of the Drive C manual is a full specification sheet — and the Executive manual's §6-2 is the same page with one word changed. These are the numbers a reproduction should be measured against:

	Original (384K unit)
Dimensions, no cables	5.6 × 5.9 × 1.0 in (14 × 15 × 2.5 cm)
Weight / shipping weight	1 lb (0.45 kg) / 3 lb (1.4 kg)
Operating temperature	+5 to +60 °C (+40 to +140 °F)
Storage temperature	−20 to +100 °C
Power	11–14.5 VDC at 0.3 A max, **2.5 W nominal**
Latency	none — solid state, no moving parts
Max data throughput	**55,000 B/s** (445,000 b/s)
Random-sector access	≈40 µs (software access time)
Load a 32K `.COM` and run it	1.1 s max (1.0 s on the Executive)
TPA RAM used on the host	1152 B; +1024 B with the print buffer

The throughput figure is the one the firmware has to live inside: 55,000 B/s is about **4.6 µs per byte sustained**, against a transport that clocks one byte per frame. §3.2's timing has to stay within it.

And one design served two machines. The Osborne 1 and Executive variants differ only in software — the Executive's book describes an `ECL`/`ECN`/`ECU` loader for CP/M+ where the Osborne 1 uses `DCL`/`DCN`/`DCU` under CP/M 2.2 — while the mechanical, electrical and performance appendices are otherwise word for word. So this is **one design, not two**, and a board built to this spec serves either host. `docs/drive-c-executive.md` §1.

2. Electrical design

2.1 The bus

P3 is a 26-pin card edge (**pins 25/26 unused**) carrying the 6821's lines (`docs/o1-memory-io-map.md` §6, `docs/o1-mainboard-schematic.md` sheet 9). The host drives it through **7406 open-collector drivers** with **3.3 K pull-ups** (UD6/UD8–UD11, RN7/RN11–RN16), and **all signals are active-low**. Three consequences, and they are the whole electrical design:

1. **Never drive a line high.** Every line is a wired-OR: the box pulls a line low or it releases it. A push-pull driver on this bus fights the O1.
2. **Level-shift.** The bus is 5 V; the RP2040 is 3.3 V and not 5 V tolerant. The same **74AHCT/HCT** conclusion that governs the interposer applies (`docs/research-gaps.md` gap 2): AHCT gates on the way in (5 V tolerant, TTL thresholds), and open-drain outputs (or AHCT with a series resistor) on the way out.

3. **Present a sane load.** The O1′s pull-ups are 3.3 K; the box should not add appreciably more capacitance or loading than one more TTL device on the bus.

Which P3 pin is which signal **is** in our material, and it never needed measuring: `docs/o1-ieee488-port.md` §6 transcribes §4.1.2 of the Technical Manual, and the ROM's own comment block agrees with it line for line. §2.3 gives the mapping for the lines this design actually touches.

2.2 Power

The EXMON precedent (`hardware/drive-c/README.md`) takes power from **P5**, the 10-pin video connector, through a **passive shunt** — a Sullins MP-0100-10-DS-4 edge connector and a Hyko K688-DI sync combiner — and uses **+12 V**.

The budget is documented, not guessed. §7-2 of the Drive C manual gives the original unit's own supply requirement:

> Power Requirements: **11 to 14.5 VDC at 0.3 A max.** (2.5 Watts nominal, 384K unit). Source: Draws power from the **O1 EXT. VIDEO connector** or from the optional Drive C BackPac backup power supply.

So the vendor ran the whole 384K unit — 384 KiB of static RAM, the printer pass-through, all of it — off P5 at **0.3 A worst case**, from the same rail that feeds the CRT. An RP2040 plus flash is a small fraction of that (a Pico is ~50 mA at 3.3 V, well under 0.1 A at the rail after a buck). **No new measurement is needed:** the shunt has to pass less than the original did, and the EXMON shunt is only a connector plus a sync combiner, so it does not change the arithmetic.

The shunt is a **mod**, even though it is passive and reversible, and must be documented as one — the same way the USB keyboard adapter's J6 bridge is.

2.3 Signals the box must actually watch, and where they are on P3

Per the protocol the box needs only five things, and `docs/o1-ieee488-port.md` §6 gives each of them a pin. The **P3** column is what the board is laid out against; the Technical Manual and the ROM agree on it.

Signal	PIA	**P3 pin**	Role in this design
DIO1–DIO8	PA0–PA7	**1,3,5,7,2,4,6,8**	the data byte
PB0	port B bit 0	*not on the connector*	direction: `0` = host drives, `1` = box drives — it gates the 7406s
EOI	PB3	**9**	low for the whole frame (`08h`, `88h`, `09h`) — frame framing
NRFD	PB7	**13**	high only in the operation byte (`88h`/`81h`) — what distinguishes it
IFC	**CA2**	**17**	**the byte clock** — one pulse per byte
NDAC	PB6	**15**	*optional* — the box may drive it high as "ready"; the transport never reads it

Three consequences a layout has to respect:

- **The data lines are interleaved across the edge** — `DIO1–4` on the odd pins `1,3,5,7` and `DIO5–8` on the even pins `2,4,6,8`. That is the card edge's own geometry, not a bit-order choice, and it is why a straight ribbon will not do.
- **Pin 15 is `NDAC` in this personality and *busy* in the printer one** (`01-ieee488-port.md` §6.2), while pin 11 (`DAV`) is the printer's strobe. A box that drives pin 15 has to stay harmless when the machine switches to the printer personality.
- **Pin 13 (`NRFD`) has to stay harmless as well.** The vendor's own printer-cable appendix (`docs/drive-c-executive.md` §6) warns that a cable carrying Centronics pin 12 (Paper Error) to pin 13 "will interfere with normal Drive C operation if connected and MUST be removed" — so the original Drive C was watching pin 13, and a box on the same line must not fight it.

PB1, PB2, PB4, PB5, CA1, CB1 and CB2 are not needed to satisfy the transport — though `PB1` and `PB2` are *enables* for the NDAC/NRFD and EOI/DAV drivers, so they gate lines this design reads (`NDAC`) and watches (`EOI`, `NRFD`).

3. Firmware — the Drive C personality

3.1 The frame state machine

The whole personality is a decoder for the frame in `drive-c-protocol.md` §3. It has five states and no timing assumptions beyond "keep up":

```
IDLE    PB=08h                    -> HEADER1      (clear the byte counter)
HEADER1 CA2 edge, DIO present     -> store h1
        PB=00h                    -> HEADER2
HEADER2 CA2 edge, DIO present     -> store h2; decode the record address
        PB=88h                    -> WRITE
        PB=81h                    -> READ
WRITE   CA2 edge                  -> latch DIO into record[n++]
READ    CA2 edge                  -> drive DIO with record[n++]
        n == 128, or PB=09h       -> IDLE
```

Notes that matter:

- **The header bytes arrive on port A, clocked by CA2** — not on port B. This is the single easiest thing to get wrong, and it is measured (`drive-c-protocol.md` §9).
- **The direction is PB0**, sampled from the operation byte (`88h` ⟶ host drives, `81h` ⟶ box drives). Release DIO whenever PB0 = 1.
- **The frame is self-contained.** No state has to survive a frame boundary, so a reset mid-frame costs nothing.
- **PB6** may be driven high continuously as "ready". Nothing in the transport reads it, so it is optional — but it is the only line the host leaves as an input after every frame, so driving it is the safe choice.

3.2 Timing

The host clocks a byte every **~52 T-states (~13 µs at 4 MHz)** and, on a read, samples DIO **~6 T-states (~1.5 µs)** after the CA2 edge. The safe design is to keep the next read byte **already on DIO** and advance on each CA2 edge, rather than to drive the bus on demand. An RP2040 does this comfortably in PIO or even in a tight GPIO loop; the numbers are two orders of magnitude inside budget.

CA2 is sampled, not generated, so the box needs one edge-triggered input per byte. A PIO state machine with a wait-on-pin is the natural implementation, with the DMA feeding read data and draining write data.

3.3 Where the code lives

The personality is the same device as the interposer's IEEE-488 storage personality (#22) — one implementation, two hosts. `firmware/drive-c-gpib/` already exists as the home; this spec is what it implements.

4. Storage and geometry

Addressing. The host sends `DE`, the CP/M record address in 128-byte units, as the header pair, and `drive-c-protocol.md` §5 decodes it. `DE>>1` is the box's **track** (256 bytes), `DE&1` is the sector inside it, and `DE>>9` selects a 64 KiB bank. The box should decode the pair back to a flat byte offset — `DE * 128` — which is the simplest correct model and the one `drive-c-box.lua` uses.

Size: 384 KiB, and the ceiling is 448 KiB. The DPB is found (`drive-c-protocol.md` §8.1): `SPT = 2`, 2 KiB blocks, `DSM = 191` ⟶ **192 blocks = 384 KiB**; `DRM = 127` ⟶ 128 directory entries; `AL0 = C0h` ⟶ 4096 bytes of directory. That is the 384K unit the archived disk belongs to, and it

is what the box must present. The host's own code caps the addressable range at `n ≤ 6` ⟶ **448 KiB** (`drive-c-protocol.md` §8.3), so a **512 KiB flash** holds any Drive C that can ever be driven, with room for a copy of the original image beside it.

What the box does *not* do. It does not implement CP/M. The directory, allocation map and file structure live on the host side, in the CBIOS that `DCL.COM` patches in. The box is a paged 128-byte record store, and the boot-time format it sees (`0xE5` written across two regions and read back, `drive-c-protocol.md` §8.5) is CP/M formatting *its own* disk through the box.

One thing to leave open in the firmware. `f(n)` is identical for `n = 0, 1, 2` (`drive-c-protocol.md` §5), so the header alone cannot tell tracks 0, 256 and 512 apart. A reproduction that treats `f` exactly as the driver does is unaffected — it aliases exactly as much as the original does, if it aliases at all — but the firmware should key storage on the **raw `(h1, h2)` pair** rather than pre-decoding to an address, so that whichever answer turns out to be right can be implemented without touching the frame decoder.

5. Coexistence: the pass-through printer, and address 0

Drive C duplicates the IEEE-488 connector on its front panel so a printer can plug through it, and the manual is explicit that **GPIB protocol is not supported** on that connector — it duplicates the *parallel port function*. So a reproduction has to decide what the pass-through connector is:

- **Minimum viable:** a mechanical/passive pass-through of the connector (a short ribbon), so a Centronics printer attached there still works with the machine's own parallel driver. That is what the original did electrically for the printer path, and it needs no firmware.
- **Full:** snoop the parallel traffic and spool it (the print buffer, `DCU`, `PRN.TST`). That is a separate feature and a separate issue.

Bus arbitration. Three devices now claim this port in the archive — Drive C, the RT-60A clock (`docs/rt60a-analysis.md`) and Trantor's hard disks (`docs/drive-c-protocol.md`) — and the O1′s **own address is 0**, so nothing else on the bus may use it (`docs/o1-ieee488-port.md` §1). A reproduction that only answers when it sees the Drive C frame pattern is compatible with all of them by construction: it stays released until it sees its own framing.

6. Test plan

1. **Protocol (done).** The frame, measured on the real software (`drive-c-protocol.md` §9).
2. **Box model (done, short runs).** `tools/research/drive-c-box.lua` makes the 1983 loader complete and PIP write to `C:`. Extend it into a proper harness once the DPB is known — that is the fastest place to iterate on firmware logic before any hardware exists.
3. **DPB (done).** `drive-c-protocol.md` §8.1 — 192 blocks of 2 KiB, 128 directory entries, 4096 bytes of directory. Give the box model that geometry and a full `PIP` copy can be checked end to end.
4. **Hardware-in-the-loop.** With a real O1: `DCL`, then `DCQ`, then `PIP C:=A:*.*[WV]` and a byte-comparison of a file moved both ways — the same acceptance test #95 used for the floppy formats, and the one the software itself provides.

7. What is left

Everything this spec was waiting on has now been answered from material already in the archive, so this is a work list rather than a measurement list. The Executive manual comparison (`docs/drive-c-executive.md`) closed two more items and confirmed the hardware is a single design for both machines.

1. **Board layout — unblocked.** The P3 pinout is documented (§2.3, `docs/o1-ieee488-port.md` §6) and independently corroborated by the vendor's own printer-cable appendix (`docs/drive-c-executive.md` §6); the power budget is documented (§2.2); and the mechanical, electrical and performance envelope is in §1.1. Nothing on the board is waiting on a measurement.
2. **Bus loading — the one real electrical unknown.** How much additional load (capacitance, pull-up) the O1′s bus tolerates before its timings move. The honest framing is that this is **probably a non-issue and should be confirmed cheaply, not designed around**: the O1′s pull-ups are 3.3 K, the transport clocks a byte every ~13 µs, and one more TTL-class load — or the AHCT input the RP2040 side will present, which is lighter still — is a leakage-and-a-few-pF question at those edge rates. What it actually decides is whether the board needs a **bus buffer at all**, so it is a confirmation step, not a blocker. A scope check with a spare TTL gate as the load settles it in an afternoon.
3. **Firmware against the model.** Turn `tools/research/drive-c-box.lua` into a proper harness (§6 item 2) and drive the §8.1 DPB geometry through it.
4. **The `f(n)` ambiguity** (`drive-c-protocol.md` §10 item 2). Not a blocker — keep the raw `(h1, h2)` key so the answer stays swappable (§4).
5. **TurboPac is out of scope for #84.** The unit is also the Trantor hard-disk host adapter (`drive-c-protocol.md` §8.4); that half needs the hard-disk side, which overlaps #8.
6. **A print buffer, if we offer one.** The manuals give its capacity and its fixed/dynamic split (`drive-c-executive.md` §5) — including that a fixed buffer is carved out of the RAM disk — but not the spooling *protocol*: how the box interleaves buffer traffic with the pass-through printer connector. Optional feature, not part of the RAM-disk contract.

Sources

- `docs/drive-c-protocol.md` — the protocol this implements (the contract), the DPB (§8.1) and the capacity ceiling (§8.3).
- `docs/drive-c-executive.md` — the Executive variant and the manuals' appendices: the envelope (§1.1), the print-buffer budget (§7 item 6) and the printer-cable corroboration of P3 (§2.3).
- `research/drive_c/Drive_C_Users_Manual.pdf` (OCR at `…ocr.txt`) — §7-2 hardware specifications (power, throughput, access time) and §7-3 CP/M parameters. Both spec pages are image-only with an OCR that loses the table alignment, so read them from the PDF: printed p. 133 = PDF p. 141, printed p. 134 = PDF p. 142.
- `research/osborne1/modem/o1techman-interface-extract.pdf` — §4.1.2 the **P3 pinout**, §4.1.1 the signal directions, §4.1.5 the same pins as a Centronics port; transcribed in `docs/o1-ieee488-port.md` §6.
- `docs/o1-ieee488-port.md` — the port's device-level behaviour, the ROM's eight drivers, and the joystick worked example.

- `docs/o1-memory-io-map.md` §6 — P3/P5 and the other connectors.
- `docs/o1-mainboard-schematic.md` sheet 9 — the IEEE-488 interface, the PIA and the open-collector drivers.
- `docs/interposer-survey.md` §5 — NODISKEMU and cbmSD, the IEEE-488 state machines to port.
- `hardware/drive-c/README.md` — the device's place in the project, and the EXMON shunt precedent.

The Drive C's Executive variant

Status: Read, compared and archived. The Executive manual turns out to be a **direct port** of the Osborne 1 manual we already held, and the diff is small enough to state exactly. It resolves several items `drive-c-protocol.md` §10 had listed as open — most importantly by showing that §10′s `DFD.SPR` item was the wrong question (§4) — and it confirms that the *hardware* is a single design for both machines (§1).

Source: `research/drive_c/Drive_C_Executive_Users_Manual.pdf`, OCR layer at `Drive_C_Executive_Users_Manual.ocr.txt`. Drive C:, 1690 65th Street, Emeryville CA. First edition July 1984, second printing 9/84; © 1984 *Drive C:, a joint venture.* 111 pages, image-only scan, OCR'd at 300 dpi per the `research/README.md` convention. The Osborne 1 manual already in the archive is `Drive_C_Users_Manual.pdf` — **second edition, June 1984**, same publisher.

Provenance (recorded 2026-10-11, closing the open item in `research/README.md`): supplied by **Jim Rayburn** from a private Facebook group, posted 2024-01-06 — two posts, …/2063148960719887 and …/2023247321376718.

One bibliographic note worth recording: the cover is stamped **PRELIMINARY**, while every running header in the body reads "(final draft)". The PDF is a draft/review copy, not a production print. Its *content* is nevertheless complete — the appendices are all present and internally consistent (§3).

1. The headline: the hardware is one design

§6-2 (Executive) and §7-2 (Osborne 1) are the same appendix. Verbatim, apart from one line:

Specification	Osborne 1 manual §7-2	Executive manual §6-2
Dimensions	5.6 × 5.9 × 1.0 in (14 × 15 × 2.5 cm)	identical
Weight	1 lb (0.45 kg)	identical
Shipping weight	3 lb (1.4 kg)	identical
Operating temperature	+5 to +60 °C (+40 to +140 °F)	identical
Storage temperature	−20 to +100 °C (−5 to +212 °F)	identical
Power	11 to 14.5 VDC at 0.3 A max (**2.5 W nominal, 384K unit**)	identical
Source	"the **O1** EXT. VIDEO connector"	"the **Executive** EXT. VIDEO connector"
Latency	none (solid state, no moving parts)	identical
Max data throughput	55,000 B/s (445,000 b/s)	identical
Random-sector access	≈40 µs (software access time)	identical
Load a 32K `.COM` and run it	1.1 s max	**1.0 s max**

The only substantive differences are the connector the unit draws power from and a 0.1 s better load time on the Executive — which is a host difference, not a Drive C one.

This is the useful result for #84. The Drive C is not two products; it is one box with two software personalities. The `docs/drive-c-protocol.md` contract is therefore the contract for *both*, and a replacement built to it serves either machine. It also means the P3 work in `docs/o1-ieee488-port.md` §6 transfers wholesale — the Executive's port is the same port.

The power figure is worth quoting precisely because it is the one number a reproduction has to beat: **2.5 W nominal** for the 384K unit, from a 11–14.5 V rail. A Pico plus flash plus SRAM is far inside that.

2. The software is not the same

The two manuals describe the same product in the same words, so the diff is clean. Everything below is a real difference:

	Osborne 1	Executive
Host operating system	CP/M 2.2	**CP/M+** (CP/M Plus), OCC BIOS 1.0/1.1
Loader	`DCL.COM`	**`ECL`**
Rename the unit	`DCN.COM`	**`ECN.COM`**
Utility program	`DCU.COM`	**`ECU.COM`**
QuickPac installer	`DCQINS.COM`	**`ECQINS`**
CCP replacement	**`DFD.SPR`** — 3K (192K) / 4K (384K)	**none**
Utility options	`S` space, **`O` optimize**, buffer, no-menu, `E` erase	`S` space, **`I` install/un-install**, buffer, **`D` file delete**, no-menu, `E` erase
System utilities appendix	COPY, SETUP, SYSGEN, **MOVCPM**	COPY, **COPYSYS**, SETUP
Floppy orientation	left / right	**upper / lower** (the Executive stacks its drives)
ZCPR appendix	§7-8	absent
CP/M tutorial section	Section 4	absent
Useful storage capacity	187 KB / 376 KB	**190 KB / 380 KB**
TPA RAM, no print buffer	1152 B	**768 B**
TPA RAM, with print buffer	2176 B	**786 B**
TPA RAM, with TurboPac	3172 B	**1024 B**
Disk overhead, 384K unit	8K min, 20K with options	**4K min, 16K with options**

Two of these are large enough to be worth stating plainly.

The Executive runs CP/M+, and that is why the driver differs at all. CP/M+ is a different system with a different BIOS and a different memory map, so the Executive's loader has to patch a different host. Everything the loader does is host-side; none of it touches the wire.

The Executive variant has no `DFD.SPR`, and uses a fraction of the TPA RAM. Both follow from the first point, and §4 explains why.

3. The CP/M parameters are identical — and the capacity figures reconcile

§6-3 (Executive, headed "CP/M+ PARAMETERS") and §7-3 (Osborne 1, "CP/M PARAMETERS") carry the same table, with the same numbers:

Parameter	192K unit	384K unit	TurboPac
Block size	1K	**2K**	8K (modifiable)
Directory entries	64	**128**	512 per logical drive
Directory space	2048 B	**4096 B**	16384 B per drive
Logical sectors	1536	3072	varies
Reserved tracks	0	6	12
Logical sectors / track	2	32	—
Bytes per sector	128	256	—

The 384K column is our reverse-engineered DPB, field for field. Block size 2K is `BSH = 4`; 128 directory entries is `DRM = 127`; 4096 bytes of directory is `AL0 = C0h` (two 2K blocks). And the capacity falls out exactly: `DSM = 191` means **192 blocks of 2K = 384 KB total**, of which the directory takes 2, leaving **190 blocks = 380 KB usable** — the Executive's own figure.

That in turn explains why the two manuals quote *different* "useful" capacities for the same unit, which had looked like a documentation error:

	192K unit	384K unit
Raw capacity	192 KB	384 KB
less the directory	−2 KB	−4 KB
= Executive's "useful"	**190 KB**	**380 KB**
less `DFD.SPR`	−3 KB	−4 KB
= Osborne 1's "useful"	**187 KB**	**376 KB**

Both books are right. The Osborne 1 figure is *net of the CCP replacement the Osborne 1 needs and the Executive does not* — which is a neat independent confirmation of §4.

4. `DFD.SPR` — what it is, and why §10's item 4 was the wrong question

`drive-c-protocol.md` §10 item 4 asked for a code-aware disassembly of `DFD.SPR`, framing it as "the resident command layer" that calls the transport. **That framing was the error, and the document half-knew it already**: its own appendix table quotes the vendor describing `DFD.SPR` as "**a substitute console command processor (CCP)**" that is "**automatically loaded**", and the `research/README.md` row for the *FOGHORN* review says the same. But §10 item 4 was written as though the transport lived there, so the two statements sat unreconciled.

The Osborne 1 manual settles it, in the vendor's own words and in more detail than the review had:

> The `DFD.SPR` file is absolutely necessary to Drive C operation. It's located in **user area 15**. DO NOT ATTEMPT TO ERASE `DFD.SPR` FROM USER AREA 15. This file is loaded each time you type `^C` and each time a program exits to CP/M. … The Drive C software automatically

loads the DFD.SPR file **to replace the O1 Digital Research CCP** which would normally be loaded from the drive A: floppy disk after each ^C.

So the picture is:

- **DCL.COM is the loader and carries the transport** — this we had right. It announces itself as DRIVEC version 2.08, carries DCN version 2.07 inside it, and creates DCN.COM on the unit when it runs.
- **DFD.SPR is the command processor** that CP/M reloads on every warm boot in place of CCP.COM. It is a *CP/M program*, not device firmware — which is exactly why its committed listing "sweeps through zero blocks": it is a CCP image, not a driver.
- **It must be present on logical drive A: during initial installation**, and thereafter lives on the unit in user area 15. Its name "has no significance, but DCL always checks for its existence each time DCL is run."
- **It costs 3K on a 192K unit and 4K on a 384K unit** — a number the review did not give, and the one that makes §3's capacity arithmetic close.

Why the Executive has none of this. The Executive runs CP/M+, whose CCP is a different animal; the O1's trick of substituting a custom CCP on warm boot has no counterpart there. That is the whole reason the Executive variant is 4K lighter on the unit and uses 384 fewer bytes of TPA without a print buffer (768 vs 1152), and 1390 fewer with one (786 vs 2176).

What this changes for #84. It *removes* an item from the work list rather than adding one. A replacement does not need to reimplement DFD.SPR: it inherits it. The CCP that ships with the product already talks to whatever driver is installed, so a box that satisfies drive-c-protocol.md §7 gets the command layer for free. The disassembly we were going to do would have characterised a CCP, not a device.

5. The disk and RAM budget, from the manuals

§7-4 (Osborne 1) and §6-4 (Executive) give the unit's overhead directly. This is the passage that answers whether a **fixed print buffer** costs RAM-disk space — it does:

	Osborne 1		Executive	
Function	192K	384K	192K	384K
Directory space	2K	4K	2K	4K
DFD.SPR	3K	4K	—	—
DCN.COM / ECN.COM	1K opt	2K opt	1K opt	2K opt
DCU.COM / ECU.COM	10K opt	10K opt	10K opt	10K opt
minimum	**5K**	**8K**	**2K**	**4K**
with options	**16K**	**20K**	**13K**	**16K**

DCL.COM also creates a second system file the disk-space table does not list — the manual's screen illustration shows DCLOADED.SYS and DRIVEC.SYS on the unit, and notes that DRIVEC.SYS is "a

SYSTEM file and will not be displayed by DIR. To check for the `DRIVEC.SYS` file, use XDIR or STAT." (Those two names come from a poor OCR of a screen dump, so treat the spelling as provisional; what is solid is that there are two, they are hidden system files, and `DCL` creates them.)

RAM. The O1 needs **1152 bytes** of the 64K, plus **1024** more if the print buffer is installed — and the buffer's RAM "is not used unless you install the Buffer at the time you run `DCL.COM`". The manual even names the consequence: applications that use all of the TPA (it cites DeskTop Accountant) may need the print buffer removed to load.

Print buffer, in the unit's own memory:

- Fixed Buffer Capacity: user-selectable **16K to 128K in 16K steps**.
- Dynamic Buffer Capacity: up to the maximum space available on the unit.
- TurboPac spool buffer: **128K fixed on the 384K unit, not installed on the 192K unit**.

That last line is the 192K/384K TurboPac split, and it is also the explanation for the print buffer's disk cost: a fixed buffer is carved out of the RAM disk, so a 384K unit running TurboPac is **256K cache + 128K fixed print buffer** — which is precisely what the Osborne 1 manual says in its feature list.

6. The printer connector is P3, re-presented — and it checks our pinout

Both manuals carry the same appendix (§7-5 / §6-5) giving the cable from the unit's **PRINTER/ HARD DISK/8088** connector to a Centronics printer. That connector is a **26-pin card edge** — the same edge as P3 — and its pin assignments match our transcribed P3 table (`docs/o1-ieee488-port.md` §6, §6.2) **line for line**:

Drive C 26-pin edge	Signal (manual)	Our P3 table	Centronics
1, 3, 5, 7	data bits 0–3	`DIO1`–`DIO4` (odd pins)	2, 3, 4, 5
2, 4, 6, 8	data bits 4–7	`DIO5`–`DIO8` (even pins)	6, 7, 8, 9
11	Output Strobe	`DAV`	1
13	**NRFD**	`NRFD`	(12, PE — must be cut)
15	Printer Busy	**`NDAC`**	11
19	SRQ	`SRQ`	13 (SLCT OUT)
12, 14, 16, 18, 20, 22, 23, 24	ground	the same eight grounds	19–30

Four things this buys us:

- **The interleave is confirmed by a second, independent source.** Odd pins are data bits 0–3, even pins are 4–7 — the vendor's own cable table, not just Figure 4.1.2.
- **The ground set is confirmed exactly** — pins 12, 14, 16, 18, 20, 22, 23, 24, the same eight our table lists.

- **The dual-purpose pin 15 is confirmed.** It is NDAC on the bus and *Printer Busy* in the printer personality — precisely the "one pin, two meanings" constraint docs/o1-ieee488-port.md §6.2 already recorded, now corroborated by the product that has to live with it.
- **The Drive C watches pin 13 (NRFD).** The manual is emphatic that a cable carrying Centronics pin 12 (Paper Error) to pin 13 "will interfere with normal Drive C operation if connected and MUST be removed". So the box must leave NRFD harmless as well as NDAC — a constraint for docs/drive-c-design.md §2.3.

One discrepancy, recorded rather than smoothed over: our P3 table has pin 10 as REN (CB2), while the Drive C's cable table lists pin 10 as **ground** (tied to Centronics pin 30). Either the Drive C's cable grounds a line it does not need, or the two connectors are not bit-identical on that pin. It does not affect anything a replacement has to do — REN is unused by every personality we care about — but the two tables disagree and that should be noted rather than assumed away.

7. What the Executive manual does *not* have

Worth listing, because absence is evidence about how the product was maintained:

- **No DFD.SPR**, and no equivalent. CP/M+ needs no CCP substitution (§4).
- **No ZCPR appendix.** The O1 manual's §7-8 covers running ZCPR with Drive C; the Executive's appendix list stops at §6-6.
- **No CP/M tutorial.** The O1 manual's Section 4 teaches CP/M; the Executive manual renumbers everything up by one and drops it, so its Section 4 is the RAM-disk/print-buffer/utility section.
- **No MOVCPM rule.** The O1′s §7-1 warns that MOVCPM "must be run BEFORE running the Drive C Loader, DCL.COM" or it reports a "Synchronization Error". The Executive's §6-1 has no equivalent — CP/M+ is configured with SETUP, and GENCPM-style rebuilding is a different mechanism.
- **No fourth system utility is named.** The Executive's §6-1 opens "There are four programs supplied with your Executive" and then documents three: COPY, COPYSYS and SETUP. The O1′s §7-1 names all four of its own (COPY, SETUP, SYSGEN, MOVCPM). Whether the Executive's fourth was dropped from the draft or simply never written is not determinable from this copy.

8. What this changes

For #84, the Executive variant is not a second design to build. It is the same box with a different loader, and the contract in docs/drive-c-protocol.md is the contract for both. The reproduction target does not change.

It removes work rather than adding it. The DFD.SPR disassembly that drive-c-protocol.md §10 item 4 was carrying is no longer needed: DFD.SPR is a CCP, it ships with the product, and a replacement inherits it (§4).

It closes the fixed-print-buffer question. A fixed buffer is carved out of the unit's RAM disk, and the manuals give the budget table (§5). The 384K TurboPac split — 256K cache + 128K fixed print buffer — is stated outright.

It corroborates the P3 pinout from a second vendor document (§6), including the interleave, the ground set and the dual-purpose pin 15.

It supplies the mechanical, electrical and performance envelope (§1): 5.6 × 5.9 × 1.0 in, 1 lb, 11–14.5 VDC at 0.3 A max and **2.5 W nominal** for the 384K unit, 55,000 B/s throughput, ≈40 µs random-sector access. Those are now targets rather than unknowns in `docs/drive-c-design.md`.

What is still open

1. **How the box resolves banks 0–2** (`drive-c-protocol.md` §10 item 2) — unchanged by any of this, still the sharpest question, still not a blocker.
2. **Bus loading** (`drive-c-design.md` §7 item 2) — unchanged.
3. **The Executive driver software.** The manual tells us what the Executive's loader is called (`ECL`) and what it does, but we do not have it. If one surfaces, the interesting comparison is *why the Executive driver needs 768 bytes of TPA where the Osborne 1's needs 1152* — the answer is in how each patches its host, and it would tell us how much of the driver is host adaptation versus device work.
4. **`DCLOADED.SYS` and `DRIVEC.SYS`** (§5) — two hidden system files that `DCL` creates on the unit and the disk-space table does not itemise. Their names come from a poor OCR of a screen illustration, so even the spelling is provisional.
5. **Pin 10** (§6) — `REN` in our P3 table, ground in the Drive C's cable table.

Sources

- `research/drive_c/Drive_C_Executive_Users_Manual.pdf` and `Drive_C_Executive_Users_Manual.ocr.txt` — the Executive manual. First edition July 1984, second printing 9/84; © 1984 Drive C:, a joint venture. 111 pages, image-only; OCR'd at 300 dpi per the `research/README.md` convention. Owner-supplied local file, 2026-10-11; source URL not yet recorded.
- `research/drive_c/Drive_C_Users_Manual.pdf` and `…ocr.txt` — the Osborne 1 manual, second edition June 1984. The source of §4, §5 and most of §3.
- `docs/drive-c-protocol.md` §5 (the frame), §8 (the DPB and the ceiling), §10 (open questions this closes or leaves).
- `docs/o1-ieee488-port.md` §6/§6.2 — the P3 pinout §6 corroborates.
- `research/drive_c/extracted/dcl.com` — `DRIVEC version 2.08`, `DCN version 2.07`, and the `DFD` `SPR` / `DCN` `COM` / `DCU` `COM` name literals quoted in §4.

The hard-disk products

Source: research/harddisk/OCC1_HARDDISK.IMD (Osborne 1 SSDD, FM 10x256). Extracted to research/harddisk/extracted/ via disk-analyse + cpmtools (`osb1sssd`).

What this is

The Osborne 1 hard disk product from **Australian Computer & Telecommunications (ACT), (c) 1982** (string in BOOTHd.COM). This is the "winchester style" interface — and, significantly for us, it is one of the known **Z80-interposer** designs: the host-side hardware plugged into the Osborne's Z80 socket.

Two products, not one

The archive drop (#71) added a **second, unrelated** Osborne 1 hard-disk product, so "the OCC1 hard disk" is a family, not a single thing:

	ACT (this doc)	Media Distributing / Adaptec (**the MD-10**)
Source	`research/harddisk/OCC1_HARDDISK.IMD`	`research/adaptec/*.img`
Vendor string	`AUSTRALIAN COMPUTER & TELECOMMUNICATIONS COPYRIGHT (C) 1982`	`Copyright (c) September 1983, Media Distributing`
Versions	`Version 31.05.82`, `Version 9-02-81`	`HARD version 1.93`, `PORTCHNG version 1.92`, `PREP version 1.93`
Controller	its own; prompts `Enter disk controller address (`	Adaptec ACB-4000; prompts `Enter starting port address in hex` / `must be on a 4 port boundary`
BIOS	`HARDBIOS.HEX`, overlaid by `LOADBIOS.COM`	`HARDBIOS.SPR`, `HARDBDOS.SPR`, `HARDCCP.SPR`, `HD00BIOS.SPR`, `HD04BIOS.SPR`
Other utilities	`MOVCPM10/5/F`, `DISKEDIT`, `DISKTEST`, `RESTORE`, `SAVEFILE`	`MD10.COM`, `MD20.COM`, `PREP.COM`, `DRIVETBL.DAT`, `PORTCHNG.COM`

The second product is identified: it is the Mega-byte MD-10. Mega-byte Inc.'s advertisement in *The Portable Companion* #5 (February/March 1983, PDF p. 25; `research/technical-articles/TPC-Issue05-1983-pp25-mega-byte-md10-hard-disk-ad.pdf`) sells the **MD-10** — 10 MB formatted for $2295, "20 and 44 megabyte Winchesters also available" — and its footnote reads *"MD-10 is a registered trademark of **Media Distributing**."* The disk in `research/adaptec/` is copyright Media Distributing (September 1983) and ships `MD10.COM`/`MD20.COM`, matching the 10 and 20 MB models, so the "Adaptec" disk is **MD-10 software**. Two details from the ad bear on the interface question below: the MD-10 "does not tie up existing OSBORNE ports" and is "significantly faster than other winchester subsystems **which use the IEEE port**" — so it is **not** an IEEE-488 device, consistent with a controller that claims its own addresses (hence `PORTCHNG.COM`) rather than borrowing the serial/video ports. Its own parts list — "drive controller, case, power supply, cabling, **Z-80 interface**" — points at the same class of host attachment as the ACT unit: a **Z80-socket or dedicated-port** design, not the IEEE bus.

Both are **port-relocatable** controllers: the ACT disk asks for a "disk controller address", the Adaptec disk for a "starting port address … on a 4 port boundary", and it ships `PORTCHNG.COM`

to change it. That is consistent with the rest of the O1, whose I/O decode only looks at the low address lines.

It also means the port claim in "Hardware interface" below should be re-derived rather than trusted: `0x2A00` and `0x2C01` are the serial ACIA and the video PIA on this machine, so either the ACT adapter is mapped elsewhere or that disassembly read the wrong operand.

So there are **three** routes to more storage than 2 × 182 KB — the two hard disks here, and Drive C (`docs/drive-c-protocol.md`) — and only Drive C is buildable without unobtainable hardware (#84).

Files

File	Role
HARDBIOS.HEX	Intel HEX of a full replacement BIOS, loads at 0xDF00-0xEB00 (17-jump BIOS entry table at 0xDF00)
LOADBIOS.COM	Overlays HARDBIOS.HEX onto the running BIOS ("Your BIOS has been overlayed"; validates 17 JMPs, checksums, warm-boot vector)
BOOTHd.COM	Hard-disk boot ("ERROR DURING BOOTING TO HARD DISK")
MOVCPM10/5/F.COM	MOVCPM variants (10MB / 5MB / full? volume sizing) for relocating CP/M under the new BIOS
DISKEDIT.COM, DISKTEST.COM	Winchester utilities
RESTORE.COM, SAVEFILE.COM	Backup/restore to floppy
AUTOST.COM, SYSGEN.COM, PIP.COM, XDIR.COM	Standard

Hardware interface (from HARDBIOS.HEX disassembly)

- Same Osborne bank discipline as Drive C: **OUT (0x00) = bank 2 (IO), OUT (0x01) = bank 1 (RAM)**, DI/EI bracketed (subroutines at 0xEA6A / 0xEA75 wrap all IO accesses).
- Device registers are memory-mapped in bank 2:
  - **0x2A00** — command/data register (written with a command byte after bank select)
  - **0x2C01** — status register (read; bit 5 (0x20) = status flag tested after rrca; merged into a saved status byte)
- Also `OUT (C),A` with port value cached at 0xEF08 — indirect port write, target TBD (possibly NMI control or bank-related).
- The BIOS preserves a full register environment (pushes IX/IY/HL/DE/ BC/AF, switches to a private stack at 0xEFC1) around hard-disk calls — it hooks deep into the 17-entry BIOS jump table as a full BIOS replacement, not just a disk-driver wedge.

Comparison with Drive C

	Drive C (RAM disk)	ACT hard disk
Physical interface	IEEE-488 port (6821 PIA bit-bang @ 0x2900-0x2903)	Custom host adapter, regs @ 0x2A00 / 0x2C01
Host code	Driver wedge (DCL/ DFD.SPR)	Full BIOS replacement @ 0xDF00
Boot	QUICKPAC autoload from floppy	BOOTHd.COM + SYSGEN to winchester

Implications for the interposer project

1. **The ACT product validates the whole interposer concept on this exact machine** — a Z80-socket interposer was a shipping commercial Osborne 1 peripheral in 1982.
2. Its register interface (0x2A00 command/data, 0x2C01 status) is a clean, tiny target: a Pico 2 W could emulate the ACT host adapter purely in bank-2 memory space, no PIA bit-banging needed.
3. For the mass-storage personality we therefore have two documented paths: emulate the Drive C GPIB RAM-disk protocol (docs/drive-c- protocol.md), or emulate the ACT adapter (this note). The ACT path looks simpler electrically but needs the interface hardware reverse-engineered (photos / unit access); the Drive C path is fully observable from software alone.

Open questions

1. Which ACT product exactly (model no./capacity)? MOVCPM10 vs MOVCPM5 suggests 10MB and 5MB volume support.
2. Command set at 0x2A00 (sector read/write/format, geometry reporting) — needs annotated disassembly of the HARDBIOS driver routines around 0xE000-0xEA0A.
3. The `OUT (C)` target port at 0xEF08.
4. Whether any ACT hardware photos/schematics survive online.
5. **Re-derive the port claim.** `0x2A00`/`0x2C01` are the serial ACIA and the video PIA, and both hard-disk products describe their controller port as *configurable* — so the addresses in "Hardware interface" above need checking against the ACT disk's own prompt and stored value before they are relied on.
6. **The second product** (Media Distributing / Adaptec ACB-4000): its command set, its default port, and what `DRIVETBL.DAT` contains. See "Two products, not one" above.

The RT-60A real-time clock

Status: Partially solved, and **the contract below is the deliverable** (#7). The hardware handshake stays a *known unknown* (reopen only if an RT-60A unit or a good `CLOCK.COM` dump ever turns up); the **memory ABI is settled and is what `firmware/rt60a-rtc/` implements** (#18). See §"The contract" and §"The third shape: the O1′s own clock" — read those two before writing any clock code.

Source: research/rtc/RT-60A_Manual.pdf (JG Communications, Tucson AZ). The TD0 on bitsavers is a damaged dump (27 bad sectors, extraction failed); this doc reconstructs the interface from the manual per user direction: "virtually rebuild from descriptions".

Hardware

- Plugs into the Osborne 1 **IEEE-488 port** (card-edge), battery backed (DL2420/DL2430/CR2320 lithium, ~1 yr), runs standalone.
- Based on **OKI Semiconductor MSM5832 / MSM58321** CMOS RTCs (BCD time/date chips, 4-bit bus, address-selected registers).
- **Pass-through**: an output connector + extra circuitry keeps the IEEE-488 / Centronics / general-purpose port usable. The clock's outputs are normally **high-impedance**; a "special low voltage CMOS circuit" recognizes a **unique handshake sequence** from the software to activate clock outputs for reading. The handshake is designed not to disturb IEEE/Centronics devices.
- Prerequisite: the port must be in **source-handshake mode** (Centronics output mode). If not found (or an interrupt hits mid-handshake), the read is deferred to the next 1/60 s interrupt.
- Crystal-controlled with a trimcap for speed adjustment.
- Clock reset only by deliberately shorting two vertical pins on the board (prompted by CLKSET).
- Bulletin #2 documents a **Drive C modification** — relevant since Drive C also uses the IEEE port (6821 PIA bit-bang); protocol clash management matters for our interposer coexistence too.

Host software architecture (CLOCK.COM)

- Installed by changing the boot disk to call **CLOCK.COM instead of AUTOST.COM** on both cold and warm boot; OS relocated **1 KB lower** to make room for resident clock code above the BIOS.
- Hooks the **interrupt vector word at 0xEFF8** (O1 60 Hz interrupt; CTC-driven). Each tick decrements MEMBUFF from 60; at 1 it reads the clock (~once/second), translates time to binary + ASCII, stores in registers, chains to the original handler.
- **Phantom Display**: on-screen clock updated every 60th via Z80 block-transfer screen sensing (costs ~15% CPU).
- **At Trap**: patches the BIOS **LIST output jump**; intercepts `@T` / `@D` in printer output and expands to time/date (char after @T/@D must be space or CR).
- File time/date stamping patches installed at boot (BDOS-level).
- Warm-boot reload needed for stamping patches; 1.4x ROM can bypass (bulletin #7).

Software-visible register map (offset from pointer at 0x0040)

Pointer word at **0x0040** points to DAY; add offsets (hex offsets shown relative to the pointer):

Offset	Name	Contents
−15 (FFF1 rel)	FLAGS	bit0: phantom on; bits3,4: at-trap use; bit5: at-trap on; bit7: software off
−14 (FFF2)	TIMEFLAG	bit0: 0=12h/1=24h; bit1: 1=AM/PM off; bit2: 1=seconds suppressed
−13 (FFF3)	DATE (7 B)	YYMMDD, DOW order; bit2 of 10s-of-days = leap year. **Only always-current date rep**
−6 (FFFA)	TIME (6 B)	raw BCD unpacked 12-hour, HHMMSS; bit2 of first byte: 1=PM. Raw from clock chip
0 (0000)	DAY	binary day (SD: FF/BB = invalid)
1	MONTH	binary
2	YEAR	binary (0x53 = 1983)
3	HOURS	binary, 24 h, 0 = midnight
4	MINUTES	binary
5	SECONDS	binary
6	MEMBUFF	60⟶1 decrementer; sticks at 1 when reads failing (port not in output mode etc.)
7	TIMEASCII	NUL-terminated ASCII time string (chosen format)
19 (0013)	DATEASCII	NUL-terminated date string

The contract — what an RT-60A-compatible time service must publish (#7)

This is the *specification*, promoted out of the analysis above. It is what `firmware/rt60a-rtc/` (the resident publisher, #18) and the interposer (#12) must agree on. The rule that makes it implementable is the one sentence that decided ##7: **the RT-60A does not put the time anywhere in memory — `CLOCK.COM` does.** The hardware is read-only; the ABI is software. So we build the *publisher*, not the *chip*.

The pointer. A word at **`0x0040`** holds the address of the block. It is the only fixed address; everything else is an offset from the pointer.

The block (offsets from the pointer, negative offsets below the pointer):

Off	Len	Field	Encoding
−15	1	FLAGS	bit0 phantom display; bit3/4 At-Trap use; bit5 At-Trap on; bit7 software off
−14	1	TIMEFLAG	bit0 0=12 h / 1=24 h; bit1 1=AM/PM off; bit2 1=seconds suppressed
−13	7	DATE	YYMMDD+DOW, BCD; bit2 of tens-of-days = leap year. **The only always-current date**
−6	6	TIME	raw BCD HHMMSS, 12-hour; bit2 of the first byte = PM. Straight from the chip
0	1	DAY	binary
1	1	MONTH	binary
2	1	YEAR	binary (0x53 = 1983)
3	1	HOURS	binary, 24 h, 0 = midnight
4	1	MINUTES	binary
5	1	SECONDS	binary
6	1	MEMBUFF	60⟶1 decrementer; sticks at 1 when reads fail
7	12	TIMEASCII	NUL-terminated ASCII time string, chosen format
19	12	DATEASCII	NUL-terminated ASCII date string

The cadence. CLOCK.COM hooks the interrupt vector **word at 0xEFF8** and decrements MEMBUFF once per 60 Hz tick; at 1 it reads the clock (~1/second), translates to binary + ASCII and refills the block, then chains to the handler it replaced. 0xEFF8 is the stock ROM's keyboard-interrupt

vector: `INTBL = 0xEFF0` and the ROM sets `INTBL+(4*2)` (`rom144.asm` line 235) — so the vector slot is `INTBL` entry 4, and a replacement publisher must save and chain, not simply overwrite.

Sentinel and invalidation rules (the "no clock" vs "clock stopped" question)
These are the rules a reader must be able to rely on — #7 asked for them explicitly, and they are what distinguishes the three states:

State	How a reader tells	What a reader must do
No clock software installed	pointer word at `0x0040` is `0xFF` or `0xBB`	treat time as unavailable; do not POKE `0x0040` yourself
Clock software installed, reads failing	pointer is valid but `MEMBUFF` has stuck at 1 for more than ~2 ticks	time is *stale* — show it, do not trust it for stamping
Clock running	valid pointer and `MEMBUFF` decrementing	`DAY…SECONDS` and `DATE/TIME` are current

`MEMBUFF` is therefore not just a countdown: **it is the health signal.** A publisher that cannot read the clock must leave `MEMBUFF` at 1 (the documented behaviour) rather than let it run free, so that a reader can distinguish "no clock" from "clock stopped". `DATE` at −13 is the one field the manual calls always-current even when the clock read is deferred, so it is the field to trust when `MEMBUFF` is stuck.

What this buys. Any software written against the documented RT-60A ABI — `CLOCK.COM`'s consumers, the `@T`/`@D` At Trap, the BDOS file-stamp patches — works unchanged against our publisher. What we deliberately do **not** reproduce is the undocumented IEEE-488 wake-up handshake, which buys nothing because the only thing that speaks it is `CLOCK.COM` itself.

Conditions that block clock reads (from HELP)

1. Disabled interrupts; 2. peripherals off/off-line; 3. port set as input for extended periods; 4. DAV/ATN held low; 5. printer status lines wired to NRFD (IEEE pin 13) — Centronics cables should have **no connections on pins 13 and 21**; 6. long periods not in IEEE source-handshake mode / long data transfers.

WordStar WSMODS patch (DD systems)
UCNSTA JMP → 0DD06H, INCON JMP → 0DD09H, third JMP → 0DD0CH (SD: 0E106H/0E109H/0E10CH; original DD values 0E506H/0E509H/0E50CH).

Prior art: a *software* date facility, before the RT-60A (#18, #28)
Before any clock hardware, the machine's users kept the date in **page zero**. "**Dating your computer — a system's date facility**" by **Walter B. Novinger**, *The Portable Companion* #7 (June 1983, PDF pp. 50–51 = printed 48–49; extract `research/technical-articles/TPC-Issue07-1983-pp50-51-dating-your-computer-date-in-memory.pdf`), describes a three-part scheme:

- **`DATE.BAS`** prompts `MMDDYY` at boot, validates month/day/year, and pokes the date into **`0050H–005BH`** — a **12-byte block the *Osborne 1 User's Guide* "Memory Allocation" chart (p. R-243) marks "not used"**. It is written twice over so a reader can take either form: **binary** (`POKE 80,D`, `POKE 81,M`, `POKE 91,Y`) and an **ASCII month name** at `82H–8AH`. `DATE.BAS` then `NEW`s itself, leaving BASIC pristine.
- **`GETDATE`** is a BASIC subroutine that reads the block back (`PEEK(80)` … `PEEK(91)`) — no machine code, no `CALL`/`USR`.
- **`AUTOST.COM`** is modified (per the User's Guide ch. 8 listing) so boot runs `MBASIC DATE`: `LXI B,10` → `LXI B,13` (command longer than 8 characters, so `length + 2`) and the command string becomes `'MBASIC DATE',0`.

The buffer survives a **warm start (RESET) and diskette swaps** but **not power-off** — the same "no hardware, so it dies with the power" limitation the RT-60A exists to fix. That is why it is the right prior art for #18: it is what a user did *before* buying a clock, and it fixes the **page-zero ABI** that any emulated date must not disturb. `0050H–005BH` is also worth remembering as a **free 12-byte scratch block** for the interposer (#12) — see `docs/o1-memory-io-map.md` §2.

The third shape: the O1′s *own* clock, and what OZROM 1E does with it (#81)

##7′s stated worry was "read the OZROM manual before we fix our own ABI, so we do not invent a third shape". Reading it — and the OCC source — turned up something better than a warning: **the Osborne 1 already has a native software clock, and OZROM did not invent an ABI either; it adopted the O1′s own reserved cells.** There is no third *invention*; there is a third *convention*, and it predates both third-party products.

The machine already ticks a clock — `UPTIM`

The stock ROM's 60 Hz keyboard interrupt handler calls `UPTIM`, which keeps the time as BCD and handles the bell and disk-active countdown:

```
GKEY:                       ; rom 1.3 source, BMKEY.ASM
    CALL UPTIM              ;UPDATE SYSTEM CLOCK
UPTIM:
    ;Update time via 60hz interrupt
    ;  Time is kept as hhmmss (3 locations)
    ...
    DEC [hl]                ; HL => SEC6  (sixtieths)
    RNZ                     ; if not 1 sec transition
    STO 59,[hl]             ; 60th tick is NOW
    XOR A                   ; A:=0, CY:=0 for DAA
    DEC HL                  ; HL => SECS
    LD A,[hl] ! INC A ! DAA ! STO A,[hl]
    CMP 60h
    RC                      ; if not 60 secs
    STO 0,[hl]
    DEC HL                  ; HL => MINS
    ...
    CMP 25h                 ; 24-hour wrap
```

Measured two ways, and they agree:

- the increment code walks **SEC6 → SECS → MINS → HOURS**, so the cells run downward in address from SEC6; and
- the CBIOS RAM map (occram25.asm) places them with HOURS: DS 6 / MINS=HOURS+1 / SECS=HOURS+2 / SEC6=HOURS+3, and its DACTVE = HOURS+4, BELCNT = HOURS+5 match the ROM's own absolutes DACTVE = 0EF6AH, BELCNT = 0EF6BH (rom144.asm lines 128–129).

So the addresses are fixed:

Cell	Address	Meaning
IDAY / IMONTH / IYR	EF63 / EF64 / EF65	calendar date, binary
HOURS	**EF66**	hours, 2-digit BCD
MINS	**EF67**	minutes, 2-digit BCD
SECS	**EF68**	seconds, 2-digit BCD
SEC6	**EF69**	sixtieths, 2-digit BCD
DACTVE	EF6A	disk-active countdown (ticked by the same routine)
BELCNT	EF6B	bell countdown

The clock **starts at midnight on every boot** (the ROM clears EF00–EFEF), nothing in the stock machine *sets* it, and nothing displays it. It is a running counter with no reader and no setter — which is precisely why both third parties built on top of it rather than beside it.

One caveat, stated honestly: the ROM 1.3 source's UPTIM comment says the time is "kept in video memory", which is inconsistent with the address map (video is F000+, the clock is EF66+). The addresses are confirmed by two independent files, so the comment is stale, not the map. **ROM 1.44′s source does not tick the clock**: its GKEY was rewritten ("fixed bell bug", rom144.asm line 16) and now handles the bell and DACTVE but no longer decrements SEC6, and the file contains no SEC6/UPTIM reference at all. So under a 1.44 machine the EF66–EF69 cells are *reserved but not ticked.*

OZROM 1E adopts those cells and adds the setter, the display and the hook

OZROM's documented memory map (manual Appendix C-5) is the same cells, named:

OZROM's name	Address	**the O1 cell**
Start of system clock — hours, 2-digit BCD	EF66	HOURS
Minutes, 2-digit BCD	EF67	MINS
Seconds, 2-digit BCD	EF68	SECS
Sixtieths of a second, 2-digit BCD	EF69	SEC6

and it adds EFDF (time-display toggle), EFE8 (clock position on the first two display lines), and its own interrupt hook at **EFF2–EFF4** (preset to RET, 0C9h) with 192 bytes of interrupt-routine RAM at FCC0–FD7F. OZROM's TIME sets the clock, ctrl-shift-T shows it, and ALARM/TIMER/QUIKTIME ride the hook. **So OZROM's "time-of-day clock" is the machine's own EF66–EF69, finally given a setter and a reader.** Its hook is *not* the RT-60A's 0xEFF8: it is a fixed call OZROM's own SYSINT makes with registers already saved, at EFF2, which is a different INTBL slot and a different calling convention. Full map: docs/ozrom-1e.md.

What this means for our ABI (the #7 decision)
Three conventions now, and they do not conflict:

Convention	Cells	Who reads it
RT-60A / CLOCK.COM	pointer at 0x0040 → binary DAY…SECONDS + ASCII + MEMBUFF	software written to the documented RT-60A ABI
O1 native / OZROM	fixed BCD EF66–EF69 (+ EF63–EF65 date)	OZROM's display/ALARM/TIMER, and anything that reads the machine's clock cells
(ours)	**both**, published together	whichever the software expects

The cheap, correct move is that our resident publisher **maintains the native EF66–EF69 cells as well as the 0x0040 block**. They are four bytes apart in effort and they make our clock visible to OZROM-style readers *and* to RT-60A software, without inventing anything. That is the whole of the "third shape" question: not a new ABI, but two existing ones kept in step.

- We can emulate the RT-60A faithfully WITHOUT its exact hardware handshake: implement the MSM5832 register semantics and the 0x40 pointer ABI, plus the IEEE-port handshake activation observed at the PIO lines. For software compatibility, matching the **memory ABI + interrupt cadence + At Trap/BIOS hooks** matters more than the internal CMOS circuitry.
- MSM5832 register set (from datasheet, 4-bit regs): sec 1/10, min 1/10, hr 1/10 (with AM/PM + 12/24), day-of-week, day 1/10, month 1/10, year 1/10 — matches the BCD layouts above.
- Drive C coexistence: both devices live on the same IEEE port; the interposer can present BOTH virtual devices with proper arbitration (Drive C's 6821 bit-bang vs clock handshake) — worth an interference test in #29.

CoPower-88: the host protocol

Sources: research/copower88/kaypro/extracted_cpm/ (copwrcpm.td0) and research/copower88/zorba/extracted/ (swpdos.td0). Disassemblies in */disasm/ (z80dasm, org 0x100). Schematics: 611-0003-1..4.pdf (Zorba site) — TODO: extract port decode details (#5).

The headline facts

1. **Two adjacent I/O ports**: a STATUS port (E) and a DATA/COMMAND port (D), consecutive addresses. All transfers use Z80 block I/O (`OUTI`) with bit-0 polling of the status port.
2. **The port base is software-configurable**: RAMDISK.COM prompts "CoPower_88 port address or to use default". Defaults are stored in the .COM header:
   - Kaypro: status **0x7F**, data **0x7E** (bytes at 0x107: `7F 7E`)
   - Zorba: status **0xFF**, data **0xFE** (bytes at 0x107: `FF FE`)
   - header byte at 0x103 stored to a work cell (function TBD; 0x0C?)
3. **Transfer protocol** (identical code at 0x0B82 in both RAMDISKs):
   - poll status port bit 0 (wait while set)
   - write command byte 0x05 to data port
   - poll, write 0x01
   - poll, then a sequence of `OUTI` block transfers (sector data) Inter-command handshaking on both directions via the same bit.
4. RAMDISK asks: drive letter (A-P), erase dir Y/N, load address (page-aligned, "??00"), port pair. Installs a BIOS wedge ("not standard CP/M system, cannot install" if the BIOS jump table isn't where expected).

Machine glue differences (Kaypro MSDOS.COM vs Zorba PCDOS.COM)

- Kaypro MSDOS.COM touches Kaypro motherboard system ports (0x10-0x14 SIO/CTC/system, 0x1C/0x1D/0x1F heavily) IN ADDITION to the CoPower ports — screen/keyboard/clock glue for the "BIOS calls only" PC emulation environment.
- Zorba PCDOS.COM error strings show the watchdog protocol: "INVALID 8088 REQUEST #XX AT XXXX", "8088 DOES NOT RESPOND", "8088 SYNC ERROR, PRESS R TO RETRY" — the Z80 host polls the 8088′s request queue; the 8088 ROM monitor services requests asynchronously.
- Both drivers are the same vintage SWP codebase (identical io subroutine offsets) — the machine-specific parts are: port defaults, console/screen BIOS glue, and install-time BIOS vector patching.

Implications for the Osborne 1 port (#26)

1. The Osborne CoPower-88 board likely decodes yet another port pair; candidates: any free O1 port pair (O1 uses few ports; the DD upgrade uses 0xF1? TBD from schematics #5 / photos #9). The configurability in RAMDISK.COM means once we know the Osborne board's decode, the existing driver may work AS-IS with a port override.
2. MSDOS.COM's Kaypro system-port access must be mapped to Osborne equivalents (bank-2 memory-mapped ACIA/PIA) — the BIOS-call translation layer is where the porting effort lives.
3. The 8088-side firmware is on the board's ROM (not on these disks); the DOS disks (copwrdos.td0, dos211-*.td0) are PC-side payloads.

Open questions

- Exact status-bit semantics (bit0=busy both directions? other bits?)
- Command set beyond 05/01 (sector read/write, identify, sync)
- Header byte at 0x103 (0x0C): drive count? IRQ enable?

- Osborne port decode on the actual CoPower-88 (from 611-0003 schematics + board photos)

CoPower-88: the schematics

Sources: research/copower88/zorba/611-0003-{1..4}.pdf ("SCHEMATIC 8088 BOARD", Software Publishers Inc., Arlington TX).

Sheet 1 — 8088 CPU

8088 (U11) in minimum mode (MN/MX hi), 8284 clock generator (U5) with X'TAL1 + reset conditioning, address demux via LS373 (U19, A0-A7) and LS375 (U12, A16-A19), data bus transceiver LS645 (U20, DIR/EN), control buffering via LS244 for RD/WR/IO-M/DEN. HOLD/HLDA + INTA/INTR brought out. So: standalone 8088 system, 20-bit address space.

Sheet 2 — RAM and ROM array

16 x 4164 = 128 KB DRAM on-board ("M1 THRU M16"), header "TO 128K RAM EXPANSION BOARD" (256 KB total option). 2732A 4 KB boot/monitor ROM (U28) with ROMCS + RD/OE gating. Address muxing via LS257 (U26/U27), refresh address counter LS244 (U18) driven by REF1-7/REF-ENA. ROM data buffered by LS244 (U23).

Sheet 3 — Memory controller

Refresh timing: LS393 (U4) + LS161 x2 (U9/U10) generate REF1-REF7 from CLK under REF-RST. Refresh arbitration state machine (LS175s U2 + LS153 U3) arbitrates CPU vs refresh, generating CPU-ENQ, REF-ENQ, RFSH and — critically — **HOLD**: the board asserts HOLD to seize the HOST Z80′s bus during refresh/arbitration windows. RAS decode: LS138 (U31) CAS0-CAS3, LS138 (U32) + LS08 (U33) RAS0-RAS3 + ROMCS, qualified by IO/M, A16, A17, A19, HLDA. MUXC + LS74s (U30) for address mux phase.

Sheet 4 — Z80/8088 PORTS (the host interface)

Host connector P1 carries: BIORQ, RDB, WRB, BA0 (Z80 A0), BD0-BD7 — i.e. the board snoops ONLY IORQ cycles, one address bit, and data.

Port decode

- 4077 XNOR comparators (U3) + 4021 shift register compare A1-A7 against an 8-position **DIP switch** (100K networks) ⟶ DECODE. **The I/O base address is DIP-switch selectable in units of 2.** This matches RAMDISK.COM's "CoPower_88 port address or " prompt: Kaypro factory default 0x7E/0x7F, Zorba 0xFE/0xFF.
- LS139 (U6): BIORQ + DECODE + BA0 ⟶ PORT0 (A0=0) / PORT1 (A0=1).

Mailbox registers

- U24 + U25 (LS373/LS373): two 8-bit latches forming a bidirectional data mailbox between BD0-7 (host) and D0-D7 (8088). PORT0 WRB latches host⟶8088 data (also clocking D0 flag LS74 U7 ⟶ STATIN visible to 8088); RDB+PORT0 reads the 8088⟶host latch.
- **PORT1 write sets INTR (LS74 U14) ⟶ interrupts the 8088**; cleared by 8088-side ack / RESET. This is the "command doorbell": host writes status/command port to kick the 8088 monitor.

- IRQOUT (WR + LS32 U15): 8088-side write toggles the host-visible status flag; host polls STATIN (RD + LS32) — **bit 0 of the status port**, exactly the `rra / jp c` poll loop in the drivers.
- RDX (from RD8088 via 40HC32 one-shot-ish network) heads to "PIN 21 OF BERG PINS" — read-strobe toward the RAM expansion header.
- HC245 (U1): 8088-side data buffer BD0-7 ↔ D0-7 (daughter board).

Protocol synthesis (hardware + drivers agree)

1. Host polls PORT1 bit0 (mailbox free), writes command byte(s) to PORT0, writes PORT1 (any value) to doorbell the 8088.
2. 8088 monitor (2732A ROM) services the request, fills the 8088⟶host latch, toggles IRQOUT ⟶ status bit.
3. Host sees bit0 change, streams data via OUTI/INI on PORT0. Commands seen in RAMDISK.COM: 0x05 then 0x01 prefix a block transfer sequence.

Implications for the interposer (#16/#19)

- The CoPower-88 is itself a **bus interposer precedent**: it sits on the host IORQ space with a minimal signal set (IORQ, RD, WR, A0-A7, D0-D7, RESET) and steals bus cycles via HOLD for refresh. Our RP2350 design needs the same signals, plus WAIT for finer-grained interposition (the SWP board couldn't stretch cycles; it used HOLD).
- Emulating a CoPower-88 (#26): trap IORQ at the DIP-configured base, implement PORT0/PORT1 + status bit0 semantics, and run the 8088 side virtually (or stub the monitor protocol).
- For finding the Osborne CoPower-88's DIP setting without the board: scan even/odd port pairs for the mailbox signature (write PORT0, read back after 8088 echo; or observe INTR behavior). MSDOS.COM's "8088 DOES NOT RESPOND" path gives a detection oracle.

80-column boards: SCREEN-PAC to TRA

Nuevo Electronics and Osmosis each shipped **two** Osborne 1 upgrades — a double-density board and an 80-column video board — and **neither follows the pattern OCC used for the same two functions**. That matters twice over: it is prior art for this project's interposer, and it means a video personality (#24) cannot assume the ScreenPac's interface.

This doc covers the **video** side. The DD side is in `docs/dd-upgrade-formats.md` and `docs/research-gaps.md` gap #11.

> **Scope note: "Executive" means a different machine.** In Osborne's world the Osborne **Executive** is the O2-class machine, not a variant of the 1. Upgrades, software and advertisements aimed at it are therefore **out of scope here** and must not be folded into Osborne 1 conclusions — for example the $595 double-sided-drive upgrade advertised in *The Portable Companion* #14, PDF p. 47, which is an Executive product. The three machines this project cares about are the Osborne 1, the Osborne 1 with OCC's own upgrades, and clones of them (`docs/research-gaps.md` gap 12).

The three implementations at a glance

	OCC **SCREEN-PAC** (1983)	**Nuevo** (1984)	**Osmosis** (1983)
What it is	add-on board, hardware only	replacement monitor ROM + patched CBIOS + own chargen	**Z80 interposer board** + supplied monitor ROM
How it attaches	board inside the machine (`2F00040` service manual §6.12)	swap the monitor ROM; patch the CBIOS with DDT	**Z80 moves onto the board, board into the Z80 socket**
ROM change	**none** — MAME models `osborne1sp` as a clone	Nuevo **1.51** monitor ROM + BIOS 1.5 objects	supplied ROM marked **"Osborne 1.44 GM"**
Widths	**52 / 80 / 104**	80	80 (and back to 52)
Selected by	**BIOS variable** — poke `E1A5H`, call `E4E5H`	patched CBIOS entry (`02C9` ⟶ `JP 02D4`)	**escape sequences** — `ESC 8` / `ESC 5` / `ESC F`
Detected by software	`2400H` bank 2, **bit 2 LOW**	—	—
Horizontal scroll	32-column blocks	—	—
Character generator	stock late	**its own** (adds underline via row 9)	stock 1.44

Sources for each row are quoted below. "—" means the material in the archive does not say; it is not a claim that the feature is absent. A **fourth** implementation, The Research Associates' TRA board, is not in the archive at all — it is covered separately below from its advertisement.

OCC SCREEN-PAC — hardware only, driven by the stock BIOS

The one that needs no ROM: `docs/o1-memory-io-map.md` §2a has the full statement. The parts that make it the odd one out are that the **stock BIOS already contains the width-switching routine** (`E1A5H` / `E4E5H`), that the board announces itself at **`2400H` in bank 2**, and that the SCREEN-PAC's own 2 KiB ROM is *the same image as the stock late character generator* (`docs/o1-rom-variants.md`) — so the capability genuinely is not in a ROM.

The Research Associates (TRA) — 80/100 columns, over the IEEE port

A **fourth** 80-column answer, and the only one that reaches the screen through a *different connector*. An advertisement in *The Portable Companion* #5 (February/March 1983, PDF p. 98 = printed 84; extract `research/technical-articles/TPC-Issue05-1983-pp98-tra-80-100-column-video-and-winchester-ads.pdf`) announces "**80 and 100 column video**" from **The**

Research Associates (Forest Hill, Dunstable, Massachusetts) as "an add-on to the Osborne 1", providing:

- an **80×24 window** and a **100×24 window** "into a 128 character by 32 line video format" — the same 128-wide buffer this project already models (`docs/o1-memory-io-map.md` §2a), so TRA widens the *window* exactly as the others do;
- **soft scrolling**, **reverse video and blinking**, **composite and TTL video monitor outputs**;
- connection "by any Osborne 1 user. **Simply plug into the IEEE port** and into your external monitor";
- "**parallel printer connection is not forfeited**" — the printer moves to a **parallel port on the TRA**;
- "a one-time, automatic disk configuration process", with the **driver software supplied by the vendor**.

Two things make it worth recording. First, **the interface is the IEEE port**, not the video bus — so a video personality (#24) cannot assume an 80-column upgrade lives on the video side of the machine, and the IEEE port becomes contended between a video device, Drive C (#6/#84), the RT-60A (#18) and Trantor storage. Second, its **on-board printer port** is the same "duplicated port" pattern as the Drive C panel (`docs/drive-c-protocol.md`): the add-on *re-presents* an existing connector. Neither its width-switching mechanism (no ROM or BIOS patch is mentioned — a driver plus a disk configuration step) nor its presence detection is documented in the ad, and **unlike SCREEN-PAC and Nuevo it is not in the archive**: the ad is all we hold.

Nuevo — a new monitor ROM, and a CBIOS patched on disk

Nuevo's video upgrade is **software**, and the archive holds the installer. The system disk `research/nuevo/OS1NUEVO.IMD` (Osborne DD: 40 tracks × 5 × 1024 B, 250 kbps MFM) contains 19 files, and two of them are the installers — not SUBMIT scripts despite the extension, but **DDT patch scripts** run through `XSUB`:

```
$ cpmcp -f osborne1 OS1NUEVO.IMD 0:SYS-80C.SUB -
B:XSUB
B:DDT B:MOVCPM.COM      ; patch the CP/M system image in memory
IBIOS15.OBJ             ; insert Nuevo's BIOS 1.5 object
R1F00                   ; ...at 1F00
IBIOS15BM.OBJ           ; and the second module
R29E0                   ; ...at 29E0
S2C9                    ; set the byte at 02C9
C3                      ;   to C3 (JP)
D4                      ;    D4 02 = JP 02D4  <- the 80-column hook
02
.
FA10,A17,00             ; clear the width/variable block
G0
SAVE 43 MOVE-CPM.COM    ; 43 pages = the new system
MOVE-CPM 59 *           ; write it to the system tracks
```

So the Nuevo mechanism is: **patch `02C9` to jump into new code at `02D4`, carrying a replacement BIOS (1.5) and its variables.** `SYS-DDEN.SUB` is the same script with the DD patches added — four sites set to `64` (100 decimal, the width value used for the wide display elsewhere) — plus an `ERA AUTOST.COM`.

Two more files patch an application rather than the system:

```
$ cat DB23PTCH.SUB
XSUB
DDT B:DBASE.COM
S130 / 01        ; patch dBASE's screen handling
S154 / 50        ; 50 hex = 80 columns
S395A / C9       ; RET
G0
SAVE 70 B:DBASE.COM
```

That is byte-for-byte the same three-byte patch Osmosis's manual gives for dBASE (below). Two independent vendors arrived at the same fix, because what needs patching is dBASE's assumptions, not the board.

Plus the ROM side: Nuevo **1.51** is an OCC ROM carrying Nuevo's copyright and patches (`docs/o1-rom-variants.md`), and the Nuevo char-gen differs from stock — it forces row 9 to `0xFF` when bit 7 of the character code is set, which is the underline attribute made real in the font.

The Nuevo product line, from its own advertising

The Portable Companion #15 (Jan/Feb 1985, PDF p. 39) advertises four upgrades, and the 80-column one is not just a wider window:

> **80 COLUMN UPGRADE** — Increase your screen size to a full 80 columns (& still be able to scroll horizontally if you choose). Excellent resolution, with a **choice of six cursor styles and reverse video capability** for easy viewing. Optional **baud rate generator provides 16 baud rates from 50 to 19.2K baud**. $250.00 (add $30 for the optional baud rate generator)

Three things worth taking from that:

- **"Six cursor styles"** — Nuevo's video board treats the cursor as a *feature*, which is the same circuit the block-cursor modifications (#24) were chasing on the stock machine and on the SCREEN-PAC. A personality that renders only the O1′s underline cursor is rendering the worst of the available options.
- **Horizontal scroll in 80 columns** — the same window-sliding behaviour the SCREEN-PAC gives, here described as a user choice rather than a side effect.
- **A baud generator to 19.2K** — well beyond the 9600 Hogan's modification reaches on the stock board (`docs/o1-memory-io-map.md` §2a), and an *option* rather than a cut track. It is the other end of #21′s range.

The disk products from the same advertisement are in `docs/dd-upgrade-formats.md`.

The Nuevo boards themselves (2024 photo drop)

No Nuevo schematic is in the archive, so both products have only ever been known through their software. Two field photographs change that: they are the **first hardware views of either Nuevo board**. Source: **Mike Loewen**, private Facebook group, 2024-10-14 (post); archived as `research/nuevo/photos/`.

80-column board — silk `NUEVO ELECTRONICS CORPORATION © 1984`, `80 COLUMN UPGRADE REV. D`, **serial 4348**. It is not a video tap; it is a **self-contained video subsystem**:

Device	What it is
NEC D780C-1 (PI9455-138)	a Z80 — the board has its own CPU
Nuevo `SY6545A-1` (8325)	a 6545/6845-class CRTC, so the board makes its own video timing
`SRM2016C25` ×2	2 K × 8 SRAM — about 4 KB of video RAM
socketed **`CHARACTER GENERATOR`** ROM, `88332 SINGAPORE`, dated **6-29-84**	the board's own font, under a Nuevo printed label
`NYMPH 12.288 NMP122`	12.288 MHz oscillator (dot clock)
`74S157` ×3, `74LS194A`, `74LS161B`, `74LS273`, `74LS374B` ×2, `74LS166`, `74LS174`, `74LS86B`, `74LS74A`, `74LS20`, `74LS10`, `74LS02` ×2, `74LS00`, `7406` ×2	glue: multiplexers, counters, shift registers, latches, drivers
`VR1`, `Y2`	video-level trimmer, crystal
`J1`–`J6`, **`J9`–`J12`**	configuration jumper blocks
`P2`, `P3` (10-way edge fingers), `P4`	host/display interface
`BAUD` section, 14-pin `P2111-2`	the **optional** baud-rate generator the advertisement prices at +$30

Two consequences, and both *corroborate* what the software already said:

- **The 80-column mode is a second video system, not a wider window.** Its own CPU, CRTC, video RAM and font are why the advertisement can sell "six cursor styles" and horizontal scroll as *features* — those are properties of a real CRTC. It is also why `docs/01-rom-variants.md` §"one late image, five names" concludes Nuevo's 80-column design is not the OCC one: the SCREEN-PAC widens the *existing* video path, Nuevo builds another one.
- **The socketed `CHARACTER GENERATOR` chip is dated 6-29-84** — the date in MAME's `character_generator_6-29-84.14`, the Nuevo-era char-gen already tracked in `docs/01-rom-variants.md`. So the font the board carries is an image the project holds, not an unknown ROM.

Double-density board — silk `NUEVO ELECTRONICS CORPORATION © 1984 MADE IN USA`, `DAL DENS` (dual density), **serial 20011**. This one is an *encoding* board, not a controller:

Device	What it is
`STD DRIVE` silk; `J1`/`J2`; 34-pin IDC header (pins 1–34)	the drive-cable side of the interface (the O1′s own 34-pin floppy header is `P8`)
`WRITE PRE COMP` silk; `74LS195A` (U4), `SN74LS14N` ×2 (U3, U5), `SN74LS00N` ×2 (U2, U6), `SN74LS02N` (U7)	a discrete **write-precompensation delay line** (shift register + Schmitt inverters), and the **data separator**
U1, 16-pin DIP marked `VD9216 / CA00 / 8402`	unresolved. `9216` is the number Standard Microsystems uses for its **FDC9216 floppy data separator**, which would make U1 the separator and put the precompensation in the LS chain — but the marking needs a higher-magnification re-read before either reading is asserted
`RN1`, `RN2`; `P4`	resistor networks; 4-pin power

The likely division of labour — that the DD upgrade supplies the **MFM data separator and write precompensation** the stock machine lacks, and leaves the controller alone — fits the rest of the archive: the controller is `MB8877` ≡ `WD1793` on the mainboard, and the separator is a *separate board on the drive side* rather than part of it (`docs/o1-mainboard-schematic.md` §2, sheets 9/9 and 2/3, U8 and U12–U22). **That is an inference, not a finding** — Nuevo's own installation instructions are the missing piece.

Does any of this bear on Drive C? No, and it is worth saying plainly. Both boards are host-side — video, and floppy encoding — while Drive C is an IEEE-488 peripheral that draws power from the EXT. VIDEO connector and speaks over the GPIB port (`docs/drive-c-protocol.md`). The photo drop is recorded here because it is the first hardware view of either Nuevo product, not because it changes the Drive C contract.

Osmosis — a Z80 interposer with an escape-sequence interface

Osmosis's 80-column board is the one that matters most to this project, because of *how it attaches*. From `research/osmosis/Osmosis_80_Column_Board.pdf` §3:

> Locate the monitor ROM (fig 2.1). This ROM can be carefully removed … Now take the ROM we supplied … Fit the ROM we supplied (marked **Osborne 1.44 GM**) … Locate the Z80 processor … Remove the 40 pin Z80 … **Fit the Z80 into the blue 40 way header on the Osmosis 80 column board** … Fit the Osmosis 80 column board onto the Osborne PCB by inserting the 40 pins … into the socket on the Osborne PCB from which you have just [removed the Z80].

The CPU is moved onto the board and the board takes the CPU's socket. That is a Z80 bus interposer — the same technique this project's interposer uses (#11/#12) — which is why #73 calls Osmosis direct prior art. It is also why the board can be ROM-based: it ships its own monitor ROM.

Its interface is a **terminal's**, not the BIOS's. From §4, the applications guide:

> B3. Press escape (ESC), 8 and return. The screen should now be showing a full 80 columns.
> … B5. Try pressing escape (ESC), 5 and return to take you back to 52 columns.

and, for software:

> B7. … The board will react to a command such as: `10 PRINT CHR$(27)+"8"`

So Osmosis switches width with **escape sequences** (`ESC 8` = 80 columns, `ESC 5` = 52, `ESC F` = auto-CR/LF) — the shape of a real terminal's control codes, reachable from BASIC, rather than a memory variable. The manual also notes that the board "has been well proven in Europe".

The manual's own dBASE instructions are the same patch Nuevo's disk applies:

> (c) Enter S130 (ret) … Enter 01 (ret) … (e) Enter S154 (ret) … Enter 50 (ret) … (g) Enter S395A (ret) … Enter C9 (ret)

The whole Osmosis product line, from their own advertising

The Portable Companion #11 (October 1983), PDF p. 32 = printed p. 30, is a full-page OSMOSIS advertisement — "The Next Step Forward" — and it gives the boards **product numbers**, which the manual we hold does not:

Product	What it is	Price
OSMOS 1	Double Density, “184k per drive using existing drives”, “fully compatible with all Osborne software” — “Install it yourself and save $165 on Osborne’s charge”	$195
OSMOS 1.5	single-sided double density now on existing drives, upgradable to 370k per drive later with new drives	$335
OSMOS 2	double-sided double density, 370k per drive	$895
OSMOS 3	double-sided double density **double tracking**, 750k per drive, “an incredible 1.5MB of portability”	$1050
OSMOS 4	disk formatting and conversion — “read, write and format **over 20 other manufactures CP/M formats**”	$225
OSMOS 5	**the internal 80-column board** — this section’s subject	$250
OSMOS 6	telex: prepare telexes in WordStar, timed send, “automatic transmission/reception in background while running any standard CP/M software”, log and queue	$1195
OSMOS 7	Winchester disks, 5¼ / 10½ / 15¾ / 21 MByte units, “a 42 Megabyte max capacity”	$2475–$3650
OSMOS 8	instrumentation — analog/digital, digital/analog, digital/digital cards	—
OSMOS 10	Olivetti ET121 to Osborne RS-232 KSR interface, 2K print buffer	$500

with a bundle — "**OSMOS 1, 4 AND 5 ALL FOR $385**" against $670 separately, "offer extended by popular demand until December 31, 1983" — and a house claim that matters for the interposer work:

> All OSMOSIS upgrades are **plug-in. There is no soldering or cutting.** Our products are supplied as kits for existing units and require no modifications to the Osborne computer. No specialized knowledge is required.

That is the design philosophy this project's interposer adopts, and the opposite of the SCREEN-PAC's soldered fly-leads (`2F00040` §6.12). Vendor address for the record: 430 Pacific Avenue, San Francisco, CA 94133, (415) 864-6372.

The 80-column board carries a hard-disk interface. The OSMOS 5 entry reads in full:

> An INTERNAL 80 column board with lots of additional features and NO alteration required to existing software: - Full width screen > 80 columns on internal and external video - Normal/ Inverted video · Underline · ⅔ and ½ intensity - Additional graphics characters for screen form design - **Includes a Hard Disk interface for future plug-in expansion** - At a very low price of only $250

So the video board was also the doorway to a hard disk — "**future** plug-in expansion", i.e. the interface ships before OSMOS 7 (the Winchester product) is something you can buy. The board this project treats as a bus-interposer template is therefore a **storage** interposer too, not only a wider screen, and that is a second reason to read its manual closely for #11/#12.

Note the attribute list as well: **normal/inverted video, underline, and ⅔ / ½ intensity**. The stock machine exposes underline and dim as attribute bits (`docs/o1-memory-io-map.md` §2a/ §3); Osmosis advertises two intensity steps. The mapping between the two is not established — relevant to #24.

What this means for this project

1. **A video personality (#24) needs four interfaces, not one.** The SCREEN-PAC's `E1A5H`/`2400H`, Nuevo's patched CBIOS, Osmosis's escape sequences and the TRA's **IEEE-port driver plus disk configuration** are four different contracts. Emulating the window is not enough; the *selection* mechanism — and, for the TRA, the *connector* — differs per board.
2. **Osmosis is prior art for the interposer, not just for video** (#11/#12): a board that takes the Z80′s socket and drives the bus is the shape being designed here.
3. **The application-patch problem is universal.** WordStar and SuperCalc needed settings on the SCREEN-PAC; dBASE needed a three-byte patch on *both* Nuevo and Osmosis. Any 80-column story for this project has to answer for software that assumes 52 columns — the practical reason the O1′s upgrades shipped with application notes.
4. **The 128-character row is the common ground.** All three widen the *window*, not the memory (`docs/o1-memory-io-map.md` §2a); only the control interface differs.

Open

- Nuevo's and Osmosis's **presence detection** is unknown — only the SCREEN-PAC's `2400H` bit is documented. Worth checking the Osmosis appendix and `NEWSETUP.COM` / `AUTOST.COM` on the Nuevo disk.
- Osmosis's full escape-sequence set is in its appendix and should be extracted into this doc.
- The DD halves of both vendors' products are not covered here.

Sources

- `research/screenpac/o1screenpacusersguide.pdf` — SCREEN-PAC User Guide, p. 13.
- `research/nuevo/OS1NUEVO.IMD` and `Nuevo_Electronics_DD_Upgrade_Manual.pdf`. Scripts extracted with `cpmcp -f osborne1`; cpmtools ships an `osborne1` diskdef matching the image exactly (`seclen 1024, sectrk 5, blocksize 1024, maxdir 64, boottrk 3`).
- `research/osmosis/Osmosis_80_Column_Board.pdf` (+ `.ocr.txt`), §3 and §4.
- `research/technical-articles/TPC-Issue05-1983-pp98-tra-80-100-column-video-and-winchester-ads.pdf` — the TRA "80 and 100 column video" advertisement (video), and the Helix Winchester advertisement on the same page.
- `docs/o1-rom-variants.md` — ROM inventory, char-gen variants, Nuevo 1.51.
- `docs/o1-memory-io-map.md` §2a — the window, the widths, the bit fields.

Software

The software library

(not yet written)

Boot disks

(not yet written)

The project

Emulation and testing

How to run the three Osborne 1 machines MAME models, what ROMs each one needs, and which floppy images actually boot on which machine.

The install/run mechanics are adapted from the sibling project `Myrcurial/o1prsnt` — its `o1prsnt/docs/emulating-the-osborne-1.md` and `o1prsnt/utilities/headless-testing-harness/` — which is where the macOS specifics and the VRAM-dumping technique were worked out. This document adds the per-machine ROM requirements and the boot-disk compatibility results, both measured here. (Paths prefixed `o1prsnt/` are in that repository, not this one.)

1. Install

```
brew install mame
brew install cpmtools     # cpmcp / cpmrm / cpmls - file-level disk work
```

`disk-analyse` (Keir Fraser's Disk-Utilities) is needed to convert between the IMD images MAME wants and the HFE images a hardware floppy emulator wants. It is not packaged; build it:

```
git clone https://github.com/keirf/disk-utilities.git
cd disk-utilities && make clean && make && sudo make install
```

`samdisk` (TD0 ⟶ EDSK) is **not** currently installed on this machine and is needed for the TeleDisk images in `research/` (Osmosis, RT-60A, the Zorba/Kaypro CoPower-88 sets).

2. The three machines

MAME models three Osborne 1 variants. Two are clones of the stock machine; one has its own ROMs:

MAME machine	Description	Year	Parent	Relationship	Display (measured)
`osborne1`	Osborne-1	1981	—	the stock machine	416×240 = **52 columns**
`osborne1sp`	Osborne-1 with SCREEN-PAC	1983	`osborne1`	clone — **no ROM differences at all**	832×240 = **104 columns**
`osborne1nv`	Osborne-1 (Nuevo Video)	1984	`osborne1`	own BIOS **and** own character generator	640×240 = **80 columns**

Yes, the Nuevo machine really is 80 columns. The display width is a property of the emulated *machine*, not of the disk, and it was measured by booting the same utility disk on each of the three and reading the screen device's dimensions (`tools/research/mame-boot-probe.sh`). Screenshots, all booted from `tools/emulator-setup/floppies/`:

Screenshot	Machine	Pixels	Columns
`docs/mame-screens/osborne1-52col.png`	`osborne1`	416×240	52
`docs/mame-screens/osborne1nv-80col.png`	`osborne1nv`	640×240	80
`docs/mame-screens/osborne1sp-104col.png`	`osborne1sp`	832×240	104

That is also why the disk prefix does not matter for booting (§4): the same image displays 52 columns on stock hardware and 80 on Nuevo, because the video window is the machine's, while the *text layout* is the CBIOS's.

The distinction matters: the SCREEN-PAC is modelled as *hardware only* (the 80/104-column logic lives on the board, not in a ROM), whereas Nuevo shipped a replacement monitor ROM. That is independent evidence that the two 80-column upgrades are different designs — see #73 and `docs/o1-rom-variants.md`.

The SCREEN-PAC has three modes, and MAME models one of them
The measured 104 columns is the SCREEN-PAC's *default*, not its only width. Its own user guide gives the full set, and all of them are selectable **by the stock BIOS**:

> SCREEN-PAC has preset video contrast in **80 and 104 column format** … In these same formats, SCREEN-PAC provides **horizontal scrolling by 32 column blocks** … At BIOS location **E1A5H**, POKE or store **0 for 52 columns / 1 for 104 columns / 3 for 80 columns**. CALL BIOS location **E4E5H**. — *Osborne 1 SCREEN-PAC User Guide*, p. 13 (PDF p. 17)

So 52 / 80 / 104 are three widths of **one window** (`o1-memory-io-map.md` §2a), and the stock ROM already contains the routine that switches between them — which is why "no ROM differences at all" is the right description of `osborne1sp`. MAME models the 104-column window; the **80-column mode should be reachable by poking `E1A5H` = 3 and calling `E4E5H`**, which is worth testing precisely because it is the other documented mode and the one nobody quotes.

Presence flag. The same page records how software detects the upgrade with no ROM change:

> reading 2400H (Bank 2) returns data **bit 2 LOW**. Without SCREEN-PAC, this bit is read HIGH. This feature allows different screen responses and menus depending on whether the Osborne is fitted with SCREEN-PAC.

`2400H` in bank 2 sits inside the memory-mapped I/O window and was not in the map until now: it is a read-only presence bit, not a register (`o1-memory-io-map.md` §3).

3. ROM requirements
Every ROM either machine needs, with the CRC32 MAME asserts against. Full inventory and provenance: `docs/o1-rom-variants.md`.

ROM file	Size	CRC32	`osborne1`	`osborne1sp`	`osborne1nv`	Role
`rev1.40.ud11`	4096	`3d966335`	✓ (BIOS 1.4)	✓	—	monitor ROM 1.4
`rev1.43.ud11`	4096	`91a48e3c`	✓ (BIOS 1.43)	✓	—	monitor ROM 1.43
`3a10082-00rev-e.ud11`	4096	`c0596b14`	✓ (BIOS 1.44)	✓	—	monitor ROM 1.44 — **the default**
`char.ua15`	2048	`5297c109`	✓ (BIOS ≤ 1.4)	✓	—	early character generator
`7a3007-00.ud15`	2048	`6c1eab0d`	✓ (BIOS 1.43/1.44)	✓	✓	late character generator
`monrom-rev1.51-12.ud14`	4096	`298da402`	—	—	✓	Nuevo monitor ROM 1.51
`character_generator_6-29-84.14`	2048	`6c1eab0d`	—	—	✓	Nuevo-era char-gen — **same bytes** as `7a3007-00.ud15`
`osba.bin`, `osb12.bin`, `osb121.bin`, `osb13.bin`	4096	—	NO_DUMP	NO_DUMP	—	BIOS A / 1.2 / 1.2.1 / 1.3 — **never dumped**

One `osborne1.zip` serves all three machines. The repository's `tools/emulator-setup/roms/osborne1.zip` holds the six available files, and MAME's hash fallback satisfies `character_generator_6-29-84.14` from `7a3007-00.ud15` because the bytes are identical. Verified:

```
$ mame osborne1nv -verifyroms -rompath tools/emulator-setup/roms
romset osborne1nv [osborne1] is good
$ mame osborne1 -verifyroms -rompath tools/emulator-setup/roms
osborne1    : osba.bin (4096 bytes) - NOT FOUND - NO GOOD DUMP KNOWN
... (the four undumped BIOSes only)
romset osborne1 is best available
```

"best available" is the correct and expected result for `osborne1`/`osborne1sp`: the only missing ROMs are the four that have never been dumped anywhere. The default BIOS is **1.44** (`3a10082-00rev-e.ud11`), which is what the banner shows.

Where to get them

The upstream collection used by `o1prsnt` is archive.org/details/osborne-1-roms (two files: `occ-7a3007-00-reva.rom`, `occ-v1.44.rom` — enough for `osborne1` and `osborne1sp`, **not** enough for `osborne1nv`). This repository carries the full set in `tools/emulator-setup/roms/`, archived with provenance in `research/roms/`.

Rompath layouts

MAME looks for machine ROMs in `<rompath>/<machine>.zip` or `<rompath>/<machine>/`, **and it will also fall back to a loose file in the rompath root** when a required ROM is missing but a file with the right CRC is present. Measured, both ways:

```
$ mame osborne1nv -verifyroms -rompath tools/emulator-setup/roms
romset osborne1nv [osborne1] is good          # loose 7a3007-00.ud15 satisfies
                                              # character_generator_6-29-84.14 by
CRC
$ # the same files in a plain <rompath>/osborne1/ directory:
osborne1nv  : character_generator_6-29-84.14 (2048 bytes) - NOT FOUND (osborne1)
```

So the CRC fallback works for **loose-root files and for files inside a zip**, but **not** for files inside a romset *directory*. The wrapper's temporary `<rompath>/osborne1` symlink (used when the rompath has no `osborne1.zip`) turns a working loose rompath into a failing directory — which is why `tools/emulator-setup/roms` keeps `osborne1.zip` alongside the loose files. This matters when substituting a third-party ROM: build a **zip**, see §5.

4. Boot disks: what actually boots on which machine

Measured, not assumed. Each cell is a headless run of the harness in §6 that presses RETURN at the BIOS prompt and waits for an `A>`.

Image	osborne1	osborne1sp	osborne1nv
tools/emulator-setup/floppies/52-blank.imd	**BOOTED**	**BOOTED**	**BOOTED**
tools/emulator-setup/floppies/80-blank.imd	**BOOTED**	**BOOTED**	**BOOTED**
tools/emulator-setup/floppies/104-blank.imd	**BOOTED**	**BOOTED**	**BOOTED**
research/nuevo/OS1NUEVO.IMD	no-boot	—	no-boot
research/nuevo/OS1NUEVO.IMD (pristine upstream copy)	—	—	no-boot

Correction: the `52-`/`80-`/`104-` prefix is a label, not a gate

`tools/check_o1_map.sh` and `tools/README.md` previously implied that an 80-column image "needs a Nuevo Video board" and will not boot a stock machine. **That is not true of these images.** All three boot on all three drivers, because all three are the *same* CP/M system disk — they differ by exactly **two bytes**:

```
$ cmp -l 52-blank.img 80-blank.img
  5737 377   0        # 52 = 0xFF, 80 = 0x00
  5798   0   3        # 52 = 0x00, 80 = 0x03
$ cmp -l 52-blank.img 104-blank.img
  5737 377   0        # 104 = 0x00
  5798   0   1        # 104 = 0x01
```

Both offsets are in the boot area (track 1, sector 1), which is where a CBIOS would keep a video-mode selection. Whatever those two bytes select, they do not change which machines the disk boots on.

So what *does* gate bootability is the **CBIOS on the disk**, not the filename prefix. The prefix is a useful human label — keep using it — but do not rely on it to decide the driver. (Worth revisiting in `check_o1_map.sh`, which currently derives the machine from the prefix.)

What the `-blank.imd` disks actually are

Despite the name, they are **not blank** — they are Osborne 1 double-density CP/M 2.2 system disks, IBM-MFM, 40 tracks × 5 × 1024 bytes, single-sided:

```
autost.com  cbas2.com  copy.com  crun2.com  mbasic.com  movcpm.com
pip.com  setup.com  sysgen.com  xdir.com  xref.com
```

The system area identifies them as `Osborne Computer System / 59k CP/M vers 2.2 / CBIOS 1.4`, with `SETUP VERS. 2.6` and `PIP VERS 1.5`. `AUTOST.COM` runs XDIR (the `Extended Directory version 3.5` banner) rather than booting to a bare prompt — the same behaviour `o1prsnt`'s `DISCOVERY.md` records.

`research/nuevo/OS1NUEVO.IMD` — the Nuevo 80-column system disk

This *is* the Nuevo boot disk. Its label is `CP/M 2.2 system disk for Osborne 1 w/ Nuevo DD & 80 column mods`, and its system area carries a CBIOS that is explicitly the 80-column one:

```
Osborne 1 Computer / CP/M vers 2.2 / CBIOS vers 1.5
*  THIS IS AN EIGHTY (80) COLUMN DISPLAY FOR THE OSBORNE ONE.  *
*  (3) FULL SCREEN REVERSE VIDEO CAPABILITY (JUMPER SELECT).
*  (4) CURSOR PROGRAMMABLE AS UNDERLINE OR REVERSE VIDEO BLOCK.
*  (6) REVERSE VIDEO BLOCK CAPABILITY (INSTEAD OF UNDERLINE; JUMPER SELECT).
```

It does not boot under MAME. On `osborne1nv` it loads, prints the CBIOS 1.5 banner, and then sits there — screen unchanged over 260 emulated seconds, with RETURN and space nudges. On the stock `osborne1` it does the same.

That is not a bad copy: the pristine upstream file from Dave Dunfield's `osborne1.zip` behaves identically, and our copy differs from it only in the IMD comment block. So either MAME's `osborne1nv` does not reproduce whatever the Nuevo CBIOS waits on, or the disk needs the physical Nuevo card to get past initialisation. **Closed as not-needed — see #79.**

Decision: the Nuevo path is not pursued

The video personality (#24) does **not** depend on answering the above. The decision taken is to build it from the **ScreenPac design plus OZROM's soft 52/80/104-column switch**, and to treat the Nuevo board as prior art (#73) rather than as a target. So the Nuevo CBIOS hang is no longer on the critical path, and #79 is closed with the question recorded rather than chased — nothing in the Nuevo design has yet turned up a feature worth the validation cost.

Practical answer for Nuevo work

- To get a **prompt** on `osborne1nv`, use any of the three `-blank.imd` disks.
- To exercise the **Nuevo 80-column video path**, `OS1NUEVO.IMD` is the right disk but it cannot get past CBIOS initialisation; per the decision above this is no longer a blocker for #24.
- Material for *building* a working Nuevo system disk, if anyone ever wants it: `OS1NUEVO.IMD` carries the Nuevo CBIOS 1.5 in its system tracks, the `-blank.imd` disks carry `SYSGEN.COM` (and `MOVCPM.COM`) for writing a new system, and `research/nuevo/` has the Nuevo DD upgrade manual and the Nuevo BIOS ROM (`monrom-rev1.51-12.ud11`). A custom build is therefore plausible.

5. Running it

Directory layout

`o1prsnt` settled on a stable layout so that launching MAME from odd places does not scatter `cfg/` and `snap/` directories everywhere. Our tooling defaults to the same paths, so both projects share one tree:

```
~/Documents/Osborne1/
  roms/           osborne1.zip
  floppies/       .imd / .hfe disk images
  o1prsnt/        working copies
  presentations/  slide sources
```

`tools/check_o1_map.sh` and `tools/emulator-setup/` read `~/Documents/Osborne1/roms` and `~/Documents/Osborne1/floppies` by default; override with `O1_ROMPATH` and `O1_FLOPPY`.

Windowed launch

```
cd ~/Documents/Osborne1
mame osborne1   -flop1 ~/Documents/Osborne1/floppies/52-blank.imd
mame osborne1sp -flop1 ~/Documents/Osborne1/floppies/104-blank.imd
mame osborne1nv -flop1 ~/Documents/Osborne1/floppies/80-blank.imd
```

The machine name is explicit — do not let the image name imply it (§4).

macOS: MAME forces fullscreen. LEFT Option + Return toggles out of it, and Cmd-Q is not honoured — use the red window button.

File-level disk work (`cpmtools`)

```
cpmls -f osborne1 IMAGE                          # list
cpmcp -f osborne1 IMAGE host.txt 0:CONTENT.TXT   # copy in
cpmrm -f osborne1 IMAGE 0:CONTENT.TXT            # remove
```

The `osborne1` disk definition matches the double-density format these images use (40 × 5 × 1024).

IMD ↔ HFE

For a hardware floppy emulator:

```
disk-analyse in.imd out.hfe
disk-analyse in.hfe out.imd
```

Running with a third-party ROM (OZROM 1E)

MAME cannot be told about a BIOS that is not in its ROM definitions, but it *will* load a file whose bytes do not match the expected hash — it warns and carries on. So a third-party monitor ROM is run by presenting it under one of the stock BIOS filenames. Build a **romset zip**, not a plain directory:

```
mkdir -p /tmp/ozbuild/osborne1 && cd /tmp/ozbuild/osborne1
cp <repo>/tools/emulator-setup/roms/*.ud11 \
   <repo>/tools/emulator-setup/roms/*.ud15 \
   <repo>/tools/emulator-setup/roms/*.ua15 .
cp <repo>/research/roms/OZROM-1E/OZROM_1E.BIN 3a10082-00rev-e.ud11
cp 7a3007-00.ud15 character_generator_6-29-84.14    # see note below
mkdir -p /tmp/ozset && zip -q -X /tmp/ozset/osborne1.zip *

O1_ROMPATH=/tmp/ozset tools/research/mame-boot-probe.sh \
    osborne1 tools/emulator-setup/floppies/52-blank.imd 120 /tmp/ozrom.png
```

Why a zip and not `mkdir .../osborne1`. The simpler directory recipe boots `osborne1` and `osborne1sp`, but **fails on `osborne1nv`**: that machine needs `character_generator_6-29-84.14`, which MAME satisfies by CRC fallback from `7a3007-00.ud15` only when the files are loose in the rompath root or inside a zip — *not* when they sit in an `osborne1/` directory. Copying the chargen under its own name (it is byte-identical, `o1-rom-variants.md`) makes the fallback deterministic for all three. Verified: `mame osborne1nv -verifyroms -rompath /tmp/ozset` reports *good*.

OZROM 1E boots and runs on all three machines. Measured:

```
osborne1    416x240  (52 columns)  ->  A>
osborne1sp  832x240  (104 columns) ->  A>
osborne1nv  640x240  (80 columns)  ->  A>
banner: OZROM 1E #00187 / (c) 1984 Micro Management
```

MAME prints *"the machine might not run correctly"* because the ROM hash is not the one it expects — that warning is the point, not a problem. **The full OZROM treatment — memory map, ROM jump table, keytable, column switch, clock, memory tester — is now `docs/ozrom-1e.md` (#81)**, which supersedes the summary that used to live here.

Why this matters to us. OZROM 1E is not a ROM swap with a nicer font; per its manual (`research/roms/OZROM-1E/OZROM_1E_Manual.pdf`, Micro Management, Manual Rev. A 84/09/01) it adds capability we would otherwise have to build:

- **User-selectable 52 / 80 / 104-column display from the keyboard** — a documented *software* video-mode switch. Directly relevant to the video personalities (#24) and to how the Nuevo and ScreenPac paths are driven. On a machine with no 80-column upgrade it can also toggle between the left and right 52 columns of an 80-column display.
- **A system memory tester** that reports which chip to replace — a validation tool for #29.
- **A software-redefinable keyboard** — individual keys at any time, plus eight locally-redefinable function keys of up to 63 characters each — relevant to #45/#47.
- **Time-of-day clock support** (`TIME`, `ALARM`, `TIMER`, and a way to set the clock) — relevant to #7/#18.
- **Documented interfaces**: an interrupt hook for programmers, the ROM jump table, the OZROM memory map and keytable, plus notes on 1793/1797 floppy controller operation (#20) and truer Televideo 912/920 emulation.
- Removes the "Centronics printer not connected" lockout on double-density systems (#19).

What it gives up, stated precisely in its own Appendix C (the manual is now OCR'd to `research/roms/OZROM-1E/OZROM_1E_Manual.ocr.txt`):

> **IEEE-488 functions.** The OZROM 1E does not support the IEEE-488 functions. Although Centronics printer operation is not affected, some printers … and some data acquisition and analysis devices … require the IEEE-488 routines to operate. In this instance, extra software would be required …

The jump-table appendix is blunt about it: the entries at `013F–0156` *were* the IEEE-488 routines and **calling them now produces a controlled system crash**, and *"IEEE-488 AS IOBYTE DEVICE 3 NOW NULL DEVICE"* — the status entries return `0FFH`, input returns zero, output does nothing. The parallel/Centronics entries at `0193+` survive.

So OZROM and Drive C are mutually exclusive. Drive C is an IEEE-488 device and OZROM removes the routines it talks through; the manual's "extra software would be required" is exactly the driver we have the source for in `rom144.asm` — so it is recoverable, but it is not free.

One more difference worth knowing, because it constrains any 2.0 ROM (#83): OZROM **shrinks the video window to 3 KB** (`F000–FBFF`) and uses the last 1 KB (`FC00–FFFF`) for its redefinable keyboard and function keys. It therefore scrolls by moving the 23 lines above rather than by moving the window. The stock ROM's 4 KB video RAM map in `docs/o1-memory-io-map.md` §2 does not apply to OZROM.

6. Headless testing

The technique, from `o1prsnt`'s `utilities/headless-testing-harness/` — which is where it was worked out, and which is worth reading in full (`README.md` for the API, `DISCOVERY.md` for how it was derived):

```
SDL_VIDEODRIVER=dummy mame MACHINE -flop1 IMAGE -video none -sound none \
  -nothrottle -autoboot_script RUN.lua -seconds_to_run N -rompath ROMS
```

- `SDL_VIDEODRIVER=dummy` is **mandatory** from a pure terminal session on macOS; without it MAME aborts with "Could not initialize SDL".
- `-video none -sound none` — no window, no audio. The emulated screen still renders into VRAM, which is all a test reads.
- The screen is read straight from video RAM: base `0xF000`, **128 bytes per row**, 32 rows. Character codes are readable directly; the 9th (attribute) bit lives in a separate bank and can be ignored for text.
- `-nothrottle` runs the 4 MHz Z80 as fast as the host allows (~20–25×) but **`emu.wait()` and `-seconds_to_run` count emulated seconds**, so any timing you measure is a true 4 MHz Z80 timing and transfers to hardware.
- End the script with `os.exit(0)`; letting MAME hit the budget often ends in a harmless but noisy macOS teardown segfault.

The two harnesses in this family

Harness	Where	Purpose
`tools/o1_mame_probe.lua` + `tools/check_o1_map.sh`	this repo	asserts `docs/o1-memory-io-map.md` (12 checks) against the running machine
`tools/research/mame-boot-probe.{lua,sh}`	this repo	boot one image on one machine: `A>` or not, screen dump, **screen geometry**, optional PNG
`o1harness.lua` + `run-test.sh`	`o1prsnt`	keyboard injection (`H.boot_cpm()`, `H.type_line()`) and screen assertions (`H.wait_for()`, `H.find()`, `H.dump()`)

The boot probe is the one that produced the geometry table in §2 and the screenshots in `docs/mame-screens/`. It is deliberately dependency-free; for scripted tests that need to *type* at the machine, use `o1prsnt`'s harness.

Screenshots. MAME's Lua screen device has `screen:snapshot(path)`, which works headless (`-video none`) and needs no display:

```
manager.machine.screens[":screen"]:snapshot("/tmp/shot.png")
```

The §4 boot matrix was produced with the second one plus a ten-line script:

```
cat o1harness.lua probe.lua > /tmp/run.lua
SDL_VIDEODRIVER=dummy mame osborne1nv -flop1 IMAGE -video none -sound none \
  -nothrottle -autoboot_script /tmp/run.lua -seconds_to_run 120 \
  -rompath tools/emulator-setup/roms
```

Gotchas worth not rediscovering

1. **VRAM is a ring buffer.** The machine scrolls by rotating a hardware start register, not by moving memory, so a given screen line moves between VRAM rows after any scroll. Assert with whole-buffer substring searches, never fixed row reads.
2. **Pace your typing.** `natkeyboard:post()` drops characters if a second post arrives while the first is still being typed, and the guest flushes its input buffer around disk I/O. Wait for the prompt, then type (~0.15 s/character).
3. **MBASIC's `Ok` prompts are stale-prone** — count them (`H.ok_count()`) rather than matching `"Ok"`.
4. **MAME writes `cfg/` and `snap/` into the current directory** — run from a neutral directory so neither the repo nor your home dir collects litter.
5. The BIOS **waits for RETURN** ("Insert disk in Drive A and press RETURN.") before it loads anything; every automated test must press it.

7. Open items

1. **#79** — why the Nuevo CBIOS 1.5 hangs (`OS1NUEVO.IMD`). Blocks validating the Nuevo video path (#24) in the emulator.
2. **The prefix → driver inference** in `tools/check_o1_map.sh` needs revisiting (§4): the `52-`/`80-`/`104-` prefix does not determine what a disk can boot.
3. **Identify the two differing bytes** at offsets 5737 and 5798 in the `-blank.imd` set — presumably a CBIOS video-mode selector, but unverified. A small disassembly job against the CBIOS in the system tracks.
4. **ROM 1.3 is undumped** and we hold its source disk — see `docs/o1-rom-variants.md`.
5. **OZROM 1E as a tool (#81).** It boots on all three machines (§5) and its manual is now indexed in **`docs/ozrom-1e.md`** — memory map, ROM jump table, keytable, the 52/80/104-column switch, the clock and the memory tester. The one measurement still missing is driving the ctrl-shift column keys from the harness (`ozrom-1e.md` §7/§9).

8. Sources

- Adaption base: `Myrcurial/o1prsnt` — its `docs/emulating-the-osborne-1.md`, `utilities/headless-testing-harness/{README,DISCOVERY}.md` and `scripts/validate-emulator.sh`.
- Machine/ROM structure: MAME `src/mame/osborne/osborne1.cpp` (`ROM_START( osborne1 )`, `ROM_START( osborne1nv )`, the `COMP()` lines).
- ROM inventory and CRCs: `docs/o1-rom-variants.md`.
- Archive provenance: `research/README.md`.
- Disk pipeline and machine checks: `tools/README.md`.

The interposer and its virtual peripherals

How the RP2350 interposer sits on the Z80 bus and where each virtual peripheral lives in the Osborne 1 memory and I/O maps.

1. Where the interposer physically sits

The Osborne 1 exposes the Z80 on a 40-pin DIP socket. Like the CoPower-88 and the OCC ScreenPac, the interposer **plugs into the Z80 socket** and the Z80 plugs into the interposer — so it sees (and can drive) every bus cycle.

```
flowchart TB
    subgraph MB["Osborne 1 mainboard"]
        ZSKT["40-pin Z80 socket<br/>(A0-A15, D0-D7,<br/>MREQ/IORQ/RD/WR/M1,<br/
>RFSH, HALT, INT, NMI, RESET, CLK)"]
        ROM["4KB BIOS ROM<br/>0000-0FFF"]
        RAM["64KB DRAM<br/>(banked)"]
        VRAM["Video RAM<br/>F000-FFFF window"]
        IO["Onboard I/O<br/>(FDC, PIO, SIO, KBD)"]
        CG["Char-gen ROM socket<br/>UA15, 24-pin<br/>(font data + scan lines)"]
        ZSKT --- ROM & RAM & VRAM & IO
    end

    subgraph INTP["RP2350 Interposer (in Z80 socket)"]
        direction TB
```

```
        Z80["Original Z80A<br/>(NMOS, 4 MHz)<br/>reinstalled here"]
        BUF["74LVC245 buffers /<br/>level shifters<br/>(5V Z80 <-> 3.3V RP2350)"]
        PIO["RP2350 PIO + DMA<br/>bus sniffer / driver"]
        MCU["RP2350 cores<br/>cycle decoder +<br/>peripheral firmware"]
        WIFI["WiFi / USB<br/>stream out"]
        VID["Video renderer<br/>(shadow VRAM + font)<br/>DVI/HDMI via RP2350
HSTX"]
        HDMI["HDMI monitor out"]
        Z80 <--> BUF <--> PIO <--> MCU --> WIFI
        PIO --> VID --> HDMI
    end

    ZSKT <-."A/D/MREQ/IORQ/RD/WR<br/>tapped + can be driven".-> BUF
    CG <-."char-gen socket tap<br/>(ScreenPac-style, #24)".-> VID
```

Second tap point (#24): like the ScreenPac, the interposer taps the Z80 socket **and** the character-generator ROM socket (UA15, 24-pin). The Z80 tap gives it every bus cycle (including all video-RAM writes); the char-gen tap captures the font. Together they feed the HDMI video-output personality described in §5.

Level shifting note (#11): the host Z80A is NMOS (confirmed, date 8408). NMOS outputs are TTL-ish; NMOS inputs have ~2.0–2.4 V high thresholds. Use **74AHCT/74HCT toward the Z80** (TTL-compatible inputs) and **74LVC245 host-side**. ScreenPac used raw 74HC; we prefer AHCT/LVC for margin across all three target machines.

2. Z80 memory map and where virtual devices attach

The O1 banks ROM, RAM, and video in a 64KB space. The interposer claims a small decode window by **snooping MREQ and either passing through or overriding** the onboard response.

```
flowchart LR
    subgraph MAP["Z80 64KB memory map"]
        direction TB
        R0["0000-0FFF<br/>BIOS ROM<br/>(bank 0)"]
        R1["1000-EFFF<br/>System RAM<br/>(shared both banks)"]
        R2["F000-FFFF<br/>Video RAM<br/>(bank-switched window)"]
    end

    subgraph VW["Interposer-claimed window (config)"]
        VP["Virtual-peripheral<br/>mailbox RAM<br/>(e.g. EF00-EFFF)"]
    end

    MCU2["RP2350 firmware"]
    MAP -->|"MREQ+A15-A0 snooped"| DEC{"Decode<br/>match?"}
    DEC -->|"no: pass through"| MB2["Onboard ROM/RAM<br/>respond normally"]
    DEC -->|"yes: assert override,<br/>tri-state host, drive bus"| VP
    VP <--> MCU2
```

Banking caveat

The O1 video RAM and BIOS share the top of the map via a bank-select I/O bit. Any memory-resident virtual device must **(a)** live in an address range the current bank exposes and **(b)** defer to the real video RAM when it is mapped in. Safest is a small mailbox page in the always-present RAM region, or an I/O-port device instead (below).

3. Z80 I/O map and the virtual-peripheral ports

I/O-mapped devices are cleaner: they never collide with the memory banks and are how every real O1 peripheral (and the CoPower-88) already works.

```
flowchart TB
    subgraph IOP["Z80 I/O space (A7-A0 decoded)"]
        direction TB
        ONB["Onboard ports<br/>FDC / PIO / SIO / KBD<br/>(existing decode)"]
        CP88["CoPower-88 mailbox<br/>0x7E data / 0x7F status<br/>(or
0xFE/0xFF)<br/>*real hw being reverse-eng*"]
        V1["VIRT: net disk<br/>port base B+0..B+3"]
        V2["VIRT: WiFi modem<br/>port base B+4..B+7"]
        V3["VIRT: RTC<br/>port base B+8..B+B"]
        V4["VIRT: virtual MX-80<br/>printer port base B+C..B+F"]
    end

    Z802["Z80<br/>IN/OUT (IORQ+RD/WR)"] --> DEC2{"Port decode<br/>A7-A0"}
    DEC2 --> ONB
    DEC2 --> CP88
    DEC2 -->|"claim free base B"| V1 & V2 & V3 & V4
    V1 & V2 & V3 & V4 <--> FW["RP2350<br/>peripheral firmware"]
    FW --> NET["WiFi / SD /<br/>host stream"]
```

Port-base strategy: pick a base `B` in a region the onboard decode leaves free **and** that does not collide with the CoPower-88′s 0x7E/0x7F (Kaypro) or 0xFE/0xFF (Zorba) — so the interposer can *emulate or coexist with* the real CoPower. The comparator-based decode on the real CoPower board (the "16-2103" IC, see photo-survey) is the model for a clean single-port-pair decode.

4. The virtual peripherals (from the roadmap)

Virtual peripheral	Bus type	Address/Port	Host interface	Back end
Network disk	I/O ports	base+0..3 (cfg)	command/data/status FIFO	SD image / WiFi block server
WiFi modem	I/O ports	base+4..7 (cfg)	8250-ish UART regs	TCP/Telnet over WiFi
RTC	I/O ports	base+8..B (cfg)	RT-60A-compatible regs (see #7)	RP2350 RTC / NTP
Virtual MX-80 printer	I/O ports	base+C..F (cfg)	Centronics-style data+strobe	WiFi / USB to host spooler
Video / HDMI output (ScreenPac personality, #24)	memory snoop + char-gen tap	VRAM `F000-FFFF` (passive) + UA15 socket	none — transparent to host	DVI/HDMI via RP2350 HSTX (§5)

The four I/O peripherals are each an independent decode window; firmware registers a `{base, mask, read_fn, write_fn}` entry and the cycle decoder dispatches. The video personality is different: it never answers a bus cycle at all — it only *listens* (and taps the char-gen socket), so it cannot collide with any onboard or virtual device.

5. Video output — ScreenPac personality and HDMI out (#24)

The interposer's second personality is video. The OCC ScreenPac is the template: it taps **both** the Z80 40-pin socket **and** the character generator ROM socket (UA15, 24-pin), and shadows the display through dual-port video RAM (6116s + 74HC157 muxes). The interposer reproduces that function in firmware and extends it to a modern output:

1. **VRAM snoop.** Video RAM lives at `F000-FFFF` (128×32 cells, 52×24 visible). Sitting in the Z80 socket, the interposer already sees every bus cycle; writes into the VRAM window update a shadow copy in RP2350 RAM.
2. **Font capture.** The 2716 character ROM (128 chars × 8×10 pixels) is dumped once at init through the UA15 tap — or a known-good font image is preloaded in firmware — and frames are re-rendered from shadow VRAM + font. Tapping UA15 like the ScreenPac did also tracks machines running replacement/upgrade char ROMs. The tap is **read-only by necessity**: on the mainboard `/CE` (18) and `/OE` (20) are bussed with GND and grounded, so the ROM is permanently selected and always driving `O0–O7` (`o1-mainboard-schematic.md` §4a). Reading costs nothing; *replacing* the font would mean intercepting those two pins between socket and ROM, so the design keeps the fitted ROM as the source of truth and preloads a font image for the 80/104-column modes instead.
3. **Render + output.** The second RP2350 core renders the shadow frame and drives DVI/HDMI via the RP2350 **HSTX** peripheral (or PicoDVI-style bitbang). O1 timing is gentle: 15.9744 MHz

master, 8 MHz dot clock, 60 Hz frame — comfortably inside DVI 640×480 margins with integer upscale.

Prior art — Pico-based bus video cards (candidate boards to study)

- **A2DVI** — `github.com/rallepalaveev/A2DVI`: Apple II digital video card, RP2040-based, HDMI/DVI out. Closest analogue: a Pico-class MCU on a vintage bus rendering HDMI in real time.
- **V2RetroComputing "analog"** — `github.com/V2RetroComputing/analog`: Pico-based Apple II card outputting DVI video over an HDMI connector.
- **AppleII-VGA** — `github.com/markadev/AppleII-VGA`: Pico-based VGA card; same snoop-and-render architecture, VGA instead of DVI.
- **PicoDVI** — `github.com/Wren6991/PicoDVI`: the RP2040 DVI bitbang library; on the RP2350 the HSTX hardware does the heavy lifting instead.

All four prove the pattern this personality needs: a Pico-class MCU snooping a vintage bus and emitting stable modern video. Unlike the Apple II cards, the interposer sits *in the CPU socket* (ScreenPac-style), so it also sees bank-switch writes and can track which bank currently owns `F000-FFFF` (video RAM vs the dim-bit attribute bank — see `docs/o1-memory-io-map.md`).

Relationship to the real ScreenPac: the reverse-engineering phase sniffs the real board like the other targets (#9 photos; service manual safety copy in `research/screenpac/`). The HDMI personality is then the "virtual ScreenPac" — plus a display output the original never had.

6. Coexistence with the real peripherals being reverse-engineered

```
flowchart LR
    subgraph REAL["Real hardware (reverse-eng targets)"]
        CP["CoPower-88<br/>0x7E/0x7F"]
        DC["Drive C<br/>IEEE-488 ramdisk"]
        RTC["RT-60A<br/>RTC"]
        ACT["ACT hard disk<br/>OCC1"]
    end
    subgraph VIRT["Interposer emulation"]
        VCP["virtual CoPower"]
        VDC["virtual Drive C"]
        VRTC["virtual RTC"]
        VACT["virtual HD"]
    end
    REAL -->|"sniff real cycles<br/>to learn protocol"| LEARN["RP2350<br/>capture"]
    LEARN --> VIRT
    VIRT -->|"later: replace real hw<br/>on same ports"| HOST["O1 runs<br/>unchanged<br/>software"]
```

The interposer first **passively sniffs** each real device to pin its protocol (the reverse-engineering phase), then can **emulate** it on the same decode so the original software runs against the virtual version.

7. Personality selection — the gate, and where each personality stands

No decisions have been taken here. This section is the *input* to that decision: the rule we apply, what it misses, and the resulting map — recorded so the choice is argued once rather than re-derived per personality.

The current rule, and its gap

`firmware/README.md` states the project rule: *"If we hold the drivers for a piece of period hardware, we plan a personality for it."* That is necessary but not sufficient — on its own it says nothing about whether a personality can be *tested*, whether it collides with another, or whether the fitted ROM still supports it.

The five tests

1. **Driver held?** — the existing rule. A driver means no Osborne-side software work and a known-good test suite.
2. **Interface reachable?** — Z80 socket, char-gen socket, or a port (P1/P2/P3/P5), without cutting traces.
3. **Testable?** — is there software we can *run* against it (MAME or hardware)?
4. **Address-space arbitration** — does it collide with another personality?
5. **ROM-gated?** — does the fitted ROM still support it? (#86 is closed; the gate is built)

Stacking is not the constraint — addresses are

Worth stating plainly, because it is easy to get wrong: **virtual interposers stack.** The maintainer's own machine carries a real SCREEN-PAC *and* a real CoPower-88 at the same time, which is the existence proof — and in the virtual case it is easier still, because every personality is firmware rather than a board. A personality that would have been a Z80-socket interposer in period (the ACT hard disk, the CoPower-88, the TRA video board) is therefore **not** excluded by occupancy: the interposer is already in that socket, so it can simply *be* the device.

The real constraint is **decode**. Two personalities cannot claim the same addresses at once, and the O1′s decode is famously loose — the "4 port boundary" the hard-disk controllers demand is the same coarse two-line decode the rest of the machine uses. So the arbitration question for each candidate is: *which addresses does the driver touch, and who else wants them?*

The map

Personality	Driver	Interface	Addresses it claims	ROM gate	Tier
`bus-analyzer`	—	Z80 socket	none (passive)	—	**0 — prerequisite for all**
`rt60a-rtc`	ABI documented (#7)	mailbox; **no IEEE handshake**	the `0x0040` block is *software's*, not ours	fine	**1**
`drive-c-gpib`	`DCL`/`DFD.SPR` ✓	IEEE-488 (P3), PIA `2900–2903`	GPIB **device address**	**not under OZROM**	**1**
`virtual-printer`	ROM IEEE driver ✓	IEEE-488 (P3)	**device 3**	Centronics survives OZROM	**1**
`floppy-emulator`	—	floppy bus	none (interrupts the Shugart bus)	—	**2**
`screenpac-video`	—	Z80 + UA15 tap	video window + `2400H` bit	—	**2**
`zimodem-fork`	—	P1 (TTL)	ACIA `2A00`/`2A01`	—	**2**
`occ1-harddisk`	`HARD`/`PORTCHNG` ✓	*virtual* Z80 interposer	a relocatable port window (#8)	?	**3**
`copower88-shim`	drivers + ROM ✓	*virtual* Z80 interposer	`0x7E`/`0x7F`	?	**3 — last**
`kvm-console`	composition	video + keyboard	none	—	**3**

The open question that decides the IEEE row

Three personalities want the IEEE-488 port: Drive C, the virtual printer and the RT-60A. **GPIB is an addressed bus** — the printer is *device 3* (OZROM's appendix: *"IEEE-488 as IOBYTE device 3 now null device"*) — so if Drive C and the RT-60A answer different addresses, all three can coexist, and the interposer can multiplex because it already sees the PIA at `2900–2903`. **That is a deliverable of #6, and it is the difference between "one personality at a time" and "coexist".**

What this section is not

It is not a decision record, and nothing above has been chosen. The tier ordering is a proposal for sequencing. The storage question in particular (#8) is still open: Drive C is the easiest to build **standalone** (#84), but that is a different claim from "the only one worth emulating".

Prior art: Z80 bus interposers

Input to the RP2350 Osborne 1 interposer (Rev A). Prior art for a Raspberry Pi Pico 2 W interposer between a 4 MHz, 5 V Z80 and its socket, with passive capture + active drive + USB/WiFi streaming. All claims cited; sources verified 2026-10-08.

---

1. RP2040/RP2350-based bus capture / CPU-emulation projects

1.1 gusmanb/logicanalyzer (24-ch, 100 Msps Pico logic analyzer)

- URL: https://github.com/gusmanb/logicanalyzer
- MCU: RP2040 (Pico / Pico W; Pico W supports WiFi streaming).
- Approach: PIO + DMA capture, up to 24 channels into RP2040 RAM; pre/post trigger; edge or pattern trigger; 100 Msps, 32K samples. Separate fast level-shifter board required for 5 V targets.
- Level shifting: dedicated shifter PCB (KiCad in repo). Documented Rev-A erratum: J1/J2 footprints exchanged (inputs<->outputs); recovered by flipping the symmetric board.
- Capture vs drive: capture only.
- Key lessons:
  - Trigger logic must live in PIO; naive GPIO-compare loops cost ~16 cycles/sample.
  - 100 Msps is burst-to-RAM via DMA, NOT continuous USB streaming; capture depth is the limit.
- Relevance: proves PIO samples >20 lines far above 4 MHz Z80 speeds; its Pico W WiFi mode is our streaming precedent.

1.2 AESilky/sd-dkr

- URL: https://github.com/AESilky/sd-dkr
- MCU: RP2040. Disk, keyboard, RTC as a device on a Z80-class parallel bus (device side, not interposer).
- Relevance: RP2040 meeting Z80 bus-peripheral timing.

1.3 lambdamikel/picoram-ultimate & picoram6116

- URLs: https://github.com/lambdamikel/picoram-ultimate, https://github.com/lambdamikel/picoram6116
- MCU: RP2040. SRAM emulator + SD interface for vintage SBCs (incl. 6116 SRAM emulation for the Z80-based Microprofessor MPF-1). Pico sits in the memory socket and drives the data bus synchronously with Z80 cycles.
- Key lesson: bare RP2040 meets ~4 MHz Z80 memory-cycle timing without glue logic; evidence RP2350 meets a Z80 T-state budget.

1.4 74HC138/RP-Floppy

- URL: https://github.com/74HC138/RP-Floppy
- MCU: RP2040. Bitbanged PIO floppy controller (early stage, GPL-3).
- Relevance: PIO MFM/flux handling reference for the floppy personality.

1.5 Flee-Time/FloppyController

- URL: https://github.com/Flee-Time/FloppyController
- MCU: RP2040 floppy controller speaking the Greaseweazle protocol, plus USB MSC. Second datapoint for PIO flux capture.

2. Prior art: interposers / bus monitors on other vintage machines

2.1 AtomBusMon (David Banks / hoglet67)

- URL: https://github.com/hoglet67/AtomBusMon (stardot.org.uk)
- Approach: "ICE" adapters piggybacking the 40-pin CPU socket (6502, Z80, 6809, 8080…); Z80 adapter uses 6x 74LVC4245 level shifters and turned-pin (Preci-Dip) headers; single-step, breakpoints, full bus trace.
- Errata accumulated across spins ("R10,R12,R13 should not be fitted", "C15 not fitted"); extra socket needed for stacking height.
- Key lesson: THE canonical socket-piggyback ICE design; its Z80 adapter BOM and shifter choice are directly copyable.

2.2 BusRaider (Peter Tandler / Rob Dobson r12a)

- URL: https://github.com/robdobsn/BusRaider
- Approach: Raspberry Pi Zero on the RC2014 (Z80) bus; can snoop and DRIVE the bus; `rawBusClockDisable` / `rawBusTake` model — stop the Z80 clock, then walk the bus at leisure.
- Key lessons:
  - Clock control converts impossible real-time problems into trivial ones. Add WAIT-stretch / clock-gate hooks to Rev A.
  - A debugging session found swapped PAGE_WR/PGEN_WR lines; a comparative trace against a known-good board would have caught it earlier — build golden-trace diff tooling, not just tracing.

2.3 RetroShield (8bitforce) and skx/z80retroshield

- URLs: https://github.com/8bit-1/retroshield (Arduino), https://github.com/skx/z80retroshield
- Approach: Arduino/Teensy drives a real Z80′s clock and buses, emulating memory/IO in software (CPU-in-a-jar). skx rewrote the

3. Electrical design lessons

- Level shifting observed in the wild: 74LVC4245 x6 (AtomBusMon), dedicated fast-shifter boards (logicanalyzer), 74LVC245-class parts recommended for inputs; 74HCT245 or dual-supply parts for outputs.
- RP2040/RP2350 GPIOs are NOT 5 V tolerant; do not rely on series-resistor hacks for a product-quality board.

- Verify whether the Osborne 1's Z80 is NMOS (VIH 2.0 V -> 3.3 V drive OK) or CMOS Z84C00 (VIH ~3.5 V -> needs HCT/dual-supply) before committing the output stage. [Check against machine photos in research/pictures.]
- Power: don't feed WiFi transmit peaks from the CPU socket 5 V pin without checking regulator headroom; prefer USB-powered Pico with common ground + generous local decoupling (AtomBusMon: 12x 100 nF
  - bulk).
- Mechanical: turned-pin headers; expect stacking-height problems inside the O1 case; keep profile minimal and test-fit early.

4. PIO sampling & streaming

- Clock-synchronous sampling (sample on the clock edge qualified by MREQ/IORQ) yields compact per-cycle records; oversampling is affordable (4 MHz bus, >=25x PIO headroom per logicanalyzer's 100 Msps / 24 ch).
- USB FS budget ~1 MB/s is insufficient for raw every-cycle capture at 4 MHz. Mitigations, in order:

  a) PIO-side filtering (M1-only, address windows, triggers) — trigger logic must live in PIO;
  b) RAM burst capture with pre/post trigger;
  c) WiFi streaming as a peer path on the Pico 2 W.
- Double-buffered DMA from day one.

5. Floppy & GPIB references

- FlashFloppy/Gotek (Keir Fraser) is STM32, NOT Pico — do not plan around porting it; instead track 74HC138/RP-Floppy and Flee-Time/FloppyController for PIO MFM, and Keir's Disk-Utilities (disk-analyse) on the host side.
- NODISKEMU (https://github.com/nils-eilers/NODISKEMU) and cbmSD (http://cbmsteve.ca/cbmsd/) are AVR-based; their IEEE-488 state machines are the reference to port to PIO for the Drive C personality.

6. Documented errata / "what I'd do differently"

- logicanalyzer: J1/J2 footprints swapped in first fab run.
- AtomBusMon Z80 adapter: BOM fit/no-fit errata across spins; stacking-height fix.
- skx/z80retroshield: rewrote upstream into a standalone library — separation of concerns matters.
- BusRaider: golden-trace diffing would have caught swapped lines.

7. Synthesis: recommendations for Rev A

1. Architecture: RP2350 BETWEEN Z80 and socket (AtomBusMon-style 40-pin DIP piggyback), real Z80 in a top socket so passive mode is electrically transparent. Two PIO banks: capture (clock-edge-sync push of addr/data/status) and drive (data bus + WAIT/NMI/INT/BUSRQ).
2. Level shifting: 74LVC245 inputs + 74LVC8T245/74HCT245 outputs, following AtomBusMon precedent.
3. Capture rate is a non-problem; spend the effort on PIO triggers.

4. Streaming: PIO filtering -> RAM burst -> USB/WiFi drain.
5. Active drive: WAIT-stretch and clock-gating hooks (BusRaider model) — being able to stop the clock is a superpower.
6. Power: USB-powered Pico, common ground, heavy decoupling.
7. Mechanical: turned-pin headers, minimal profile, early test-fit in the O1 case.
8. Process: publish BOM with fit/no-fit notes; expect >=1 footprint erratum; build trace-compare tooling early; keep the bus-driver firmware layer standalone.

upstream sample into a standalone library after finding mixed concerns (buttons/LCD/SPI-RAM tangled with bus driving). - Key lesson: keep the bus-driver firmware layer a clean standalone module; also proves ~MHz Z80s can be driven by MCUs.

Open questions

Last reviewed: 2026-10-11 (third pass), after #6 was finished — the Drive C transport and device contract are now measured and validated, and `docs/drive-c-design.md` (#84) is the build spec. Previous review 2026-10-11 (later pass), after #95 was answered from the OCC CBIOS source, #7′s clock ABI was written as a contract, #81′s OZROM index was built, and #6′s `DFD.SPR` blocker was re-characterised. Previous review 2026-10-11, after the *Portable Companion* extract batches (#73), the issue triage that re-scoped this doc's subject list, and the Phase-0 sub-issue tree (§F). Previous review 2026-10-10, after the archive drop (#71), the ROM-source indexing (#72) and the MAME work (#79).

This is the single place that says, for every open research question, **whether it is answered, by what evidence, and what is left** — cross-checked against the `research`-labelled issues. When the two disagree, one of them is wrong.

Status words (same vocabulary as `docs/README.md`):

Status	Means
Solved	Answered, and the answer is checkable from a doc
Partially	The shape is known; a *named* sub-question is not
Open	Not answered
Superseded	Stopped mattering — a better route was found, or a decision was taken

A gap is only **Solved** if the evidence is written down. "We know this" is not a status.

Summary

	Count
Solved	6 of the original 15
Partially	6
Open	1
Superseded	2

What actually blocks the next phase — everything else can wait:

1. **#39** — the identity of the three target machines (user action; blocks gap 12).
2. **Drive C's command set** — blocks #6/#22, and therefore the standalone Drive C reproduction (#84, `hardware/drive-c/`). *Updated 2026-10-11 (later):* **the transport and the device contract are solved** — `drive-c-protocol.md` is measured from the disassembly *and* a trace of the real software, and a model of the box makes the 1983 loader complete and PIP write to `C:`. What is left is (a) the **DPB** `DCL.COM` patches for drive `C:` (its size), and (b) the **`DFD.SPR` command layer**, whose committed listing is a linear sweep through zero blocks. The build spec is `docs/drive-c-design.md` (#84).
3. **The hard-disk controller decision (#8)** — two products are characterised; pick the one worth emulating before any personality code is written.
4. ~~**The RT-60A ABI (#7) — decide the contract, then implement it in #18.**~~ **Done 2026-10-11.** The ABI is written as a contract in `rt60a-analysis.md` (pointer block, sentinel/invalidation rules) and #18 implements it. The hardware handshake is retired as a known unknown.
5. ~~**Does the DD upgrade read foreign formats? (#95)**~~ **Solved 2026-10-11 from the OCC CBIOS source**: a DPB + sector-skew table (`occbio95.asm`) whose capacities match the manual's list exactly, selected by a monitor-ROM density probe. `dd-upgrade-formats.md`; Osborne SD confirmed end-to-end in MAME.
6. ~~**OZROM 1E as tooling (#81)**~~ **Solved 2026-10-11**: `ozrom-1e.md` indexes the manual and measures it on all three machines.

Everything else is either done, or is research that can proceed in parallel. **#79 is closed and no longer blocking**: the decision taken is not to chase the Nuevo path, and to build the video personality from the ScreenPac design plus OZROM's soft 52/80/104-column switch.

A. The original gaps

#	Gap	Status	Evidence	Issues	What's left
1	Exact memory & I/O map, free decode windows	**Solved**	`o1-memory-io-map.md` §2–§7 (schematic), confirmed from the ROM source in `o1-rom-source.md`, measured on the running machine by `tools/check_o1_map.sh`	#40, #62, #64, #72	Decode *granularity* is still schematic-level. Free gaps between the five device blocks: `2300`–`28FF`, `2B00`–`2BFF`, `2D00`–`2FFF`
2	NMOS vs CMOS Z80 across the target machines	**Superseded**	Resolved by history: the CMOS Z84C00 shipped in 1985, O1 production ended in 1983 — every O1 is NMOS. Interposer standardised on 74AHCT/HCT	#39	#39's remainder: machine IDs and date codes, for the record
3	Z80 socket **and** chargen socket pinout	**Solved**	Z80: datasheet, schematic sheet 3 of 9, `tools/check_schematics.py`. UA15: read at 1200 dpi off **sheet 4 of 9** (not the video sheet) — plain Intel 2716 pinout, `/CE+/OE` grounded,	#62, #24, #64	—
[illegible]	[illegible]	**Superseded**	[illegible]	[illegible]	*system*[illegible]O.IMD)
[illegible]	vid[illegible]devices	[illegible]	[illegible]	[illegible]	[illegible]

B. Questions that appeared since the original review

B1. Hard-disk products — two in the archive, and seven routes advertised in total

The original gap 9 assumed a single OCC1 hard disk. The archive drop (#71) turned up a second, unrelated one. Confirmed from their own strings:

	ACT hard disk	Media Distributing / Adaptec
Source	`research/harddisk/OCC1_HARDDISK.IMD`	`research/adaptec/*.img`
Vendor string	`AUSTRALIAN COMPUTER & TELECOMMUNICATIONS COPYRIGHT (C) 1982`	`Copyright (c) September 1983, Media Distributing`
Versions	`Version 31.05.82`, `Version 9-02-81`	`HARD version 1.93`, `PORTCHNG version 1.92`, `PREP version 1.93`
Controller	its own; `Enter disk controller address (`	Adaptec ACB-4000; `Enter starting port address in hex, must be on a 4 port boundary`
Utilities	`BOOTHD`, `LOADBIOS`, `HARDBIOS.HEX`, `MOVCPM10/5/F`, `DISKEDIT`, `DISKTEST`, `RESTORE`, `SAVEFILE`	`MD10.COM`, `MD20.COM`, `HARD.COM`, `HARDBIOS.SPR`, `HARDBDOS.SPR`, `HARDCCP.SPR`, `HD00/HD04BIOS.SPR`, `PREP.COM`, `PORTCHNG.COM`, `DRIVETBL.DAT`

So the three routes **represented in the archive** are the **ACT hard disk**, the **Media Distributing/Adaptec hard disk**, and **Drive C** (IEEE-488). Both hard disks use a *port-relocatable* controller — which matters, because the O1′s I/O decode is sloppy, so "4 port boundary" is the same two-address-line decode the rest of the machine uses.

Four more routes appear only in the magazine archive, with no software or media of ours behind them — recorded because a survey of "what storage existed" is incomplete without them, and because none is buildable from what we hold:

- **Corvus hard disks**, reached through a link program — `CLINK2`, built from `CLINK2K2.ASM` and assembled by the owner with `ASM.COM` and `LOAD.COM`. *The Portable Companion* #11 (October 1983), PDF p. 54 = printed p. 52, "**Doubling hard disk memory**" by Bob Gras (reprinted from the OSUG newsletter, Anchorage), is a patch for exactly that program, needed because DD leaves less memory for it (`docs/dd-upgrade-formats.md`). No Corvus software or media held.
- **Osmosis OSMOS 7**, advertised as "Winchester disks … 5¼, 10½, 15¾ & 21 MByte units - a 42 Megabyte max capacity", $2475–$3650 — on the same page as the OSMOS 5 80-column board,

which itself "**includes a Hard Disk interface for future plug-in expansion**" (`docs/o1-upgrade-boards.md`). No media held.

- **Trantor TSL-5 / TSL-10** — *The Portable Companion* #5 (February/March 1983), PDF p. 28: "unpack your disk, **plug it into the IEEE port**, run a simple 5 minute installation procedure", 5 MB $2995 / 10 MB $3395, supplied with BIOS, back-up and retrieve utilities and cable (`research/technical-articles/TPC-Issue05-1983-pp28-trantor-tsl-hard-disk-ad.pdf`). This is the line `docs/drive-c-protocol.md` records the Drive C vendor as reselling at a discount; the advertisement is what pins its interface to the **IEEE bus**. No media held.
- **Helix** (Micro Computers of New Orleans) — *The Portable Companion* #5, PDF p. 98, "**Winchesters Now Available For The Osborne 1**", 5/10/15 MB at $2500/$3000/$3500, a "**Z-80 adaptor** card [that] leaves external ports free" and a ribbon cable that "disconnects for portability" (`research/technical-articles/TPC-Issue05-1983-pp98-tra-80-100-column-video-and-winchester-ads.pdf`). No media held.

One of the two archived products now has a name. The Mega-byte Inc. advertisement in *The Portable Companion* #5 (PDF p. 25) sells the **MD-10** — 10 MB formatted for $2295, "20 and 44 megabyte Winchesters also available" — and its footnote reads *"MD-10 is a registered trademark of **Media Distributing**."* That is the same Media Distributing whose September-1983 disk sits in `research/adaptec/`, and that disk ships `MD10.COM`/`MD20.COM`, matching the 10 and 20 MB models. So the archive's "Media Distributing / Adaptec" entry **is the MD-10**, with an Adaptec ACB-4000 controller inside (`docs/occ1-harddisk-analysis.md`). The ad also fixes the *architecture*: the MD-10 "does not tie up existing OSBORNE ports" and is "significantly faster than other winchester subsystems **which use the IEEE port**" — i.e. it is **not** an IEEE-488 device, unlike Trantor, Drive C and the RT-60A. Taken with the ACT unit's Z80-socket interposer and Helix's "Z-80 adaptor card", that is a pattern worth recording for #84: **several vendors reached the Osborne 1's storage through the CPU socket or a port of their own, not through IEEE.**

So the honest count is **seven** advertised storage routes beyond the floppies — ACT, Media Distributing/Adaptec (**the MD-10**), Drive C, Corvus, OSMOS 7, Trantor TSL-5/TSL-10 and Helix — of which only **Drive C** is buildable from what this project holds (#84).

B2. Character-generator ROM variants — closed

##44 asked for the early-font and Nuevo char-gen variants. Both are captured and catalogued (`o1-rom-variants.md`): `5297c109` (early, paired with BIOS ≤ 1.4) and `6c1eab0d` (late, paired with 1.43/1.44 and used by ScreenPac and Nuevo alike). Open sub-question: nobody has rendered the two fonts side by side to confirm they actually differ.

B3. The Nuevo system disk does not boot under MAME — #79, closed

`OS1NUEVO.IMD` carries the Nuevo 80-column CBIOS 1.5, loads, prints its banner, and stops. The pristine upstream copy behaves identically, so it is not a bad conversion. The three plain DD system disks *do* boot the Nuevo machine, at 80 columns (`mame-emulation.md` §2, §4).

Decision: not pursued. Nothing in the Nuevo design has turned up a feature worth the validation cost, so the video personality (#24) is built from the **ScreenPac design plus OZROM's soft 52/80/104-column switch**, and the Nuevo board stays prior art (#73). The residual question —

MAME fidelity gap, or a physical-card dependency — is recorded in `mame-emulation.md` §4 and only matters if the Nuevo path is ever reopened.

B4. OZROM 1E — a third-party ROM usable as tooling — #81
It boots under MAME (via BIOS substitution) and its manual documents a software 52/80/104-column switch, a chip-level memory tester, a redefinable keyboard, and its own ROM jump table and memory map. It also **removes the IEEE-488 routines entirely** — the vectors at `013F–0156` now deliberately crash the machine, and *"IEEE-488 AS IOBYTE DEVICE 3 NOW NULL DEVICE"* — so **OZROM and Drive C are mutually exclusive** without reimplementing the driver. It also shrinks the video window to 3 KB (`F000–FBFF`) to make room for its keyboard features, which constrains any 2.0 ROM (#83).

B5. Other open research items

Item	Issue	State
Nuevo + Osmosis upgrade boards as prior art (two products each; neither matches OCC's implementation)	#73	**Video half written up**; the DD halves of both products, and how each lets software detect the board, are outstanding
Advent Products' **Osbaud** baud-rate generator (#21)	#21	The four connection points are decoded from the review + schematic (6850 pins 4/12, 74LS32 and 74LS04 pin 7); the internal crystal/divider topology is **not** known — needs the instruction sheet or a photo. `o1-modem-serial.md`
Archive provenance TODOs (2 items)	#74	**Closed**
`cpmtools` misreads Kaypro DSDD images	#35	`misc`. Upstream identified: the primary site is `moria.de` (Michael Haardt, 2.23, **email patch** — no GitHub, no tracker), and the practical PR venue is the unofficial `lipro-cpm4l/cpmtools` fork (3 open PRs, 2.21-based). Closes **when filed**, not when merged
G2 multi-format diskette read/write	#30	**Closed** — folded into #95
Case + power research (the LFF and SFF boards, and the O1NG enclosure)	#52	**Phase 5** (#49). A design task, not Phase-0 research
Contribute OZROM 1E to MAME (BIOS slot or an `osborne1oz` machine)	#82	`misc`. The **machine definition may be worth more than the dump**; a physical OZROM 1E is available to read, which would at least give honest provenance
A local Jekyll environment	#106	`misc`. `site/` has **no `Gemfile`**, so this is partly creating the manifest, not just resolving it
The complete Osborne 1 reference (the book)	#107	Unstarted — collating the OCR'd pieces tracked as #108–#115
[illegible]	168	[illegible] face with accents

C. The `research`-labelled issues, and what closes each

After the 2026-10-11 triage the label means **reverse engineering and documentation** again. Five issues left it: **#35** and **#82** (`misc` — upstream contributions), **#97** (`misc` — site and font), **#84** (a build, now `hardware/drive-c/`) and **#52** (Phase 5). **#83** was never research-labelled and has been removed from this table.

Issue	Title	State	What closes it
#1	Phase 0 umbrella	Open	The sub-issue tree (§F) — 15 of 23 closed
#6	Drive C ⟶ GPIB protocol spec	**Largely**	The **command layer** (`DFD.SPR`, a code-aware disassembly) and how the box resolves banks 0–2; plus the GPIB **device-address** question (can Drive C, the printer at device 3 and the RT-60A coexist on the one bus?)
#7	RT-60A ⟶ a time **ABI**, not the hardware	Partially	The contract written down, then implemented in #18
#8	Choose which hard-disk controller to emulate	Partially	A written recommendation — one personality, not two
#73	Nuevo + Osmosis prior art	Partially	The DD halves of both products, and the board-detection method
#75	The book	**Ungated**	The pieces in #108–#115
#81	OZROM as tooling	Partially	The column-switch mechanism; appendices C-5…C-7
#95	DD multi-format (now also the G2, absorbed from #30)	Partially	The MAME test matrix — read **and** write

Closed, but still carrying the label — the work that got Phase 0 to where it is: **#2, #3, #4, #5, #9, #10, #30, #44, #64, #70, #71, #72, #74, #79**. They are attached to #1 as sub-issues, so the parent's progress bar counts them.

D. Triage decisions taken 2026-10-11

This section used to be a "recommended close-outs" list. **It has been done** — ##3, #4, #5, #44, #64 and #72 are all closed — so it now records the *decisions* from the issue triage instead, because they are reached by discussion and are not re-derivable from a source:

- **#30 folded into #95.** Same question one product apart: the DD half is already answered in `dd-upgrade-formats.md` from the DD upgrade manual, and the G2 format table moves to that doc.
- **#35 and #82 stay open, as `misc`.** They are upstream contributions, not reverse engineering — and #35 is a genuine courtesy fix now that `cpmfs.py` gives us a known-good oracle to diff against.
- **#7 re-scoped to an ABI.** Implement the *documented* memory contract, not the *undocumented* IEEE handshake (see §A gap 8).
- **#8 re-scoped to a decision.** Which controller to emulate — one personality, not two.
- **#97 keeps the machine's own repertoire** — the graphics-set Unicode mapping and the recovered logo art. The modern font is split out as **#107** (`Osborne1ng.ttf`).
- **#84 is a build**, filed beside the other standalones (#47, #56) in `hardware/drive-c/`, not under the interposer.
- **#52 belongs to Phase 5** (#49).
- **The book is ungated.** #75 no longer waits on every research issue; it is a rolling build (#108–#115), with Part I and the appendices buildable today.

E. What is left before hardware design starts

Ordered by what blocks what:

1. **#39** — machine identities, board revisions and ROM markings (user action; unblocks gap 12 — the issue lists the three things actually worth reading).
2. ~~**Finish Drive C's command set** from `DCL`/`DFD.SPR`.~~ **Largely done 2026-10-11** — the protocol, the DPB and the capacity ceiling are in `drive-c-protocol.md`, and the **P3 pinout** came out of the Technical Manual already in the archive (`o1-ieee488-port.md` §6). What remains is a **code-aware `DFD.SPR` disassembly** to name the command layer, and the open question of how the box resolves banks 0–2 (`drive-c-protocol.md` §10).
3. **Decide the hard-disk controller** (#8): characterise both, pick one to emulate, and correct `occ1-harddisk-analysis.md`'s port claim.
4. ~~**Write the RT-60A ABI contract** (#7), then implement it in #18.~~ **Written 2026-10-11** (`rt60a-analysis.md`); #18 implements it.
5. **#73** — the DD halves of the Nuevo and Osmosis boards.

The book (#75) is **no longer gated on all of this**: it is a rolling build whose pieces are tracked as #108–#115, and Part I plus the appendices are buildable now. ##11/#12 (interposer design) have everything they need.

F. The issue tree (added 2026-10-11)

Phase 0 was an umbrella with **no children**, so "how close is Phase 0?" was unanswerable from the issue list. Every phase parent now carries a sub-issue tree, **closed issues included**, so each one renders a progress bar:

Phase parent	Children	Closed
#1 — Phase 0: research & reverse engineering	23	15
#11 — Phase 1: interposer hardware (+ the standalone adapters #47, #56, #84)	8	0
#16 — Phase 2: firmware personalities	11	1 (#86)
#25 — Phase 3: Osborne-side software	4	0
#29 — Phase 4: on-hardware validation	0	—
#49 — Phase 5: the reproduction	1 (#52)	0
#75 — the book	8 (#108–#115)	0

Two conventions came out of this, and both are worth keeping:

- **Closed work stays attached.** #2, #3, #4, #5, #9, #10, #30, #40, #44, #64, #70, #71, #72, #74 and #79 are children of #1. The progress bar is the point: Phase 0 is two-thirds done and should read that way.
- **`misc` is the label for work that is not a phase** — upstream contributions (#35, #82), the site (#97), build tooling (#106) and the modern font (#107). Everything else belongs to a phase, and to a parent.

The rule that keeps it honest: when an issue is opened, closed or re-scoped, the tree and this table move in the same PR — the same way a doc row does.

Appendix — decisions already taken

Kept here because they were reached by discussion and are not re-derivable from a source:

- **All Osborne 1s are NMOS.** The CMOS Z84C00 shipped in 1985; O1 production ended in 1983. The interposer uses 74AHCT/HCT; the replica uses a modern CMOS Z84C00 deliberately.
- **The interposer gets its own 5 VDC feed** near the Z80 socket (~300 mA+).
- **Bank switching cannot be latched.** `ROM MODE*` is re-derived on every instruction fetch; a bank-2 access is only meaningful while the CPU is actually in bank 2 (`o1-memory-io-map.md` §4).
- **The interposer must never drive I/O addresses `0x00–0x03`** — any `OUT` there toggles the bank latch. The ROM never does it, and there is no software bank switch to model.
- **The `52-`/`80-`/`104-` floppy prefix is a label, not a bootability gate.** The display width is the machine's; the text layout is the CBIOS's (`mame-emulation.md` §4).

Appendices

Archive provenance

This directory is the project's **research archive**: the documentation, disk images, and photographs that the reverse-engineering and design work under `docs/` and the GitHub issues depend on.

These files are reserved as safety copies of the documentation necessary to complete this project. They are committed in-repo on purpose: upstream archives move, go offline, or drop files, and this project cannot finish without these specific documents. If an upstream link dies, the in-repo copy is authoritative. Please do not "clean up" large PDFs or disk images from this tree — their presence here is the point.

Where each item was sourced is recorded below, so provenance is preserved alongside the copies.

OCR convention: most of the scans here have no text layer. When a document is OCR'd for research, the text is committed **adjacent to the PDF** as `<filename>.ocr.txt` (e.g. `drive_c/Drive_C_Users_Manual.ocr.txt`) so it never has to be regenerated. Regenerate with `tools/ocr_pdf.sh` if a better pass is needed.

osborne1/ — Osborne 1 mainboard documentation

File	What it is	Sourced from
`2F00153-01_Osborne1TechnicalManual_1982.pdf` (19 MB)	Osborne 1 Technical Manual, 1982. No text layer — image-only scan. Theory of operations, banking, I/O, video, disk, keyboard.	bitsavers
`OCC1_1A2011-00_Schem_RevE.pdf` (1.2 MB)	**Two drawing sets in one 12-page scan.** Pages 1–3: DISK ELECTRONICS, DWG 1A3004-00/01 Rev C (sheets 1–3 of 3 — the *drive's* own board). Pages 4–12: MAIN PC BOARD, DWG 1A2011-00 Rev E (sheets 1–9 of 9). 600 dpi bilevel; the embedded text layer is a useless early-90s OCR pass. Indexed, and the Z80 socket and P8 floppy pinouts extracted, in `docs/o1-mainboard-schematic.md` (#62).	high-value schematic mirrored as `Osborne_1_Schematics_1A2011-00_Rev_E.pdf` on RetroTechCollection/Internet Archive)
`OCC1_1A2011-00_Schem_RevE.ocr.txt`	OCR text layer for the above (per the convention below), generated 2026-10-10 at 400 dpi. Good enough to grep for component references; **not** reliable for pin-level detail — read sheets visually with `tools/schematic_render.sh`.	—
`keyboard/osborne_keyboard_pinout.html` + `keyboard/PROVENANCE.md`	Verbatim capture of the pinouts.ru Osborne 1 keyboard page: P4 pinout (incl. pin 19 = +12V via J6+R21) and the full key matrix map. Used by #45/#47.	pinouts.ru, captured 2026-10-08 (provenance in the folder)
`floppy-adapter/` (README, LICENSE, `.sch`, `.kicad_pcb`, PROVENANCE.md)	Snapshot of WayneVisser's Osborne1FloppyAdapter KiCad project — passive Gotek adapter documenting the Osborne 1 "Shugart-like" floppy interface (P3 over the cable's unused pins). Used by #62/#63.	github.com/WayneVisser/Osborne1FloppyAdapter, captured 2026-10-09 (provenance in the folder)
`o1doublodensitytechmanual.pdf` (2.2 MB, 15 pp)	*Chapter 4 "Double Density Theory of Operation"* from the Osborne 1 Technical Manual, the §4-level source for the double-density MODEM (P1) and serial (P2) interfaces.	kev.pulo.com.au/osborne1, supplied by maintainer 2026-10-09
`o1doublodensitytechmanual-interfaces.pdf` (1 MB, 10 pp)	*Appendix D "Osborne 1 User Interfaces"* — pinouts for the IEEE-488 (P3), serial RS-232 (P2), and MODEM (P1) interfaces.	kev.pulo.com.au/osborne1, supplied by maintainer 2026-10-10

Basis for `docs/o1-memory-io-map.md` (#40) and `docs/o1-mainboard-schematic.md` (#62). Both scans are image-only; extraction is manual — OCR for text pages, and for schematics a native-resolution region render (`tools/schematic_render.sh`) read by eye. The mainboard map is independently cross-checked against the running machine in MAME; see `docs/o1-mainboard-schematic.md` §7 (#64).

inspired-by/ — adjacent projects

`inspired-by/README.md` — other Osborne 1 reinventions (MAME cyberdeck, Raspberry Pi brain transplant, Loxley's Gotek mod): what they did, and what we take from each. Contrast defines the project: **we keep the original machine intact**.

mx80/ — Epson MX-80 printer references

File	What it is	Sourced from
`mx80/Epson_MX-80_Operations_Manual.pdf`	Epson MX-80 operation manual; Appendix 4 (PDF p. 97+) has the character-font tables (6×9 matrix) and ESC code reference.	Internet Archive / manualsbase, captured 2026-10-09
`mx80/EPSON_MX-80_Fonts_v1.0.zip` + `mx80/fonts/`	Michael Walden's MX-80 font pack (CC BY-NC-SA 4.0). Convenience cross-check; see the folder README for the double-strike/2×-width caveat.	mw.rat.bz/MX-80, captured 2026-10-09

Also: `drive_c/Drive_C_Users_Manual.ocr.txt` — OCR text layer for the image-only Drive C manual scan (generated 2026-10-09 per the OCR convention above; source of the TurboPac details used by `firmware/drive-c-gpib/`).

References for the virtual Epson MX-80 printer personality (#19).

screenpac/ — OCC ScreenPac (80- and 104-column upgrade)

File	What it is	Sourced from
`2F00040_service_2ndEd_1983.pdf` (12 MB)	Osborne Field Service Manual, 2nd Edition, 1983. Covers the ScreenPac upgrade (Z80 + char-gen socket taps, dual-port video RAM). §6.12 documents the ScreenPac IC-harness fly-lead solder connections (used by `firmware/screenpac-video/`).	bitsavers
`2F00040_service_2ndEd_1983.ocr.txt`	OCR text layer (per the convention above), generated 2026-10-09	—
`o1screenpacusersguide.pdf` (2.3 MB, 20 pp)	*Osborne 1 SCREEN-PAC User Guide*. **The primary source for what the upgrade does**, and it is more than a wider window: p. 13 (PDF p. 17) gives the three software-selectable widths (52 / 80 / 104), horizontal scrolling in **32-column blocks**, and the **presence flag at `2400H` bank 2** that lets software detect the board. Quoted in `docs/mame-emulation.md` §2 and `docs/o1-memory-io-map.md` §2a/§3 (#73, #24).	kev.pulo.com.au/osborne1, supplied by the maintainer 2026-10-10

The ScreenPac is the mechanical/electrical template for the interposer and for the video-output (HDMI) personality (#24, #9 photos).

technical-articles/ — extracts from Osborne-era magazines

Only the pages that matter are kept, cut from the original and OCR'd. The full issues are not in the repo: they are 50–60 MB scans each, they are on the Internet Archive (https://archive.org/details/theportablecompanion) and on the maintainer's own backup, and keeping whole magazines for a few pages of schematics is not worth the repo weight. Each extract names the issue and the pages it was cut from, so it can be checked against the original.

Extract	What it is	Cut from
`TPC-Issue11-1983-pp32-osmosis-product-line-ad.pdf` (0.6 MB, 1 p)	OSMOSIS' full-page advertisement, "The Next Step Forward" — the whole product line **with product numbers**: **OSMOS 1** DD 184k/drive ($195), **1.5** ($335), **2** DS/DD 370k ($895), **3** DS/DD double-tracking 750k/1.5MB ($1050), **4** "read, write and format **over 20 other manufactures CP/M formats**" ($225), **5** the internal 80-column board ($250 — "**includes a Hard Disk interface for future plug-in expansion**", normal/inverted video, underline, ⅔ and ½ intensity), **6** telex, **7** Winchester to 42 MB, **8** instrumentation, **10** Olivetti ET121 interface; bundle OSMOS 1+4+5 for $385. Quoted in `docs/o1-upgrade-boards.md` and `docs/dd-upgrade-formats.md`. Relevant to #24, #11/#12 and #95/#30.	*The Portable Companion* #11, October 1983 — PDF p. 32 (printed 30)
`TPC-Issue11-1983-pp54-doubling-hard-disk-memory.pdf` (0.7 MB, 1 p)	"**Doubling hard disk memory**" by **Bob Gras** (reprinted from the OSUG newsletter, Anchorage) — patching Corvus's `CLINK2K2.ASM` so the link program works after a double-density upgrade: `DELTA EQU 500H` → `900H`, because "**more memory is required for managing the Osborne floppy drives**" under DD. The same page carries the tail of John Dilday's WordStar printer-spooler article.	*The Portable Companion* #11, October 1983 — PDF p. 54 (printed 52)
`TPC-Issue11-1983-pp41-45-[illegible].pdf` (0.9 MB, 5 p)	[illegible]	*The Portable Companion* #11, October 1983 — PDF pp. 41–45 (printed 39–43)

How these extracts are made

The maintainer reads the issues and hands over the ones worth keeping, giving the **PDF page sequence** to cut. The standing workflow:

1. Cut the pages: `pdfseparate -f A -l B ISSUE.pdf /tmp/p%03d.pdf` then `pdfunite /tmp/p*.pdf OUT.pdf`.
2. OCR the extract with `tools/ocr_pdf.sh OUT.pdf` — adjacent, per the convention above. **Wait for its `Wrote <path>` line before committing**; the file exists while it is still being written.
   - **If you drive `tesseract` by hand, do not render into `/tmp`.** On this machine it cannot open images under `/tmp` (Leptonica reports the file as not found) and, with `2>/dev/null`, fails *silently* — a 43-page sweep over `/tmp` returned nothing at all and looked like "no matches". The script is fine because it renders into a `mktemp -d`.
   - **A display-type page can defeat tesseract entirely.** Full-page advertisements set in display faces over tinted screens sometimes OCR to nothing but the folio — the #5 Mega-byte MD-10 ad (`TPC-Issue05-1983-pp25-…`) was the first. Do **not** commit an empty `.ocr.txt`: transcribe the page by eye into the file behind a `[HAND TRANSCRIPTION, not tesseract output …]` header, so the text still lives beside the scan and never has to be re-derived.
3. Add a row below, and write the findings into the `docs/` page they bear on with the extract cited.

Two things to know before cutting pages:

- **Page numbers are PDF pages, not printed pages.** Every page checked so far prints its own number two lower (`PDF = printed + 2`) — issues 11, 13, 14, 15, and now #5 p. 25 (⟶ 23) and #7 pp. 50 and 56 (⟶ 48 and 54). So an article "on page 26" is PDF page 26 and printed page 24 — and cutting printed 26 would miss the article's title page. **The offset is not uniform, though:** the #5 extract filed as `pp98` carries folio **84**, i.e. `+14`, so an issue can contain unnumbered pages. Read the folio off the page rather than assuming — `pdftotext -f N -l N -layout ISSUE.pdf - | tail -3` prints it, and the extract's own `.ocr.txt` usually ends with it.
- **Provenance for every article from *The Portable Companion* is the same collection**: https://archive.org/details/theportablecompanion. Rows therefore cite the issue and the page range rather than a per-file URL.

The **full issues are not kept in the repo** — 50–60 MB each, and they live on the Internet Archive and on the maintainer's own backup. Only the extracts are committed, and they are cut from files outside the repo (the maintainer's `~/Documents` tree), never from a copy inside it.

copower88/ — SWP CoPower-88 (Z80-coprocessor board being reverse-engineered)

Path	What it is	Sourced from
`zorba/Co-Power-88guide.pdf`, `zorba/Co-Power-88advert.pdf`, `zorba/611-0003-1..4.pdf`	SWP CoPower-88 user guide, advertisement, and SWP technical document scans (Zorba-flavored set)	zorba.z80.de — the Zorba Portable Computer archive site
`zorba/*.td0`, `kaypro/*.td0`	TeleDisk images of CoPower-88 CP/M-86 / DOS boot disks (Zorba and Kaypro versions)	zorba.z80.de (exact path not recorded in repo history)
`zorba/extracted/`, `kaypro/extracted_cpm/`, `*/disasm/`	CP/M files extracted from those images by the project pipeline (#2), and disassemblies of the CoPower RAM-disk/DOS drivers	Derived in-project — not external sources
`swp/Copower_User_guide.pdf`, `swp/Swp_ad.jpg`	Fuller CoPower-88 user guide (4.7 MB, not the Zorba set) and an SWP advert scan	kayprojournal.com `Swp_copower_88` — `File:Copower User guide.pdf`, `File:Swp ad.jpg`
`swp/1600px-73356_8088_card.jpg`, `swp/1600px-73356_mem_card.jpg`, `swp/866px-73356_cpu_card.jpg`	Board photographs of the CoPower-88 8088 card, memory card and CPU card — held as MediaWiki thumbnails (1600 px / 866 px) of `File:73356 8088 card.jpg`, `File:73356 mem card.jpg`, `File:73356 cpu card.jpg`	Same page: kayprojournal.com `Swp_copower_88`

The whole `swp/` set is from that one page. Confirmed 2026-10-10 by querying the wiki's API (`list=allimages` and `prop=images&titles=Swp copower 88`) and matching all five `File:` names exactly — not by reading the rendered page text. The wiki also carries `File:Copower sch 1..4.pdf`, a second copy of the SWP schematics we hold from zorba.z80.de.

Basis for #3/#4/#5 (CoPower-88 protocol analysis). The board's own 8088 ROM is `roms/swp-p88.rom` — see `roms/` below and #70.

drive_c/ — Drive C (IEEE-488 ramdisk, #6)

Path	What it is	Sourced from
`Drive_C_Users_Manual.pdf`	Drive C user manual (IEEE-488 virtual disk drive), **second edition June 1984**	bitsavers `bits/Osborne/Osborne1/Drive_C/` (upstream name `Drive_C_-_Users_Manual.pdf`)
`Drive_C_Executive_Users_Manual.pdf`	**Drive C user manual for Osborne Executive computers** — the same product on the other host: identical hardware appendix, `ECL`/`ECN`/`ECU` software under CP/M+ instead of `DCL`/`DCN`/`DCU` under CP/M 2.2, and **no `DFD.SPR`**. First edition July 1984, second printing 9/84; cover stamped PRELIMINARY, running headers "final draft". 111 pp, 12 MB, image-only (150 dpi grayscale re-encode — see the note below). Analysis: `docs/drive-c-executive.md` (#84, #6, #145)	**Jim Rayburn**, private Facebook group, posted 2024-01-06 — posts …/2063148960719887 and …/2023247321376718; supplied to the maintainer as `Executive_DriveC.pdf`, 2026-10-11
`Drive_C_Executive_Users_Manual.ocr.txt`	OCR text layer for the above — 300 dpi `tesseract`, per the convention below	Derived in-project, 2026-10-11
`DRIVE_C.IMD`	ImageDisk image of the Drive C utility disk (label comment `Drive C for the Osborne 1 computer`)	same folder — **not** imaged in-project; corrected 2026-10-10
`Drive_C_disk_label.jpg`	Photo of the physical disk label	same folder (15 044 B, byte-identical) — **not** a project photo; corrected 2026-10-10
`extracted/`	Files extracted from `DRIVE_C.IMD` by the project pipeline (#2)	Derived in-project
`disasm/`	`dcl.asm` + `dfd.asm` — z80dasm listings of the installer and the resident driver, for the protocol work (#6). Regenerate with the exact command in `docs/drive-c-protocol.md` §Disassembling	Derived in-project

A note on the Executive manual's size. The owner supplied a 300 dpi scan, 42.33 MB — larger than any other single file in this tree. It is committed as a **150 dpi re-encode** instead (12.13 MB), the same resolution as the Drive C manual above: the document is a typewritten mono manual, and the page images carry **no colour at all** — every pixel is neutral grey, checked across all 111 pages — so the extra resolution bought nothing but file size. All **111 pages** and the page geometry are preserved. The **300 dpi text layer** was taken from the owner's original before the re-encode and sits beside the PDF as `Drive_C_Executive_Users_Manual.ocr.txt`, so no text was lost. It is recorded as a re-encode rather than as the owner's original, so the provenance stays honest: the file here is **not** byte-for-byte what Jim Rayburn posted.

rtc/ — RT-60A real-time clock (#7)

Path	What it is	Sourced from
`RT-60A_Manual.pdf`	RT-60A RTC manual (JG Communications)	bitsavers `bits/Osborne/Osborne1/Osborne_1_Real_Time_Clock/` (upstream name `RT-60A_Real_Time_Clock_Manual_-_Osborne_1.pdf`)
`RT-60A.TD0`	TeleDisk image of the RT-60A utility disk — **a damaged dump** (27 no-id sectors); filesystem extraction fails, see `tools/README.md`	same folder

harddisk/ — ACT/OCC1 hard disk (#8)

Path	What it is	Sourced from
`OCC1_HARDDISK.IMD`	ImageDisk image of the OCC1 hard-disk system disk (label comment `act-osborne / hard disk software / rev 2`)	bitsavers `bits/Osborne/Osborne1/_floppy_images/` — **not** imaged in-project; corrected 2026-10-10
`extracted/`	Files extracted from the image, incl. `hardbios.hex`	Derived in-project

adaptec/ — Adaptec ACB-4000 hard-disk software (2026-10)

Raw Osborne 1 SSSD floppy images (102 400 B = 40 tracks × 10 sectors × 256 B, **no IMD header**) each with a photo of the disk's label. Source: Internet Archive, `md-utilities-adaptec-4000-controller-ver.-1.93-osborne`.

File	What it is
`MD Utilities Adaptec 4000 Controller Ver. 1.93 Osborne.img`	Adaptec ACB-4000 utilities
`MD Utilities Adaptec 4000 Controller Ver. 1.93 Osborne - MD-10 MD-20.img`	Same, for the MD-10/MD-20 drive pair
`Osborne Hard Disk Control Software Version 2.01 (1984).img`	Osborne's own hard-disk control software

Together with `acb4000a.rom` (Maslin's archive) this identifies the controller behind `harddisk/OCC1_HARDDISK.IMD` — issue #8.

diagnostics/ — Osborne 1 confidence tests (2026-10)

File	Label comment
`DIAG_2.04.IMD`	`2.04 / diagnostics`
`OCC1_DIAG_2.1.IMD`	`occ diag / ver 2.1 / 12/31/84`

Source: bitsavers `bits/Osborne/Osborne1/_floppy_images/`. These are the known-good boot images the validation matrix (#29) needs.

nuevo/ — Nuevo Electronics double-density upgrade (2026-10)

File	What it is	Sourced from
`Nuevo_Electronics_DD_Upgrade_Manual.pdf` (13 MB)	DD upgrade manual	bitsavers `bits/Osborne/Osborne1/` (13 810 755 B, exact match)
`Nuevo_Electronics_DD_Upgrade_Manual.ocr.txt` (919 lines)	OCR of the manual — the BIOS 1.4⟶1.5 difference list (DPBs for IBM/Xerox/DEC removed, 80-column presence check, jump table at `E100h`), the 380K double-sided claim and the single/double-density system-track rules, all quoted in `docs/dd-upgrade-formats.md`	derived in-project with `tools/ocr_pdf.sh … 300`
`OS1NUEVO.IMD`	`Osborne 1 with DD and 80 col mods` boot disk, reimaged 2026-10-10 with DiskUtilities	derived in-project
`photos/nuevo-80-column-board.jpg`	**80-column board, `REV. D`, serial 4348** — the first hardware view of the product. Its own **NEC D780C-1** (Z80), a Nuevo **`SY6545A-1`** CRTC, 2 × `SRM2016C25` SRAM, a socketed **`CHARACTER GENERATOR`** ROM dated **6-29-84**, a 12.288 MHz oscillator, and the optional `BAUD` generator. Devices, and what they imply, in `docs/o1-upgrade-boards.md` §"The Nuevo boards themselves" (#73, #24)	**Mike Loewen**, private Facebook group, 2024-10-14 — post
`photos/nuevo-double-density-board.jpg`	**DD board, serial 20011** — `DAL DENS` silk, a 34-pin IDC header with a `STD DRIVE` jumper, and a discrete write-precompensation/separator chain (`74LS195A`, `SN74LS14N` ×2, `SN74LS00N` ×2, `SN74LS02N`) plus a 16-pin 9216-family device (U1). Same analysis as above	same post
`photos/nuevo-80-column-and-double-density-boards.jpg`	Both boards in one frame — the shot the 2024-10-14 post is built around	same post

The photographs are **third-party** (Mike Loewen, a private group) rather than in-project like `pictures/`, and they are the only hardware views of either Nuevo product — there is no Nuevo schematic in the archive. Neither board bears on Drive C: both are host-side (video, and floppy encoding).

> **Removed 2026-10-10: `NUEV0151.BIN`.** It was a byte-identical duplicate of `tools/emulator-setup/roms/monrom-rev1.51-12.ud11` — same 4096 bytes, SHA-256 `aec8f5be127fdff5d30cae0b7f45ee48d13a8c1d5745a262b944b9ba67778906`, CRC32 `298da402` — verified with `cmp` before deletion. It carried no provenance of its own, so the tracked emulator-set copy, which has a recorded source, is the one we keep. The BIOS strings are unchanged by this: `COPYRIGHT 1983, OSBORNE COMPUTER CORP. / COPYRIGHT 1984, NUEVO ELECTRONICS CORP.`

Nuevo shipped **two** products — a DD upgrade and an 80-column board — and neither matches Osborne's own implementation of that function; see #73.

`OS1NUEVO.IMD` **is** the Nuevo 80-column system disk: its system area carries `CBIOS vers 1.5` and the text *"THIS IS AN EIGHTY (80) COLUMN DISPLAY FOR THE OSBORNE ONE"*. It does **not** currently boot under MAME — it prints the CBIOS banner and stops (measured over 260 emulated seconds; the pristine upstream copy behaves the same), which blocks validating the Nuevo video path in the emulator. See #79 and `docs/mame-emulation.md` §4.

osmosis/ — Osmosis Computer upgrades (2026-10)

Source: bitsavers `bits/Osborne/Osborne1/Osmosis_Upgrades/` — all seven files byte-for-byte identical.

`Osmosis_CPM_Disk_Emulator.pdf`, `Osmosis_80_Column_Board.pdf`, `Osmosis_SSDD_Upgrade.pdf`, `OSMO-EMU.TD0`, `OSMOS-DD.TD0`, plus two disk-label photos. Osmosis likewise shipped two products (a CPM disk emulator and an 80-column board). **The 1983 CPM Disk Emulation System is direct prior art for this project** — see #73 and `inspired-by/`.

gotek/FF/ — working Gotek configuration (2026-10)

`FF.CFG` + `IMG.CFG` — the project's own Gotek configuration for the O1: `interface = shugart`, `pin02`/`pin34` = `nc`, and an `IMG.CFG` default of 40 cylinders × 1 head × 16 × 128 B, FM, 125 kbit/s — i.e. O1 SSSD. Captured from the physical Gotek, so these are primary sources, not downloads. Used by #62 and #20.

roms/ — ROM images (2026-10)

Canonical inventory, CRC32s and the third-party ROM survey are tracked in #70. Files here come from two sources:

- `os1-143.rom`, `os1-144.rom`, `os1-vid8.rom`, `os1vid80.rom`, `swp-p88.rom` — Don Maslin's ROM archive on retroarchive.org
- `OZROM-1E/OZROM_1E.BIN`, `OZROM-1E/OZROM_1E_Manual.pdf`, `OCC1_ScreenPac_RevA.BIN` — github.com/BrettHallen/Osborne_1 `ROM/`

`swp-p88.rom` is **8088 code, not Z80**: the SWP co-processor board's own ROM. The board and interposer card are common to every host machine (only the driver varies), so this is the 8088-side firmware of the CoPower-88 family — #3/#4/#5.

`OZROM-1E/OZROM_1E_Manual.ocr.txt` — OCR text layer for the OZROM manual (image-only scan; generated 2026-10-10 at 300 dpi with `tesseract`, per the OCR convention above). Good enough to grep for the technical appendices — which is how the IEEE-488 removal and the 3 KB video window were found (#81) — but not reliable for exact code or pin detail.

`OZROM-1E/OZROM_1E_Utilities.IMD` — the **OZROM 1E Utilities disk**, the software half of the OZROM package, which we did not previously have. ImageDisk 1.19, label `ozrom`; 40 cyl × 10 × 256 B FM (O1 single density), 22 of 40 tracks in use. Its CP/M directory is complete and readable: 20 files, all Micro Management 1984, with `READ.ME`, `CRCKLIST.CRC` and the docs for `TIMER`, `CLKPATCH`, `DVORAK` and `TD`. **Condition: partial.** 290 of the 400 sectors read back as uniform `0xE5` and the gaps fall *inside* file bodies, so most files are truncated or holed; the directory and a minority of complete files (`SET.COM`, `MEMORIZE.COM`, `FK.COM`) are the recoverable part. It is a data disk, not a system disk — the boot tracks carry a stub whose message is "There is no CP/M system on this disk." Provenance: supplied by the project owner as a local file (2026-10-11); the original source URL is not yet recorded, so treat the chain of custody as open. Analysis: `docs/ozrom-1e.md` §10. Used by ##81.

shipped-software/ — the shipped application library (2026-10)
Fourteen ImageDisk images of the disks that shipped with (or were published for) the Osborne 1. The ImageDisk label comment is quoted because it is the disk's own identity. Source: **Dave Dunfield's ImageDisk archive** (`dunfield.classiccmp.org`, which rotates its directory to discourage deep links) — `d/osborne1.zip` and `d/o1ddsys.zip`, byte-identical to these copies.

File	Label comment
O1CPM.IMD	Osborne-1 CP/M System Disk / Double Density
O1CBMB.IMD	Osborne 1 - CBASIC/MBASIC / Double Density
O1SCALC.IMD	Osborne 1 - SuperCalc 1.12 / Double Density
O1WSMM.IMD	Osborne 1 - WordStar/MailMerge 2.26 / Double Density
OS1SYSD.IMD	CP/M-2.2 system disk for Osborne 1 w/ DD drives / SSDD 1024 byte sector
OS1SYSS.IMD	CP/M 2.2 System Disk for Osborne 1 / SSSD 256 byte sector
OS1BASIC.IMD	CP/M 2.2 Basic Disk for Osborne 1
OS1DBASE.IMD	dBase II for the Osborne 1 on sysgened disk
OS1DIAS.IMD	CP/M 2.2 Diagnostics Disk for Osborne 1
OS1MCAL.IMD	CP/M 2.2 Mocro-Call Disk for Osborne 1 (Communications)
OS1MDM7.IMD	Modem 740 communications for Osborne 1
OS1UTLS.IMD	CP/M 2.2 Utilities Disk for Osborne 1
OS1XUTLS.IMD	Extended utilities for Osborne 1 w/ 80 column video card
OS1WRDST.IMD	CP/M 2.2 WordStar 3.0 Disk for Osborne 1

The corpus for the image library (#20), the updated boot disk (#28) and the validation matrix (#29). Dunfield also publishes a 2005 single-density set (`O1CPMS.IMD`, `O1CPMU.IMD`, …) which is deliberately **not** copied here.

source-code/ — Osborne's own firmware source (2026-10)

Path	What it is	Sourced from
`BIOS-144-source/` (18 files)	**The original Osborne ROM source**: `rom144.asm` (DOUBLE DENSITY ROM REV 1.44, Roger W. Chapman, 2/4/1983), `occbio05..95.asm` (OCC CBIOS Rev 1.41), `occram15/25.asm`, `occtxt6.ast` (ACT80 assembler equates), `bios141.gen`, `release.txt`, `tran.txt`. Basis for #72.	github.com/BrettHallen/Osborne_1 `ROM/Source_Code_v144/` (all 18 file sizes match)
`OCCCBIOS.IMD`, `OCCROM.IMD`, `OCCUTIL.IMD`, `OSBIOSSR.IMD`, `OSBROM.IMD`, `OSROM13.IMD`, `OSUTLSRC.IMD`	OCC engineering **source disks** (1981–82): CBIOS, ROM rev B, utility programs, ROM 1.3. The ImageDisk comments carry the OCC part numbers (`4d2007-00 occbi00a.asm, rev b 9/12/81`).	bitsavers

Licensing: `release.txt` is Osborne's 1985 letter permitting the First Osborne Group to reprint the schematics and the BIOS listing, on condition that any money goes to the non-profit FOG and that the trademark/copyright notices are retained. FOG is long defunct and this material is already public in several places — this copy came from a public GitHub repository. The notice is retained verbatim; if the terms ever need re-examination, that is the file.

pictures/ — machine photographs

`IMG_4856.jpeg` … `IMG_4871.jpeg` — disassembly/mainboard/upgrade-board photos of the project's own machines (mainboard, double-density upgrade, ScreenPac, CoPower-88). Taken in-project for #9; these are primary sources, not copies. Further photo sets are tracked in #39.

Provenance

The two long-standing **provenance gaps are closed**: the Drive C manual and the RT-60A manual/TD0 both come from bitsavers' `bits/Osborne/Osborne1/` (`Drive_C/` and `Osborne_1_Real_Time_Clock/`), as does `OCC1_HARDDISK.IMD` and the two diagnostic disks (`_floppy_images/`). Four rows that had been recorded as in-project work or as unknown URLs were corrected on 2026-10-10.

Both remaining TODOs are closed (2026-10-10, #74):

- `copower88/swp/` — kayprojournal.com's `Swp copower 88` page; all five files matched by `File:` name through the wiki API (CoPower-88 table above).
- `nuevo/NUEVO151.BIN` — removed as a byte-identical duplicate of the tracked `tools/emulator-setup/roms/monrom-rev1.51-12.ud11` (nuevo/ table above).

No provenance TODOs remain.

The machines: a photographic survey

Status: 14/16 examined (IMG_4856–4869). IMG_4870/4871 could not be processed by the vision pipeline (tool rejects even after EXIF strip/ re-encode) — user to describe or retake.

CoPower-88 8088 board (IMG_4856–4867)

- Silkscreen: **"KAYPRO-88"**, "(C) SWP 1983", "Lic. SWP Inc.", "COPOWER-88 REV 1.8". ⟹ The user's board is the **Kaypro variant**, consistent with software default port pair **0x7E (data) / 0x7F (status)**.
- CPU: **Fujitsu MBL8088** (NMOS 8088, ceramic DIP, 8319).
- Clock: CRYSTEK 16.000 MHz oscillator can ⟶ 8284 divides by 3 ⟶ **5.33 MHz 8088**.
- Boot ROM: 2732A labelled "SWP-8088" (the 8088-side monitor — the firmware that answers the host mailbox; a dump of this ROM would fully specify the protocol. Consider reading it when hardware is accessible — candidate for the reader hardware in #14).
- RAM daughterboard: 16 x **Mostek MK4564N-25** (64Kx1, 128 KB) soldered via SIP-style carrier above the main board.
- Glue visible matches 611-0003: 74LS138 x3 (RAS/CAS + port decode), 74LS244, 74LS373/375, 74LS645, 74LS257, 74LS175, 74LS153, 74LS32, MC74HC374 x2, 74LS161/393/174 etc.
- **Blue DIP-package IC** marked "16-2103 8140" — per user correction this is a **chip, not a DIP switch**. Almost certainly an 8-bit magnitude comparator (74LS688-class) doing the A1-A7 port-base compare from sheet 4. ⟹ The host port base is **fixed by this device / its hardwired reference, not user-switchable**. Consistent with the Kaypro software default 0x7E/0x7F. TODO: confirm the exact part number + how the reference nibble is set.
- Host ribbon: IDC at board edge (to Z80 socket adapter), plus a second 16-pin(?) white Berg-type connector (matches "BERG PINS" RDX read-strobe / expansion note on sheet 4).
- Solder side (4866): single blue bodge wire, otherwise clean.

Osborne mainboard + Z80 adapter (IMG_4868, IMG_4869)

- Mainboard marked **"24187A"** (handwritten/silk), assorted 74LS, socketed 6116 x2 (video RAM), WD1793? area partially visible.
- **The Z80 lives on the CoPower-88 interposer adapter**: a small PCB plugged into the mainboard's Z80 socket carrying
  - a plastic **Zilog Z80A, date code 8408 ⟹ NMOS** (a CMOS Z84C00 would be marked differently; also 1984 predates common CMOS Z80s)
  - 74HCT-series glue (74HCT04 visible) and pads labelled **A4 A5 A6 A7** — address lines broken out for the ribbon to the 8088 board's port decode (sheet 4 A1-A7 compare feeds the blue comparator IC; adapter passes them down)
  - ribbon cable from adapter to the KAYPRO-88 board.

- **Level-shifting conclusion for #11**: this machine is NMOS Z80A (TTL-ish outputs, high input thresholds) ⟹ use 74HCT/74AHCT toward the Z80 bus or 74LVC245 with careful thresholds; per survey (#31) the safe choice remains LVC245 host-side + HCT toward host inputs. Still confirm the other two machines before finalizing.

Open items (need user)

1. IMG_4870 / IMG_4871: what do they show? (They follow the CPU-area shots.) Please describe or retake — likely the missing piece for the adapter↔mainboard ribbon routing.
2. Which of the 3 machines is this (OCC1 label? board rev 24187A)?
3. Close-up of the blue comparator IC ("16-2103") to read the full part number and confirm the 0x7E/0x7F decode hypothesis.
4. Photos of the other two machines' Z80 areas for the NMOS/CMOS matrix in #29.

The Portable Companion extract index

(not yet written)